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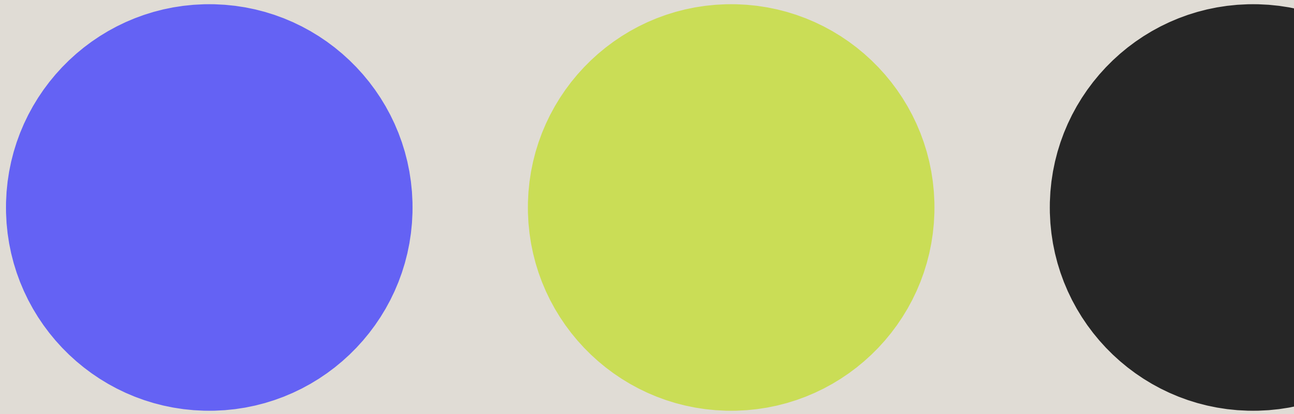
Generative AI

Prerequisites

Before starting there are things that you should know.

1. Python Programming
2. Machine Learning Fundamentals
3. Deep Learning Concepts
4. Understanding of Transformers
5. Basic NLP (tokenization, embeddings)

Here’s some important phases before starting.



Phase 1
Research layer

Phase 2
Bedrock models

Phase 3 Providers

Phase 4
This is where you comes in picture

You are going to use these LLM models and create some great AI applications.

Plan of Action

1. LLM foundation
2. RAG system
3. Agentic AI
4. project and deployment

Question 1
what is LLM ?

LLM = Large Language Model
Now break it into 3 words:

1. Large
It is trained on a huge amount of text data
(books, websites, Wikipedia, articles, code, conversations, etc.)

2. Language
It understands and generates human language
Like English, Hindi, Spanish... even programming languages like Python.

3. Model
It is a deep learning system (a neural network) that learns patterns from data.

Question 1
what is LLM ?

When you type:
“What is Python?”

The LLM does NOT:
- **Search Google**
- **Think like a human**
- **Understand meaning like we do**

Instead it:
- **Looks at patterns it learned**
- **Predicts the next most likely word**
- **Keeps predicting word by word**

Some popular LLM’s

GPT Models
Gemini Models
LLaMA
Claude
Grok
Mistral ... etc

You might have seen or heard about these large models.
In Generative AI, we don't build them from scratch
we learn how to use these pre-built LLMs effectively.

But how to use these models?

Open AI

```
from openai import OpenAI

# Create client
client = OpenAI()

# Send request to model
response = client.chat.completions.create(
    model="gpt-4o-mini",  -O
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "Explain what is a Large Language Model in simple terms."}
    ],
)

# Print output
print(response.choices[0].message.content)
```

Gemini

```
import google.generativeai as genai
import os

# Configure API key
genai.configure(api_key=os.environ["GOOGLE_API_KEY"])

# Choose model
model = genai.GenerativeModel("gemini-1.5-flash")

# Send prompt
response = model.generate_content(
    "Explain what is a Large Language Model in simple terms."
)

# Print response
print(response.text)
```

claude

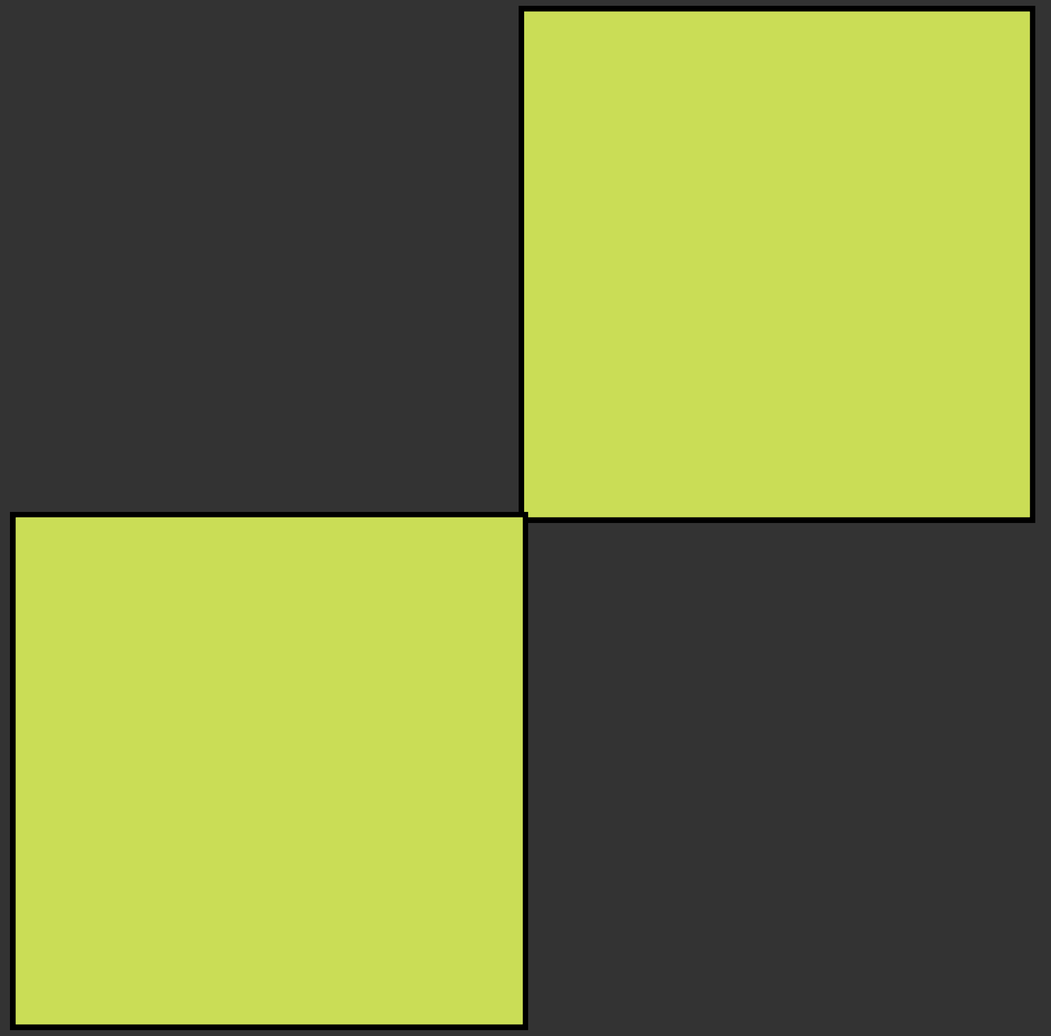
```
import anthropic
import os

client = anthropic.Anthropic(
    api_key=os.environ["ANTHROPIC_API_KEY"]
)

response = client.messages.create(
    model="claude-3-5-sonnet-latest",
    max_tokens=300,
    messages=[
        {"role": "user", "content": "Explain what is a Large Language Model in simple terms."}
    ],
)
print(response.content[0].text)
```

Problem

If every provider has a different SDK,
different syntax,
different response format...
How do we build scalable AI applications?



Langchain is the solution

Components

- Models
- Prompts
- Chains
- Memory
- Indexes
- Agents

Models

Models are the core intelligence layer.
They generate text, embeddings, or other outputs.
They connect your application to LLM providers like:

- OpenAI
- Google DeepMind
- Anthropic

Models

In practical Generative AI, there are mainly 3 important types of models:

Chat / Language Models

1. Generate text
2. Answer questions
3. Write code
4. Summarize content

Embedding Models

1. Convert text into numbers (vectors)
2. Used in search & RAG
3. Help find similar content

Multimodal Models

1. Work with more than just text
2. Can process images, audio, or files

Prompts

A prompt is the instruction we give to a language model.

It tells the model:

- What to do
  - How to respond
  - What format to use

Without prompts, the model has no direction.

Prompts

LLMs are powerful but they are not mind readers.
The quality of output depends on:
How clearly we instruct them.
Better prompt → Better output.

Prompts

Types of prompts

1. Simple Prompt

Just a direct instruction.
Example:
Explain what is Machine Learning.
Basic. Straightforward.

2. System + User Prompt

Modern chat models use roles:

- System → Sets behavior
- User → Asks question

Example:
System:
You are a helpful AI teacher.
User:
Explain LLMs in simple terms.
System controls tone and style.

Prompts

Types of prompts

3. Prompt Templates

Instead of writing prompts manually every time, we create reusable templates.
Example:
Explain {topic} in simple terms.
Now you can replace {topic} dynamically.
This is useful in applications.

4. Structured output

Used when we want:

- JSON output
- Bullet points
- Fixed format

Example:
Respond in JSON format with keys: definition, example.
This makes output machine-readable.

Chains

A Chain is a sequence of steps connected together.
Instead of:
One Prompt → One Response
We create:
Step 1 → Step 2 → Step 3 → Final Output

Chains

Real AI applications are not one-step.
Example:
User asks:
Summarize this article and translate it into Hindi.
What actually happens?

1. Summarize text
2. Translate summary
3. Return result

That’s a chain.

Memory

Memory allows the AI system to remember past interactions.
Without memory:
Every message is treated independently.
With memory:
The model can use previous context.

Memory

Example:
User: My name is Akarsh.
Later...
User: What is my name?

Without memory → Model doesn’t know.
With memory → It remembers.

Indexes

Indexes allow us to connect external data to an LLM.
LLMs only know what they were trained on.
Indexes help them use:

- PDFs
- Documents
- Databases
- Company data
- Websites

Indexes

LLMs:
❌ Don’t know your company data
❌ Don’t know private documents
❌ Don’t update in real-time

So how do we solve this?
We connect external knowledge.
That system is called Retrieval.

Agents

Agents are AI systems that can decide what action to take.
Unlike normal chains:
Chains → Follow fixed steps
Agents → Decide the next step dynamically

Agents

Normal LLM:
User → Prompt → Response
But what if the task requires:

- Searching the web
- Doing calculations
- Calling an API
- Using a database

The LLM alone cannot do that.
It needs tools.

Enough for now
Lets code

Welcome Back

Recap

- ⟶ LLM foundation
- ⟶ API keys
- ⟶ Used the Chat models
- ⟶ Used hugging face and local model
- ⟶ Used messages
- ⟶ Prompts
- ⟶ Structured output

Now you know me guys
we have to complete everything
about RAG in this video but while
teaching Generative AI I got to
know the best way of teaching
this is project based learning

So Lets start with a great project

CourseMate AI

CourseMate AI is an AI-powered study assistant designed to help students interact with their learning materials more efficiently. Modern students rely on multiple sources of study material such as lecture notes, textbooks, PDFs, and research papers. These documents are often long and difficult to navigate, making it time-consuming to find specific information.

CourseMate AI

CourseMate AI aims to solve this problem by allowing students to chat with their study materials. Instead of manually searching through pages of content, students can simply ask questions and receive accurate answers extracted directly from their documents.

CourseMate AI

By leveraging Retrieval-Augmented Generation (RAG), CourseMate AI combines document retrieval with large language models to provide context-aware explanations, summaries, and answers from the student's own study resources.

Development plan

Now to develop this kind of system we are going to follow some steps and the good thing is we are going to apply each and every step and create our project as well.

toh padhai bhi ho jaaye gi and project bhi ban jaaye ga.

Development plan

Step 1 - User Uploads Study Material

User Uploads Study Material
Students upload learning resources such as:

- PDFs
- lecture notes
- textbooks
- research papers

Development plan

Step 2 - Document Loading

The system loads documents using document loaders.
Goal:
Convert raw files into document objects that can be processed.

you may clean the document as well

Development plan

Step 3 - Text Splitting (Chunking)

Documents are usually too large for LLM context windows.
So we split them into smaller chunks.

Chunking improves retrieval accuracy.

Development plan

Step 4 - Embedding Generation

Each chunk is converted into a vector embedding.
Embedding models transform text into numerical vectors.

"Gradient Descent Optimization"
↓
[0.23, -0.81, 0.44, ...]

These vectors represent semantic meaning.

Development plan

Step 5 - Vector Database Storage

All embeddings are stored inside a vector database.
The vector database stores:

- embeddings
- original text chunks
- metadata

Development plan

Step 6 - User Asks a Question

Now the student interacts with the system.

Development plan

Step 7 - Query Embedding

The question is also converted into an embedding.

Development plan

Step 8 - Similarity Search

The vector database performs semantic similarity search.
Goal:
Find chunks that are most relevant to the question.

Development plan

Step 9 - Retriever Component

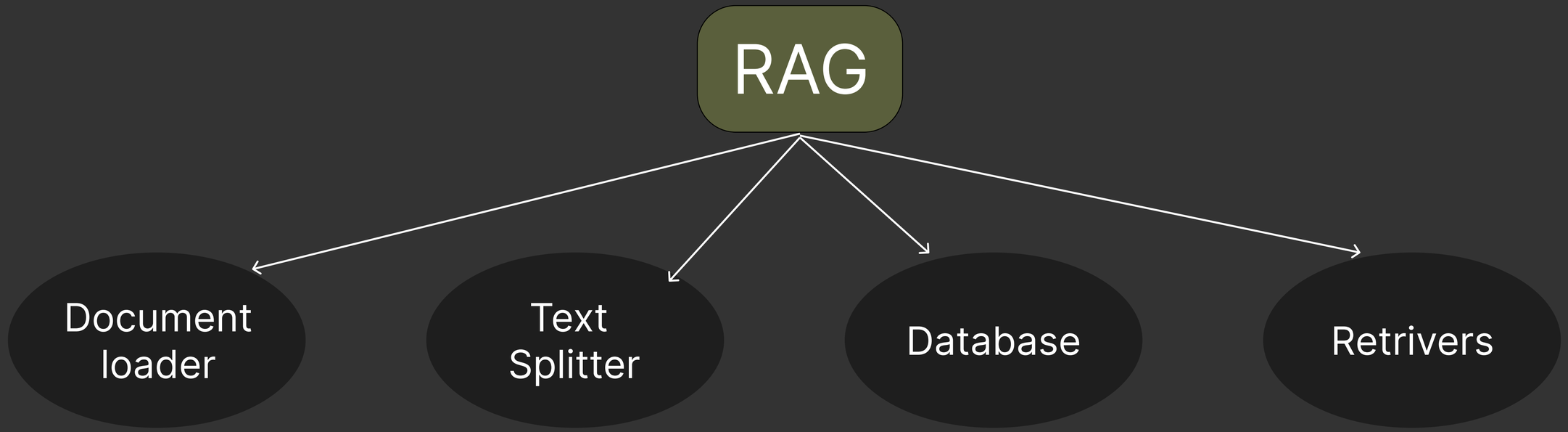
The retriever selects the top-k relevant chunks.
These chunks form the context.

Development plan

Step 10 - LLM answers

Based on the context the LLM answers.

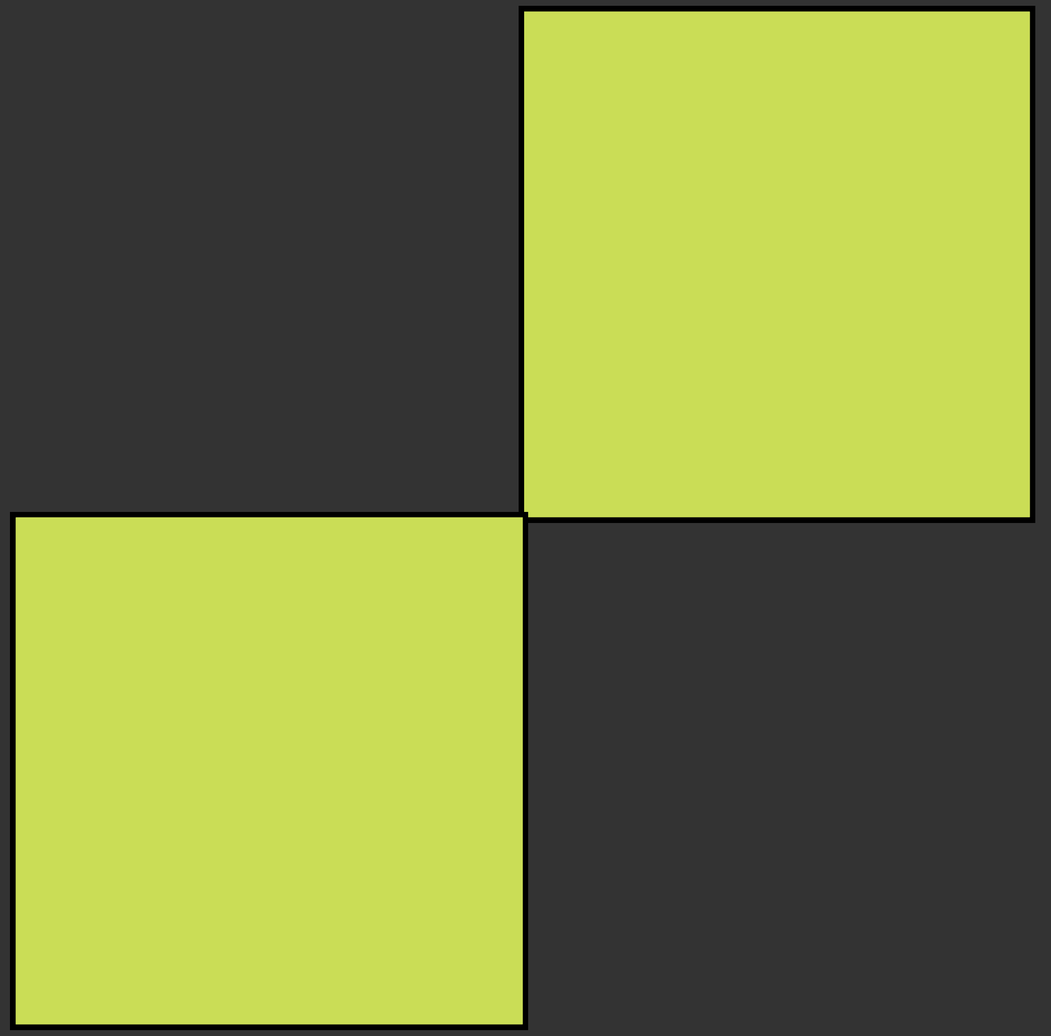
RAG



Retrieval Augmented Generation

Lets Go...

Ok so now lets create this and also I will teach each and every thing while making the project all the best.



Text Splitter

1. Most of the embedding models and language model have a context window and they cannot take infinite tokens at the same time so for that we have to use the text splitting.

Text Splitter

2. Better Retrieval in RAG

In Retrieval-Augmented Generation (RAG), we search through vector embeddings.

If the document is too large:

- embeddings become less precise

Smaller chunks allow the system to retrieve only the most relevant piece of information instead of the whole document.

Text Splitter

3. More Accurate Embeddings

Embeddings work best when the text represents one clear idea or topic.

If you embed very large text:

- multiple topics mix together
- semantic meaning becomes blurred

Chunking ensures each embedding represents a focused concept.

Text Splitter

4. Faster Processing
Working with smaller chunks:

- speeds up embedding creation
- speeds up similarity search in vector databases (Chroma, Pinecone, FAISS).

Large documents would make the pipeline slower and expensive.

Text Splitter

We are going to see 3 types of text splitting methods.

1. Character - Based Splitting
2. Token-Based Splitting
3. Semantic / Meaning-Based Splitting

Text Splitter

1. Character - Based Splitting

Character-based splitting is the simplest method of text splitting.
It divides a large document into smaller chunks based on the number of characters, without understanding words, tokens, or meaning.

Text Splitter

2. Token-Based Splitting

Token-based splitting divides a document into chunks based on the number of tokens, not characters.
Since LLMs process tokens instead of words, this method aligns better with how models actually read text.

Text Splitter

3. Semantic / Meaning-Based Splitting

Semantic / Meaning-Based Splitting divides a document into chunks based on the meaning and structure of the text, rather than characters or tokens.
The goal is to keep complete ideas together so that each chunk contains a coherent piece of information.

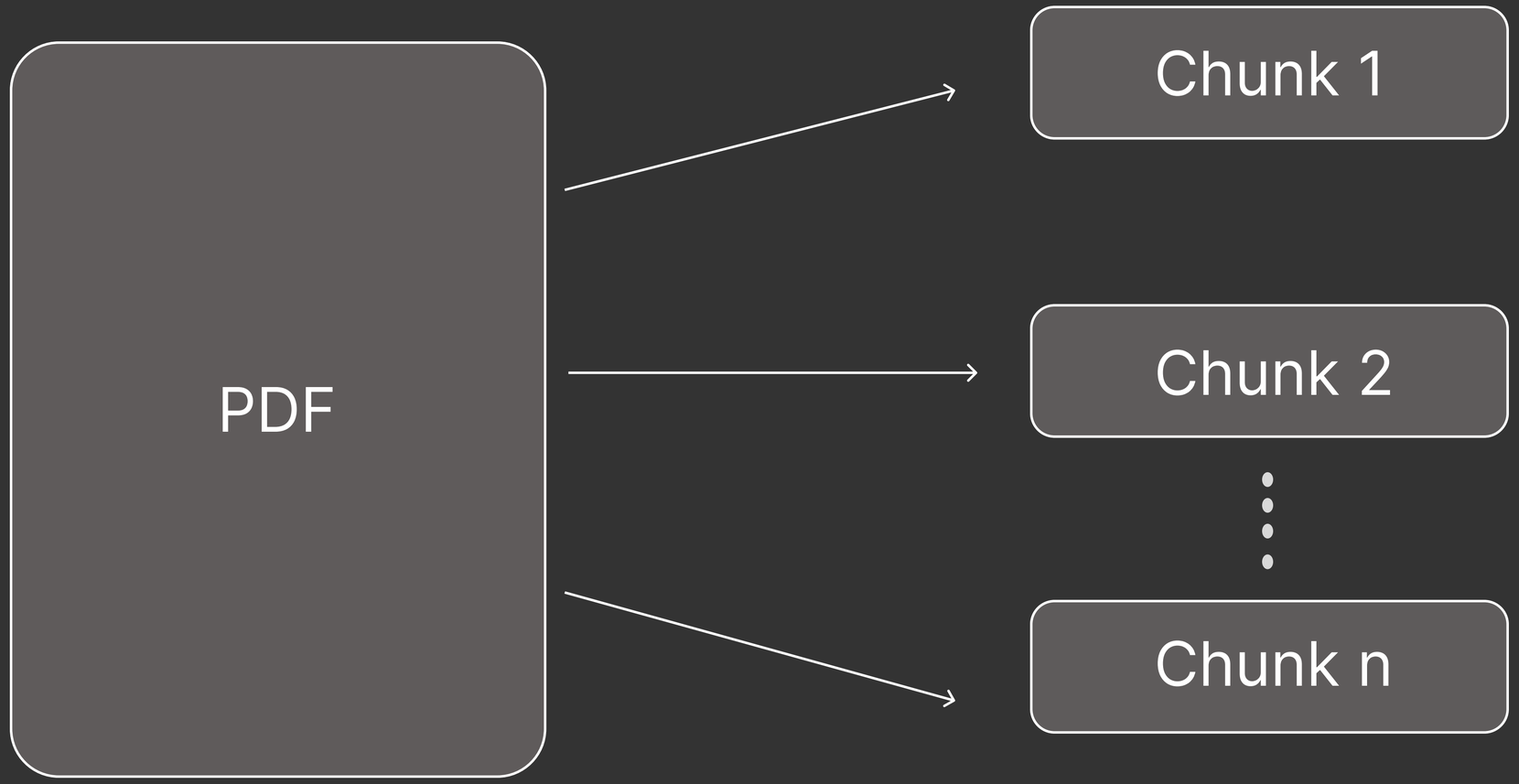
Text Splitter

Deep learning is a specialized area of machine learning that uses neural networks with multiple layers to automatically learn complex representations from large amounts of data. Natural Language Processing (NLP) is another important area of artificial intelligence that focuses on enabling computers to understand, interpret, and generate human language in tasks such as translation, chatbots, and sentiment analysis.

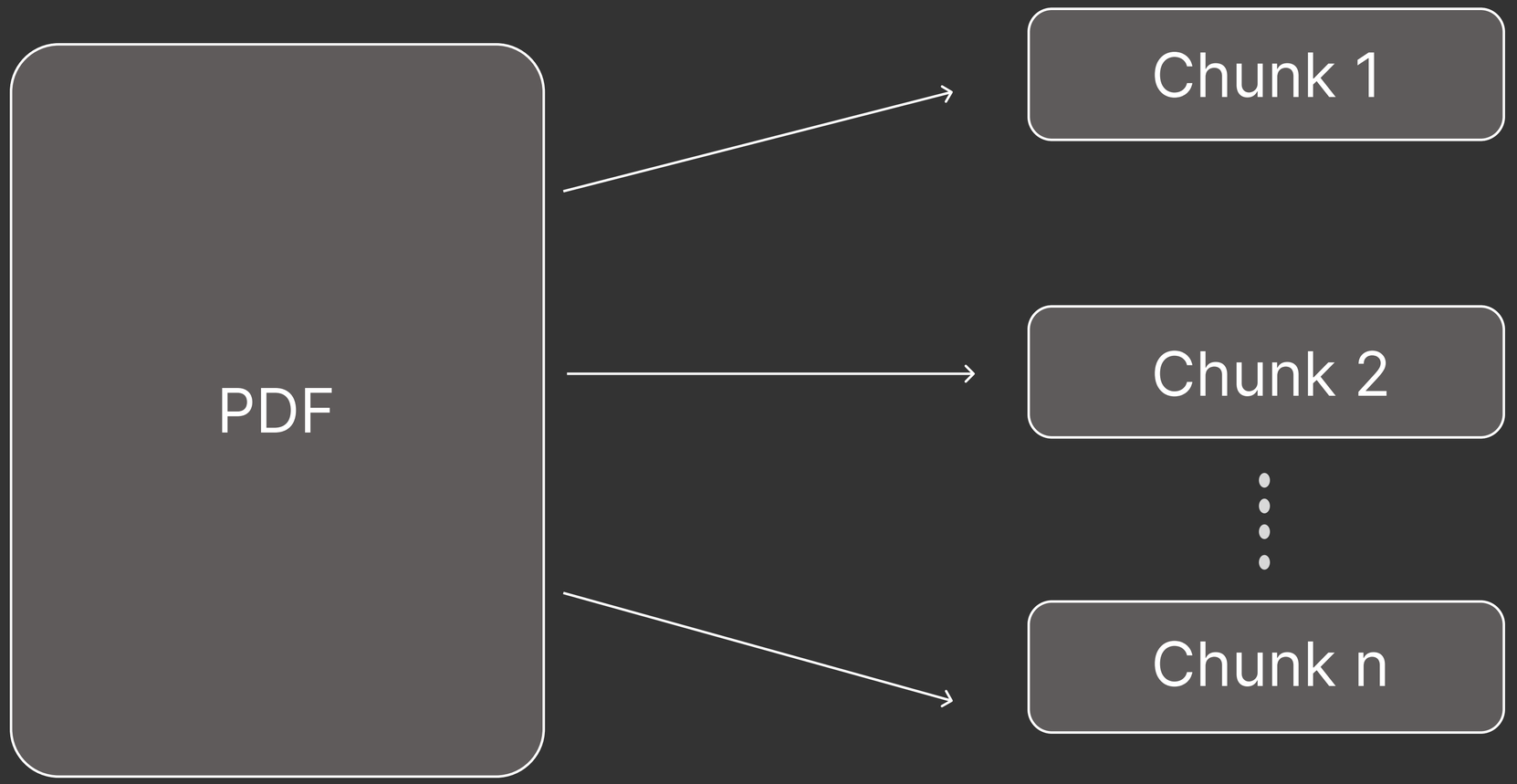
Vector Store

So first of all if we talk about our project we are creating a AI assistant which is going to search our pdf and give answers based on that.

Vector Store

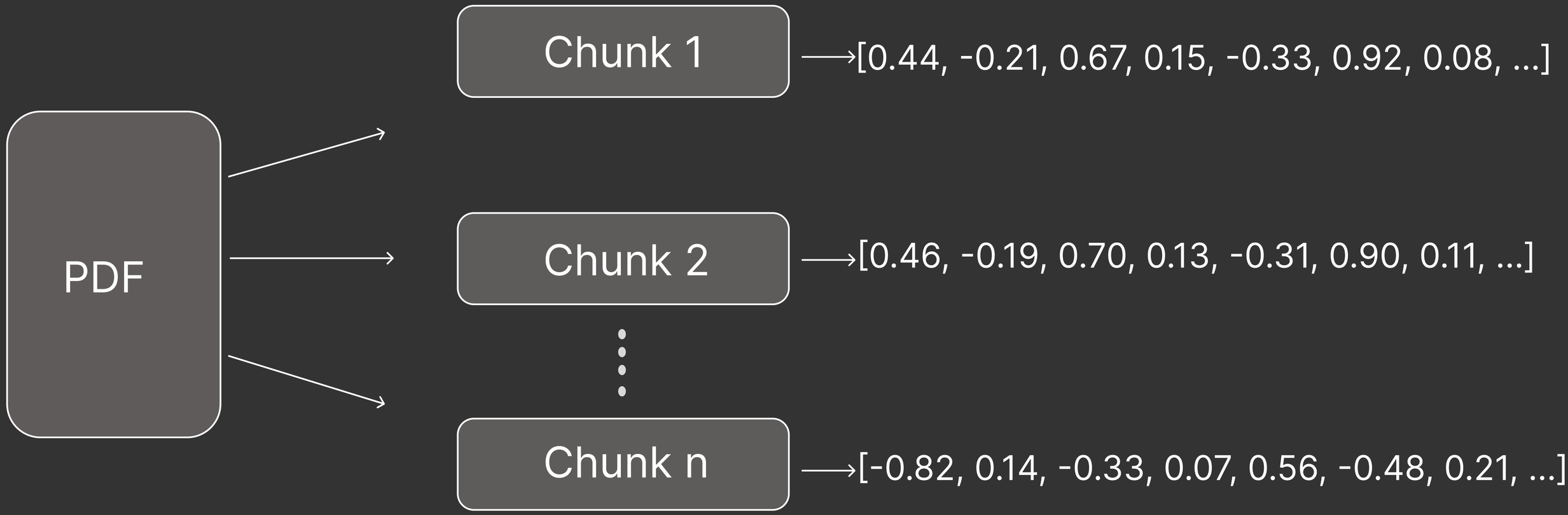


Vector Store

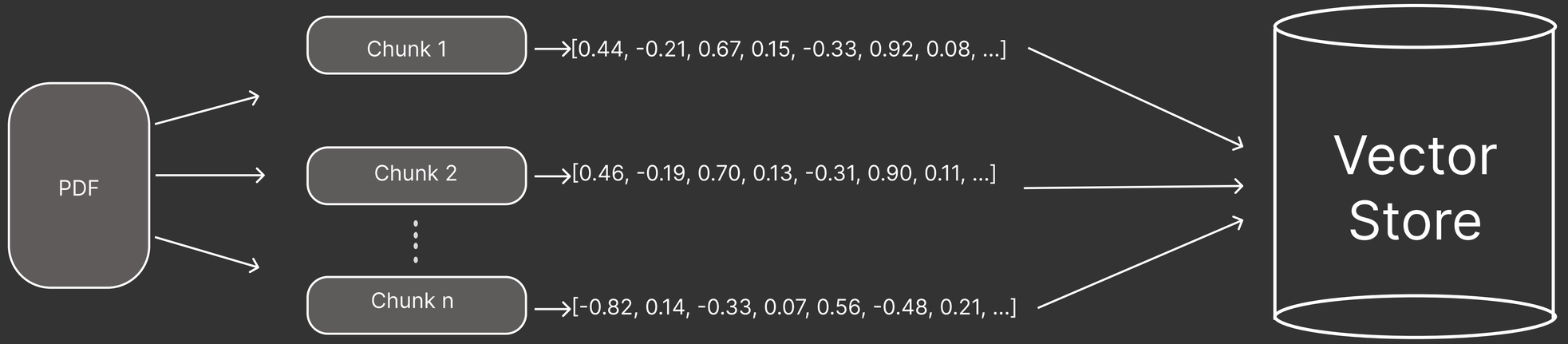


After this our next step will be to create embeddings of these chunks.

Vector Store



Vector Store



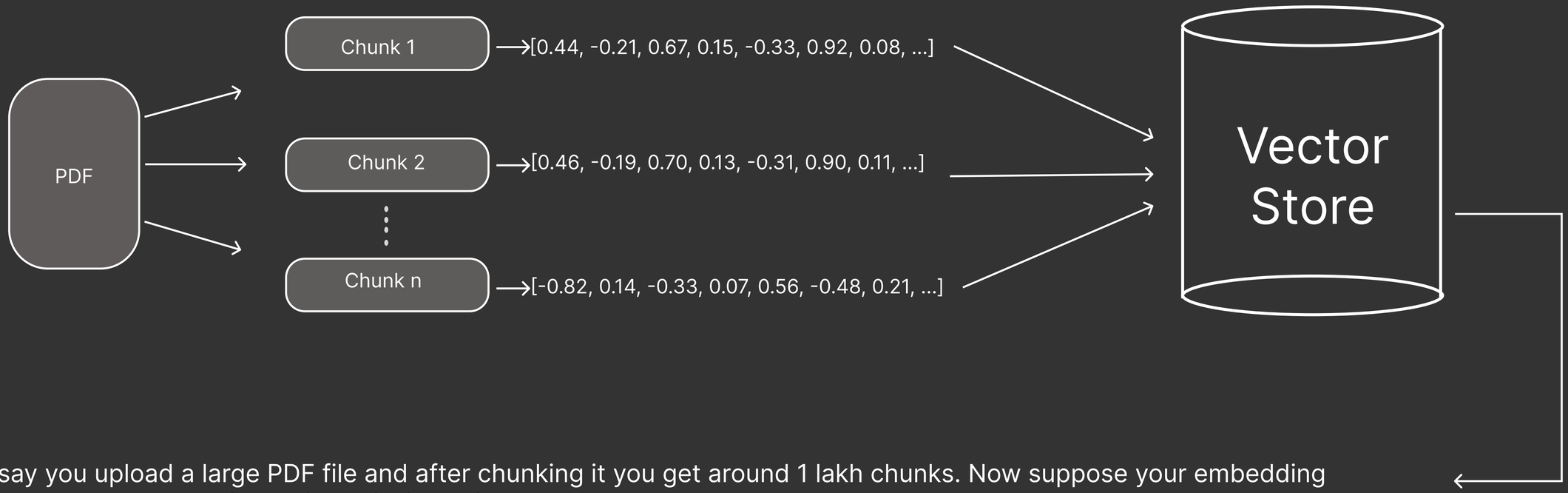
Now after this we need to store these embeddings and for that we are going to use vector store.

Vector Store

Now the question that might come to your mind is that we already have many types of databases like MySQL, PostgreSQL, MongoDB, etc. So why do we even need vector databases?

Vector Store

Answer is actually simple lets understand.



Let's say you upload a large PDF file and after chunking it you get around 1 lakh chunks. Now suppose your embedding model produces embeddings with 512 dimensions.

Vector Store

1 lakh
embedding

Now lets say this is a normal DataBase. and user give a query.

↓

What does the notes say about Natural Language processing.

Vector Store

1 lakh embedding

The Query is first converted into embedding

What does the notes say about Natural Language processing.

↓

[-0.82, 0.14, -0.33, 0.07, 0.56, -0.48, 0.21, ...]

512 - dimensions

Vector Store

1 lakh embedding

The biggest problem is this 512 dimension embedding is different from all the 1 lakh embeddings in our database so we conduct a similarity search with all the 1 lakh embeddings.

[-0.82, 0.14, -0.33, 0.07, 0.56, -0.48, 0.21, ...]

512 - dimensions

Vector Store

1 lakh
embedding

so you are working at 0(n) time complexity and you are searching for 1 lakh time and then finding out and this dataset can become more big so they are not the reliable option for similarity searching.

[-0.82, 0.14, -0.33, 0.07, 0.56, -0.48, 0.21, ...]

512 - dimensions

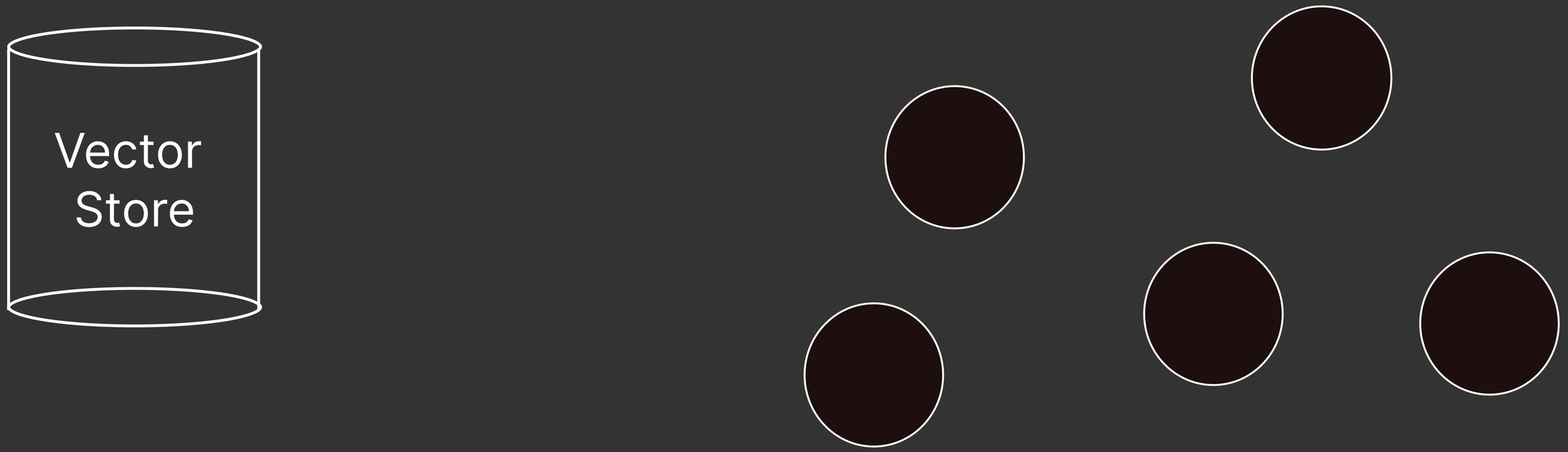
Vector Store

Where as vector store use Approximate Nearest Neighbour algorithms like -
HNSW
IVF
PQ

and all of these are indexing techniques
now we are not going to see all but lets see an example why they are fast.

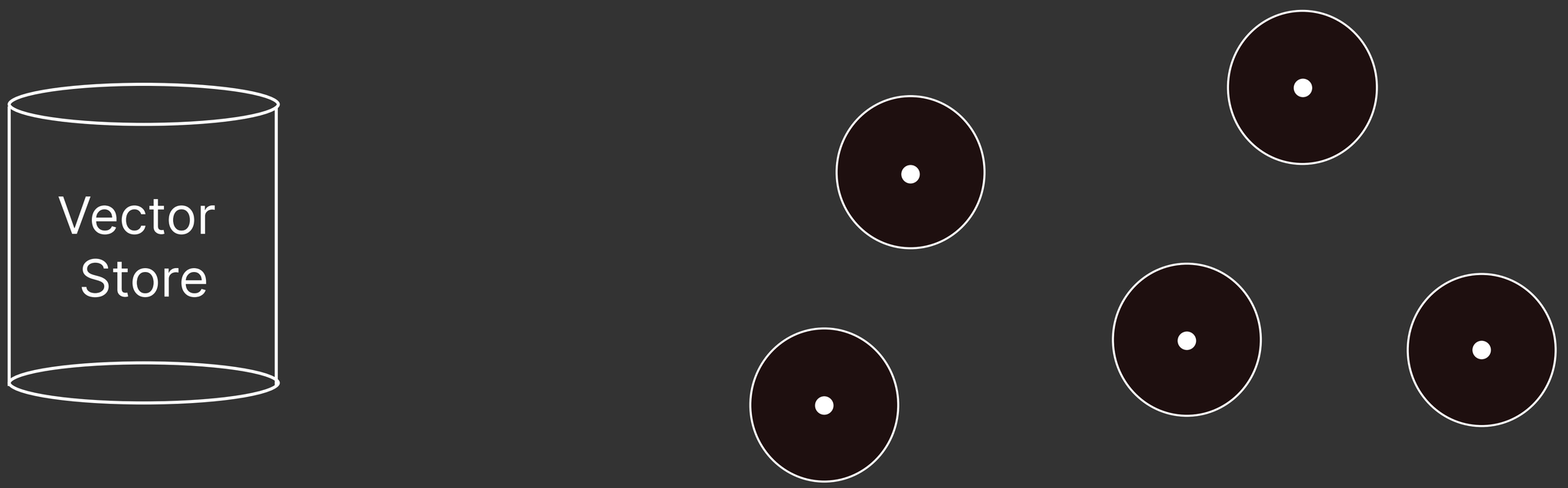
Vector Store

IVF (Inverted File Index) - Lets say we divide our Database in 5 clusters using k-means or any other algo. so each will have 20000 embeddings and all of them have some sort of similarity.



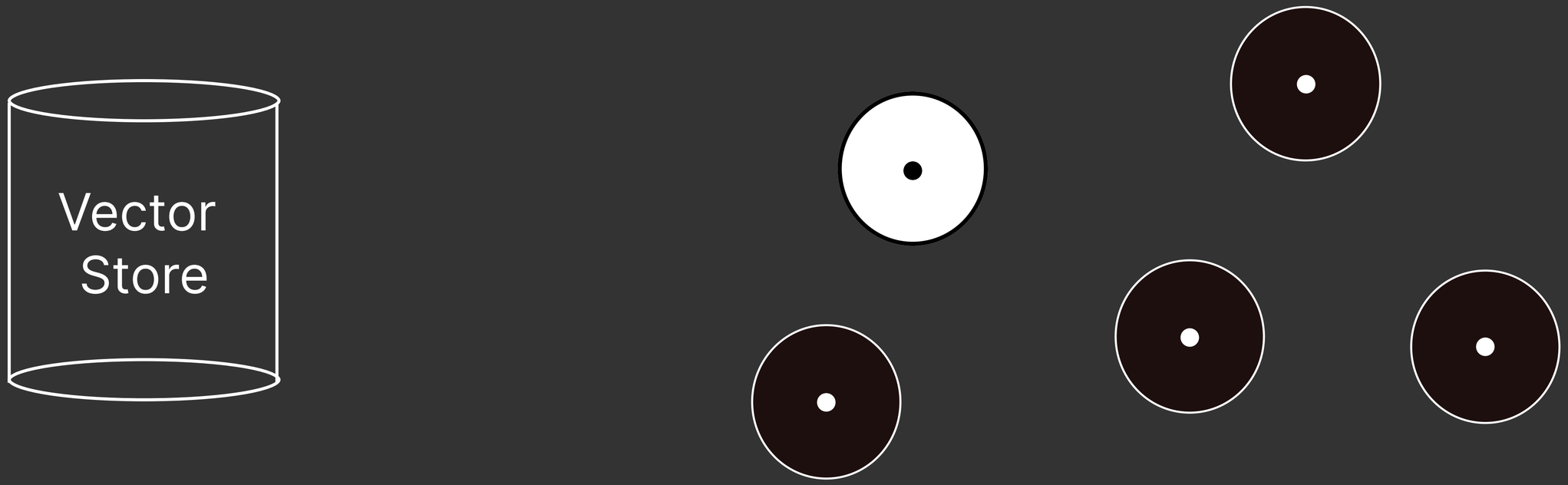
Vector Store

After that we calculate the average embedding of all the 20000 embeddings. we can call it centroid. so now we have 5 average embeddings.



Vector Store

Now for the query embedding we can find which cluster is suitable for searching and we search in those 20000 embeddings. that is actually 5 times faster that normal database



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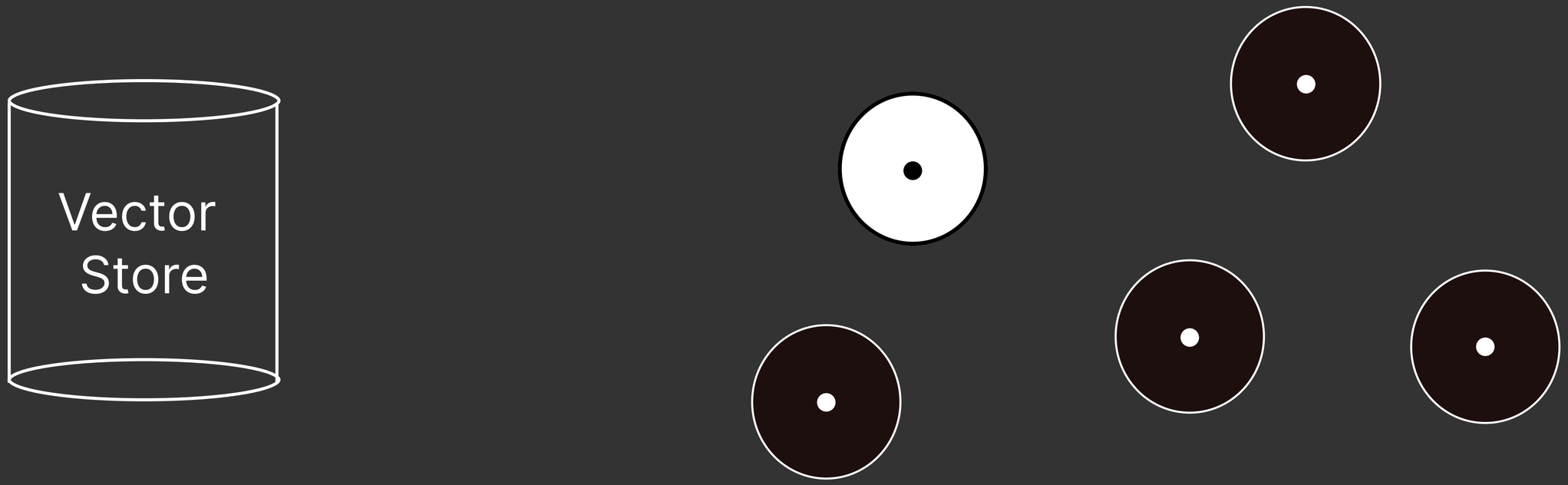


Vector Store

Feature	Normal Database	Vector Database
Data stored	Structured data (text, numbers, rows)	Embeddings (vectors like `[0.23, -0.41, ...]`)
Search type	Exact match	Similarity search
Query example	`WHERE movie = "Interstellar"`	"movies about space travel"
Indexing	B-Tree, Hash	HNSW, IVF, PQ
Use case	Banking, user records, transactions	AI search, recommendations, RAG
Comparison method	Equality or filtering	Cosine similarity / vector distance

Vector Store

Now there are many Vector stores like
These are widely used with frameworks like LangChain and LlamaIndex.

- Chroma
- FAISS
- Annoy

Vector Store

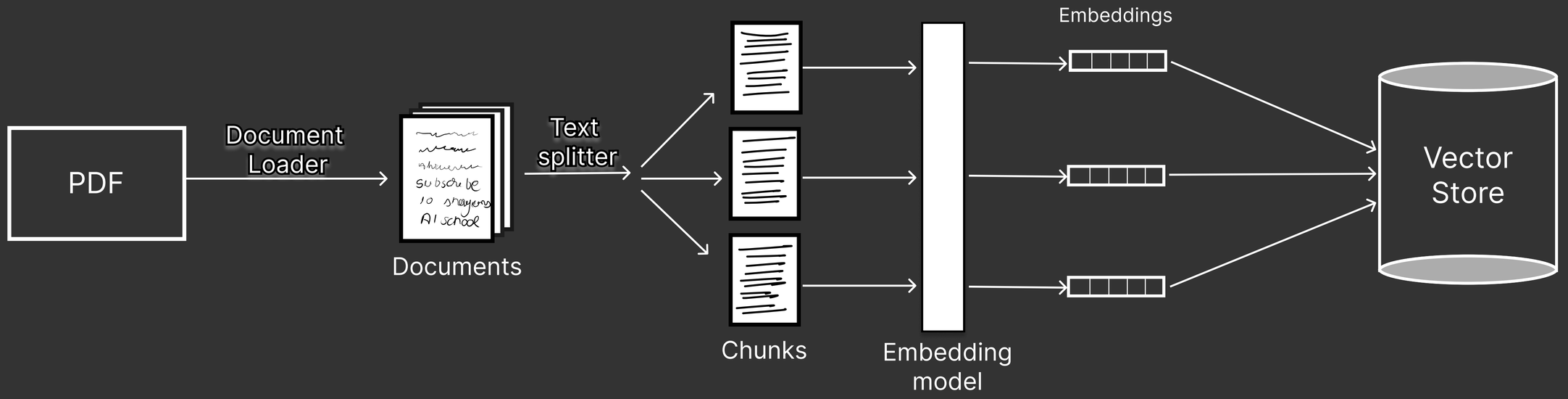
Enough Yapping lets see how to use one Vector Store.

Retrievers

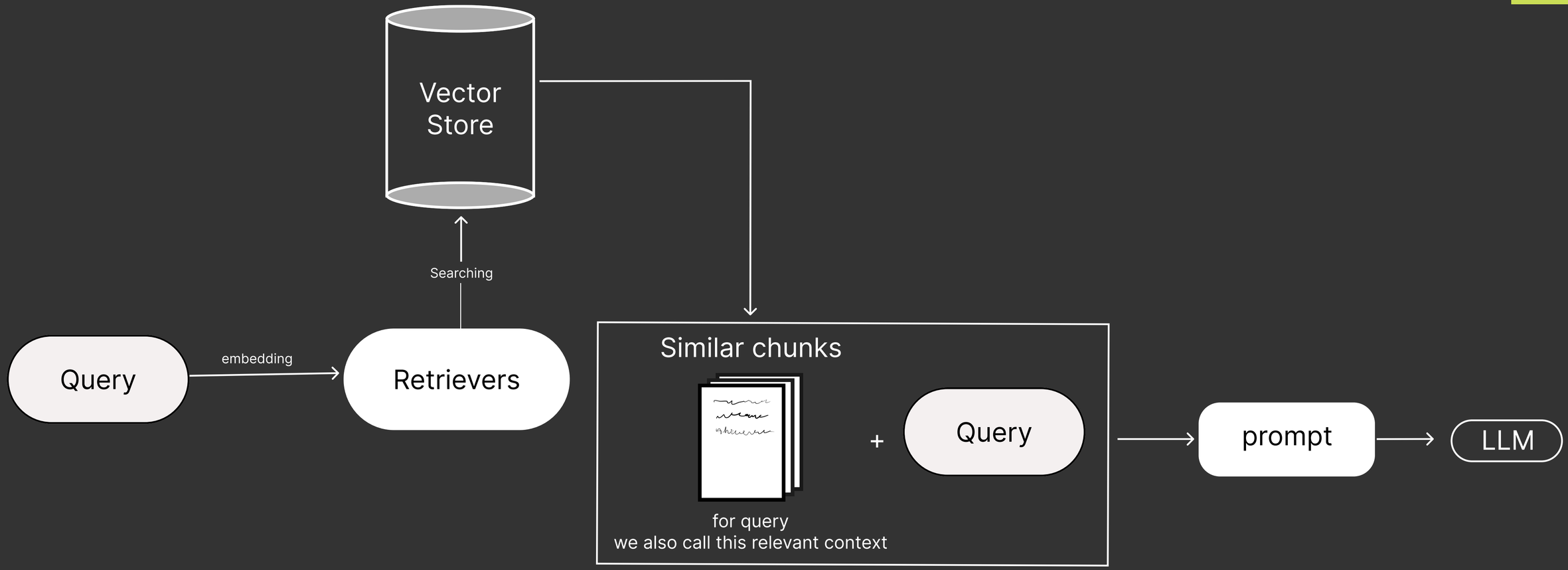
Now before understanding retrievers lets go through a diagram to understand the application that we are creating and we will also see the use of retrievers in that application.

Retrievers

Phase 1



Retrievers


Vector Store
Searching
Query
embedding
Retrievers
Similar chunks
+
Query
for query
we also call this relevant context
prompt
LLM

Retrievers

A retriever takes a user query and returns the most relevant documents or chunks from a database.

Now there are 2 types of retriever

1. by data source (Wikipedia, Arxiv, PubMed, etc.)
2. By retrieval strategy (Similarity, MMR, MultiQuery, etc.)

Retrievers

By data source

When we talk about retrievers using data sources, we mean retrievers that fetch information from specific external sources or databases instead of your own vector store. These retrievers connect to APIs or datasets and return relevant documents for the query.

Lets see an example.

Retrievers

By data source

So I hope now you get the gist how by data source works same code can generate the results from the wikipedia and other types of retrievers by data source.

Retrievers

By Retriever strategy

First go to Vector store code for an important information.

Retrievers

By Retriever strategy

First go to Vector store code for an important information.

Now you know that retrievers are not only used for basic similarity search strategies. They also provide some powerful search strategies that make them more useful, and they are runnables as well, which makes them suitable for building chains.

Retrievers

Search Strategies

In most RAG (Retrieval-Augmented Generation) systems, retrievers mainly use three core retrieval strategies. These are the ones you'll see most often in real projects and tutorials.

1. Similarity Search (Most Common)
2. MMR (Max Marginal Relevance)
3. MultiQuery Retriever

Retrievers

Search Strategies - Similarity Search

The system compares the query vector with document vectors using similarity metrics like:

- Cosine similarity (most common)
- Dot product
- Euclidean distance

The retriever finds the most similar vectors.

Top-K documents are retrieved

Retrievers

Search Strategies - MMR (maximum marginal relevance)

Lets understand this with an example :
Imagine you are building a RAG system for your students.
Your knowledge base contains many chunks about Gradient Descent.
Your database chunks look like this:

Chunk 1: Gradient descent is an optimization algorithm used in machine learning.
Chunk 2: Gradient descent minimizes the loss function.
Chunk 3: Gradient descent is an optimization that minimizes the loss function.
Chunk 4: Neural networks use gradient descent for training.
Chunk 5: Support Vector Machines are supervised learning algorithms.

Retrievers

Now the user asks
What is Gradient Descent ?

Normal Similarity search finds the most similar chunks to the query embedding.
and gives chunk 1 , chunk 2 , chunk 3

but if you see carefully all the 3 chunks are saying the same thing.

This wastes:

- context window
- token usage
- information diversity

The LLM doesn't learn new information.

Retrievers

MMR stands for:
Max Marginal Relevance

Its goal is to balance two things:
1. Relevance to the query
2. Diversity among retrieved documents

So instead of retrieving similar documents, it retrieves different but relevant documents.

Retrievers

Search Strategies - MultiQuery

Imagine you built a RAG system for machine learning notes.

Your database contains these chunks:

Chunk 1: Gradient descent is an optimization algorithm used to minimize loss functions.
Chunk 2: Neural networks are trained using optimization algorithms like gradient descent.
Chunk 3: Weight updates in machine learning models are done using gradient descent.
Chunk 4: Support Vector Machines are supervised learning algorithms.

Retrievers

Chunk 1: Gradient descent is an optimization algorithm used to minimize loss functions.
Chunk 2: Neural networks are trained using optimization algorithms like gradient descent.
Chunk 3: Weight updates in machine learning models are done using gradient descent.
Chunk 4: Support Vector Machines are supervised learning algorithms.

Now the user asks:
"What is gradient descent?"

the retriever might return:
Chunk 1
Chunk 3

Retrievers

Chunk 1: Gradient descent is an optimization algorithm used to minimize loss functions.
Chunk 2: Neural networks are trained using optimization algorithms like gradient descent.
Chunk 3: Weight updates in machine learning models are done using gradient descent.
Chunk 4: Support Vector Machines are supervised learning algorithms.

But notice something
There are other relevant chunks like Chunk 2 that might not be retrieved.
Why?
Because the query wording is limited.

Retrievers

The Core Problem
A single query might not capture all ways a concept can be expressed.
for example :
"What is gradient descent?"

could be asked as

"Explain gradient descent"
"How does gradient descent work?"
"Optimization algorithm used in neural networks"

Retrievers

The Core Problem
A single query might not capture all ways a concept can be expressed.
for example :
"What is gradient descent?"

could be asked as

"Explain gradient descent"
"How does gradient descent work?"
"Optimization algorithm used in neural networks"

Different wording → different embeddings → different search results.
This is where MultiQuery Retriever helps.

Retrievers

So Instead of using one query, it generates multiple variations of the query using an LLM. so now the pipeline becomes

User Query
↓
LLM generates multiple queries
↓
Retriever searches for each query
↓
Combine all retrieved documents

Retrievers

So now lets finally complete our project.

Made with Goodnotes

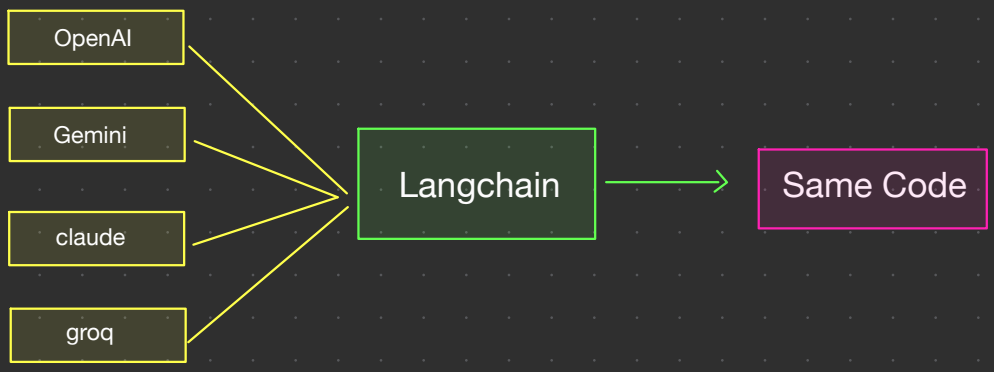
Runnables

So imagine you want to build a chatbot. For that, you obviously need an LLM.
Now earlier, there was a big problem. There were many different LLM providers like OpenAI, Anthropic, Google, and others. The issue was that each of them had their own SDK, their own syntax, and their own way of calling the model.

Because of this, if you built your project using one provider, like OpenAI, and later decided to switch to something like Gemini or Claude, you basically had to rewrite a large part of your code. That was frustrating and time-consuming.

This is where LangChain comes in and solves the problem beautifully.

LangChain provides a unified interface. That means no matter which LLM you use, the way you write your code remains almost the same. You just change the model, and everything else stays as it is.
This flexibility is one of the biggest reasons why developers love LangChain. It gives you the freedom to experiment and switch between models without rebuilding your entire project from scratch.



So once developers started using LLMs, things got interesting very quickly.

At first, it was simple. You had a model, you gave it some input, and it gave you an output. That was it.
But as people started building real applications, they realized something important a single LLM call is not enough.
Let's say you're building something like a chatbot, or a document Q&A system, or even a summarization tool. Now your application is not just one step anymore. You might need to first format the input properly, then send it to the model, and then clean or structure the output before showing it to the user.
So now instead of one step, you suddenly have multiple steps.
You have:

- • some kind of input processing
- • then a prompt template
- • then the LLM call
- • and then an output parser

At this point, you can clearly see that your application is no longer just a single model call.
Now you have multiple components involved things like LLMs, prompt templates, output parsers, and even custom logic in between.
And all of these components need to work together in a proper sequence to give you the final result.

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Some of the most common ones are:

- • **LLM / Chat Model** → the core model that generates responses
- • **Prompt Template** → formats the input before sending it to the model
- • **Output Parser** → cleans or structures the model's response
- • **Retriever** → fetches relevant data (used in RAG systems)
- • **Document Loader** → loads data from PDFs, files, etc.
- • **Text Splitter** → breaks large text into smaller chunks
- • **Embeddings Model** → converts text into vectors
- • **Vector Store** → stores and searches embeddings

```python
from dotenv import load_dotenv
load_dotenv()

from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser

# 1. Model
model = ChatOpenAI(model="gpt-4o-mini")

# 2. Prompt Template
prompt_template = ChatPromptTemplate.from_template(
    "Explain {topic} in simple words"
)

# 3. Output Parser
parser = StrOutputParser()

# Step-by-step manual flow (NO chains)

# Format the prompt
formatted_prompt = prompt_template.format_messages(topic="Machine Learning")

# Call the model (old style)
response = model.predict_messages(formatted_prompt)

# Parse the output
final_output = parser.parse(response.content)

print(final_output)
```

The base code
We and everyone
else used
to crete a basic
chatbot

So even though we now have better components like prompt templates and output parsers…
we are still manually connecting everything step by step.

So now if you look at this, even though we are using better components like prompt templates, models, and output parsers… we are still doing everything manually.
We have to:

- • format the prompt
- • call the model
- • parse the output

Now imagine you are building a large application, something like a RAG system.
In that case, you're not just dealing with one or two components anymore. You have a retriever, a prompt template, an LLM, an output parser, maybe even embeddings and a vector database.
And the problem is… you have to call each of these components manually.
First you retrieve the data, then you format the prompt, then you call the model, then you parse the output…
and you have to manage the flow between all of them yourself.
As your application grows, this quickly becomes messy and difficult to maintain.
Handling so many moving parts manually is not just repetitive, it's also error-prone.

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```python
from dotenv import load_dotenv
load_dotenv()

from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser

# 1. Model
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# 2. Prompt Template
prompt_template = ChatPromptTemplate.from_template(
    "Explain {topic} in simple words"
)

# 3. Output Parser
parser = StrOutputParser()

# 🔥 Chain (everything connected automatically)
chain = prompt_template | model | parser

# Run the chain
result = chain.invoke({"topic": "Machine Learning"})

print(result)
```

So chains were definitely a big improvement.
They made our code cleaner, reduced repetition, and allowed us to connect multiple components into a single pipeline.
But as developers started building more complex applications, a new problem started to appear.
There wasn't just one type of chain.
LangChain had introduced many different chains for different use cases.
For example:

- • **LLMChain** → for simple prompt + model tasks
- • **SequentialChain** → for multi-step workflows
- • **RouterChain** → for dynamic decision making
- • **RetrievalQA Chain** → for RAG-based applications

Now on paper, this sounds useful.
But in practice, it became confuing.
As a developer, you always had to think:
"Which chain should I use for this problem?"
And sometimes, your use case didn't fit perfectly into any one chain.
So you either had to:

- • force your problem into an existing chain
- • or combine multiple chains in a complicated way

This made things harder instead of easier.
On top of that, chains were not very flexible.
If you wanted:

- • custom logic in between steps
- • parallel execution
- • or more control over how data flows

Chains started to feel restrictive.
So instead of simplifying things, the growing number of chains actually created confusion and friction for developers.
And because of all these limitations, LangChain made an important decision.
They moved away from the concept of predefined chains and introduced a much more flexible system… Runnables

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Now this is where things become really interesting.
Instead of having different types of chains for different use cases, LangChain introduced a much simpler and more powerful idea.
They said what if every component we are using is treated the same way?
Think about it.
We already have multiple components in our application:
we have prompt templates, we have models, we have output parsers, retrievers, and even custom functions.

Now instead of treating all of these differently, LangChain introduced a unified concept…
Everything is a Runnable.
That means:
your prompt template is a runnable,
your model is a runnable
your output parser is a runnable,
even your own Python function can be a runnable.
So instead of thinking in terms of different types of chains, now you just think in terms of **runnables that can be connected together**.

Now imagine all these components as **Lego blocks**.
Each block has its own purpose. One block might be a prompt template, another might be a model, another could be an output parser, and so on.
Individually, they all do different things.
But here's the interesting part…
No matter what the block does, the way you connect one block to another is always the same.
Just like Lego.
You don't need a different way to connect different pieces. Every block follows the same connection structure.
And that's exactly what LangChain did with runnables.

Now there's one really interesting thing about runnables.
All these different components, whether it's a prompt template, a model, an output parser, or even your own function, they all expose the same kind of methods.
For example, you've already seen one of them:
👉 **invoke()**
This is used to run the component on a single input and get the output.
But that's not the only one.
You also have:
batch() when you want to process multiple inputs at once
stream() when you want to get output token by token like real-time responses

Now think about this…
Completely different components, doing completely different tasks, but all of them can be used in the exact same way.
And that is the real power here.
You don't need to learn different ways to use different components. Once you understand how one runnable works, you can use any other runnable the same way.

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Actual Runnables

These are your components/runnables:

- • LLM
- • Prompt Template
- • Output Parser
- • Retriever
- • Tools

Basically: *they do the actual job*

Runnable Sequence

Now that we understand what runnables are, let's see how we actually use them in real applications.
We're going to learn this in three simple steps.
First, we'll start with the most basic and most important pattern a simple sequence.
In this, we just connect components like a pipeline. The output of one component becomes the input of the next. So your flow becomes something like prompt to model to parser. This is the foundation, and honestly, most applications start like this.

Parallel Runnable

Now till this point, we have only seen how to build a single pipeline where the input flows step by step and produces one output. But in real-world applications, sometimes we don't want just one result. We might want multiple outputs at the same time, like a short explanation and a detailed explanation, or maybe the same answer in different formats. Instead of running the pipeline again and again for each case, we can use a Parallel Runnable.

In this approach, we define multiple pipelines inside a dictionary, and all of them run simultaneously on the same input. Each pipeline produces its own output, and in the end, we get all the results together in a structured form. So instead of one input giving one output, now one input can give multiple outputs in a single execution, which makes our applications more powerful and efficient.

Runnable Lambda

Now sometimes in our pipeline, we don't just want to pass data forward, we might want to slightly modify it or pick a specific part of it before sending it to the next step. This is where RunnableLambda comes in. It allows us to write a simple Python function and insert it inside our pipeline as a runnable. For example, if our input is a dictionary with multiple keys, and we only want to send one specific part of it to a particular pipeline, we can use RunnableLambda to extract that part. So instead of the entire input going everywhere, we can control exactly what each component receives. In simple terms, RunnableLambda lets us add custom logic inside our runnable flow, making our pipelines more flexible and powerful.

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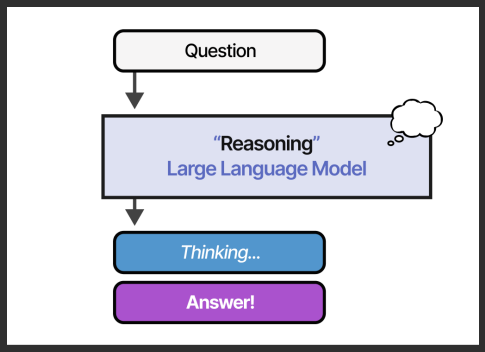
Runnable Passthrough

Runnable Passthrough is used when we want to keep the original input or some intermediate data while passing it through the pipeline. Normally, in a sequence, each step replaces the previous output, so earlier data gets lost. But in many real-world scenarios, we need to carry multiple pieces of information together. RunnablePassthrough allows us to forward the input as it is, without modifying it, so that it can be used along with other outputs in later steps. In simple terms, it helps us preserve data while still continuing the flow of the pipeline.

Tools

An LLM (Large Language Model) is an AI system trained to understand and generate human language.



Core capabilities an LLM can think / reasoning
And then generate an answer.

But our LLMs are not good with many things like

- No Real-Time Data
- Not Reliable for Calculations
- Cannot Call APIs
- Cannot Access Your Data

Cannot Perform Actions

Examples:

- • Send email ✗
- • Book ticket ✗
- • Run code ✗
- • Update database ✗

Tools are simply **functions or external services that an LLM can use to do real-world tasks**. On its own, an LLM can only generate text, but with tools it can **fetch live data, perform accurate calculations, access databases, or execute actions**. In short, tools extend an LLM from just *thinking* to actually *doing things*.

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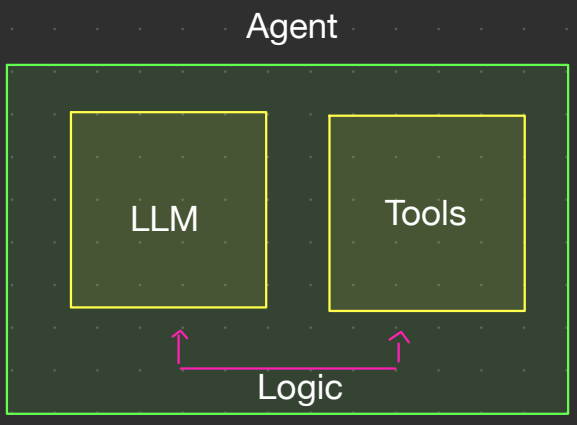
We use an LLM as the brain, tools to perform tasks, and our code to define the logic together, this creates an agent.

LLM decides *what to do*,
tools help *do the task*,
and our code controls *how everything works together*.

Agent = Brain (LLM) + Tools + Control Logic (our code)



Tools category

Build-in Tools

Built-in tools are tools that are already provided by a framework, so you don't have to create them from scratch. They are ready to use and help the LLM perform common tasks like searching the internet, doing calculations, or working with data. You just connect them to your LLM, and it can start using them easily.

Custom tools

Custom tools are tools that **you create yourself** based on your needs. They are usually simple functions written in code that help the LLM perform specific tasks, like accessing your database, processing your data, or calling your own APIs. This allows you to give the LLM **exact abilities that match your project**.

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Tool Binding

Tool binding means connecting our tools to the LLM so it can use them when needed.

We give our tools to the LLM, and now the LLM knows:

- • what tools are available
- • what they do
- • when to use them

Tool calling

In tool binding, we gave tools to the LLM…
in tool calling, the LLM chooses to use them.

The LLM does NOT execute the tool it only suggests the tool call.

Tool Execution

Tool execution is when we actually run the tool after the LLM decides to use it.
LLM selects the tool → we execute it → get the result → send it back to LLM.

AI agents

So let's say today is Sunday and you want to go on a date with your girl/boy friend now what are you going to do.

You might:

- • Check weather 🌦️
- • Order food 🍔
- • Book a cab 🚗
- • Look for a movie 🎬

You are not doing one task… you are making *decisions* and using different tools.

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Title:

Now Imagine AI Doing This
Content:
User says:
"Plan my evening"
AI should:

- • Understand intent
- • Decide steps
- • Use tools
- • Give final result

💡 This is where **Agents** come in

Agent = Brain + Tools + Decision Making

Lets Create an Agent

We are going to create a city intelligence Agent and we need some tools for that

1. Weather Tool
API: OpenWeatherMap
Purpose:

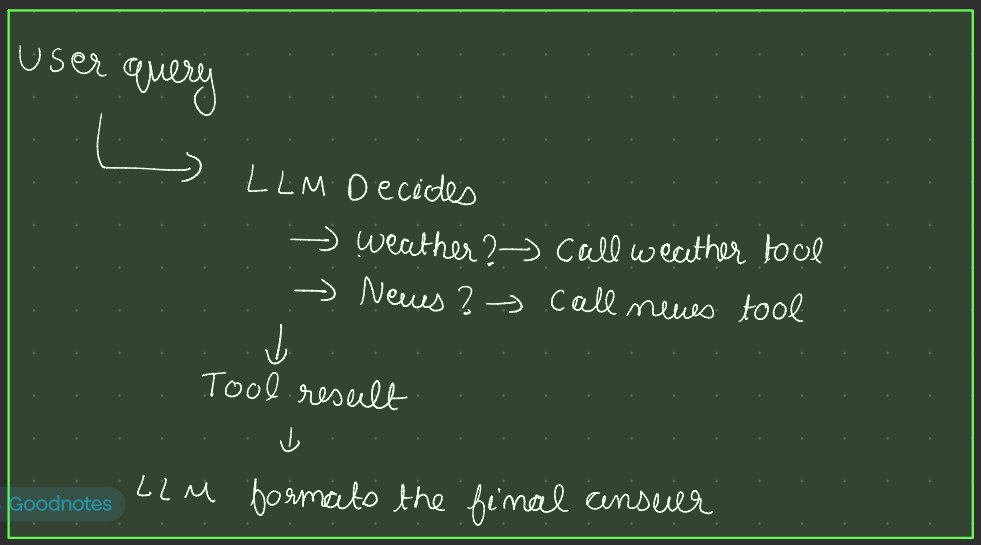
- • Get **real-time weather data**

Why it's needed:
LLM does NOT know live weather → needs external tool

2. News Tool
API: Tavily
Purpose:

- • Get **latest news about a city**

Why it's needed:
LLM does NOT know current events → needs search



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In our code, we didn't just call an API or use an LLM separately. We created a system that could understand a user's intention, decide what action is needed, execute that action using external tools, and then use the result of that action to generate a meaningful response. For example, when the user asked about weather, the LLM first analyzed the question and decided that weather information is required. Then it triggered the weather tool, which fetched real-time data from an API. After that, we fed that result back into the LLM, and only then the LLM generated the final answer.
So what we really built was not just code we built a decision-making loop where the LLM acts like a brain and the tools act like hands interacting with the real world.

In your manual code, every single step is visible:

- • when the LLM is called
- • when a tool is triggered
- • what data is passed
- • what result comes back

You can literally trace the entire flow from user input to final output.

But there are some problems with this code I am not saying the code is bad actually its really good but lets see some problems

Too Much Boilerplate Code

In your implementation, you had to write:

- • the loop
- • tool execution logic
- • message handling
- • condition checks

All of this is **repetitive work**.

Manual Loop Management is Error-Prone

You are responsible for:

- • when to stop the loop
- • when to continue
- • handling edge cases

A small mistake like:

- • forgetting `continue`
- • misplacing a condition

👉 can break the entire agent.

Scaling Becomes Difficult

Right now you have:

- • 2 tools (weather + news)

Imagine:

- • 10 tools
- • multiple conditions
- • complex workflows

Your code will quickly become:

- • messy
- • hard to maintain
- • difficult to extend

Harder to Maintain Over Time

As your project grows:

- • more tools
- • more features
- • more logic

The manual loop becomes harder to manage.

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`create_agent` is a high-level abstraction provided by LangChain that automates everything we manually implemented in our agent. Instead of writing the loop, handling tool calls, managing messages, and deciding when to stop, we simply define the model and the tools, and the framework takes care of the entire execution process internally. The core logic remains the same the LLM still decides which tool to use, tools still fetch external data, and results are passed back for final response generation but all of this happens behind the scenes. The main difference is that our manual approach gives full control and visibility over each step, while `create_agent` hides this complexity to make development faster, cleaner, and easier to scale.

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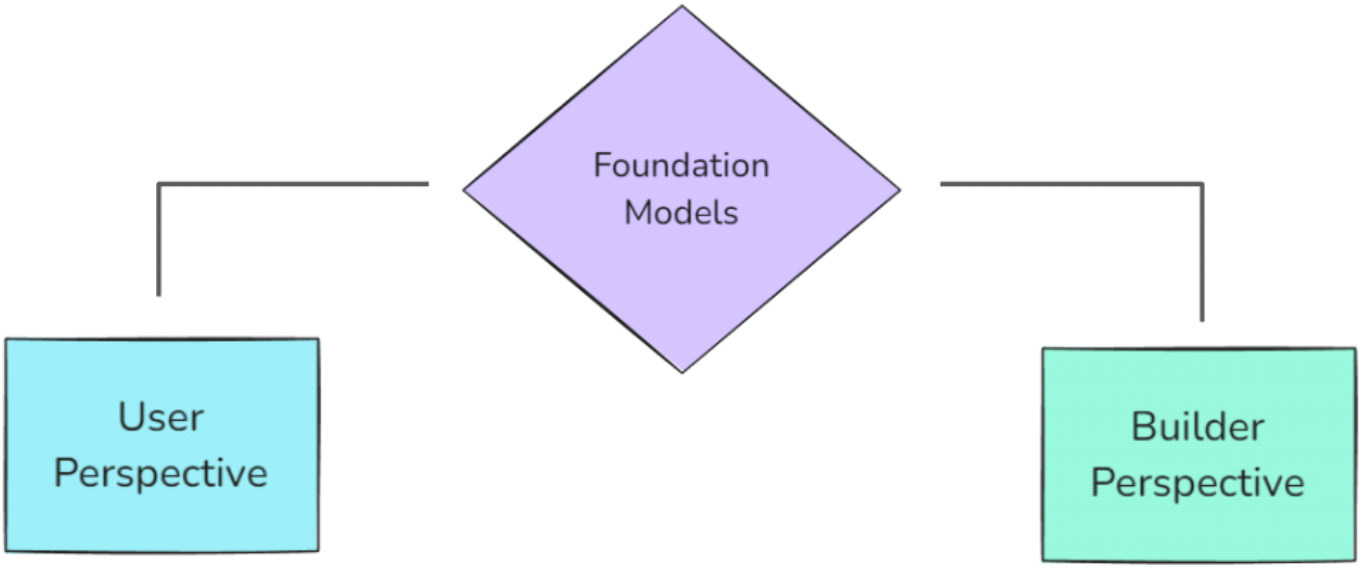
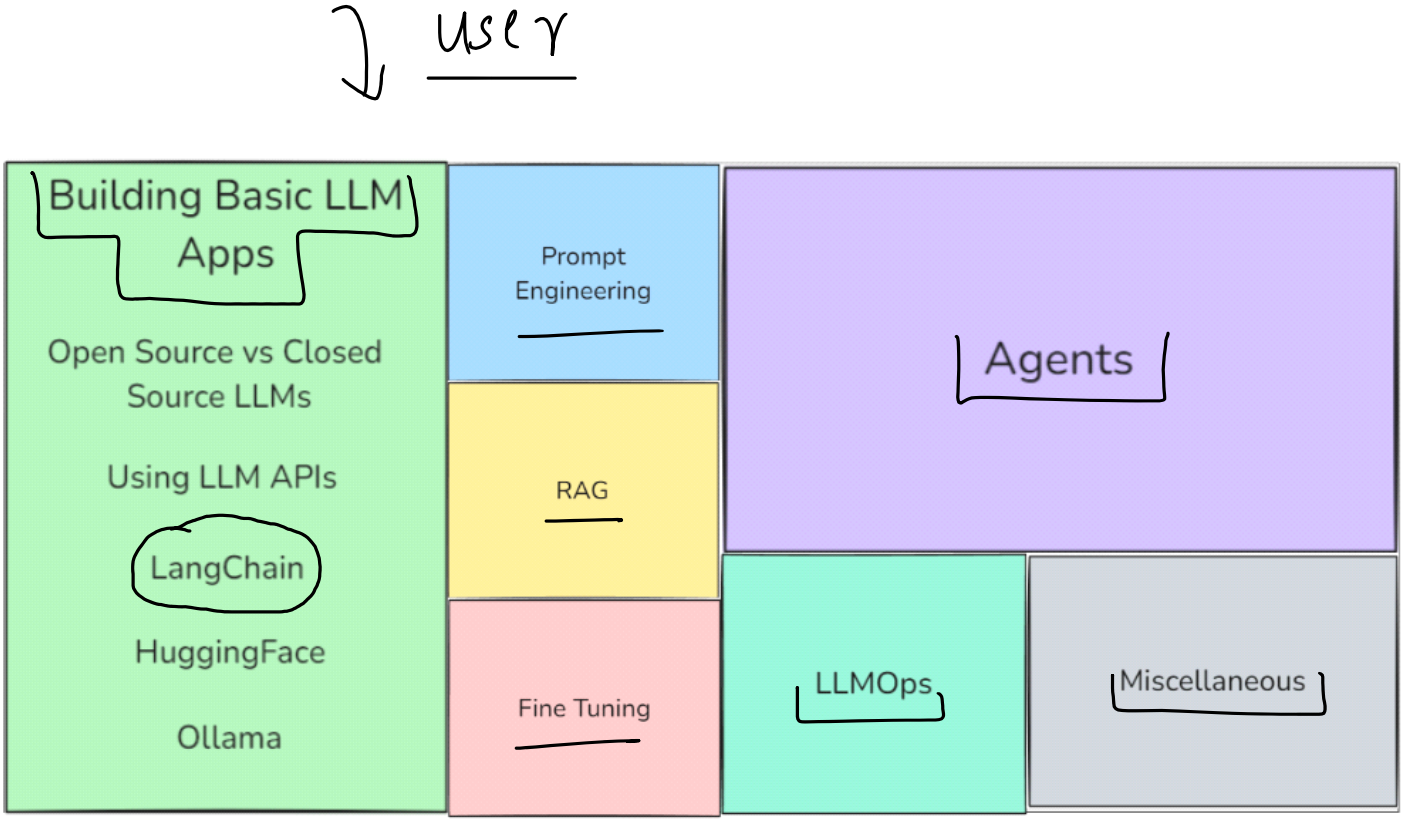
A little background!

29 January 2025 08:28





What is LangChain

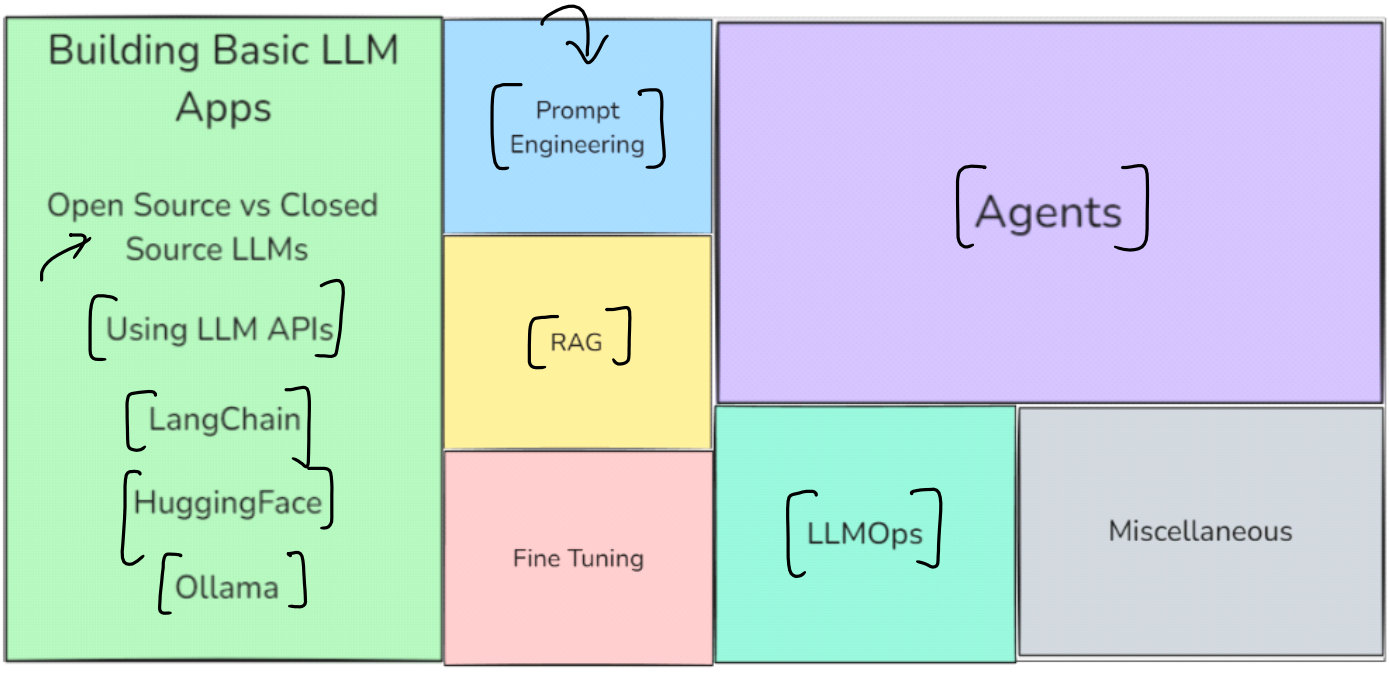
29 January 2025 08:35

LangChain is an open source framework that helps in building LLM based applications. It provides modular components and end-to-end tools that help developers build complex AI applications, such as chatbots, question-answering systems, retrieval-augmented generation (RAG), autonomous agents, and more.

1. Supports all the major LLMs
2. Simplifies developing LLM based applications → chains
3. Integrations available for all major tools
4. Open source/Free/Actively developed
5. Supports all major GenAI use cases

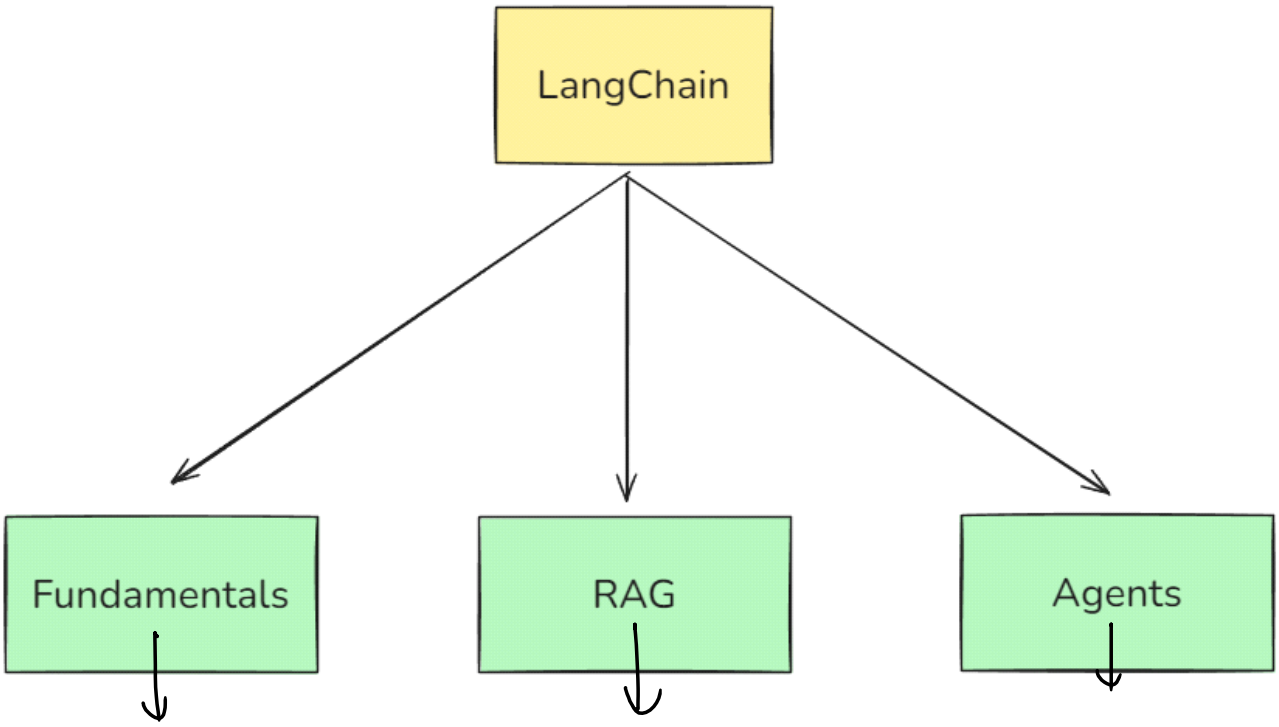
Why LangChain first

29 January 2025 08:35



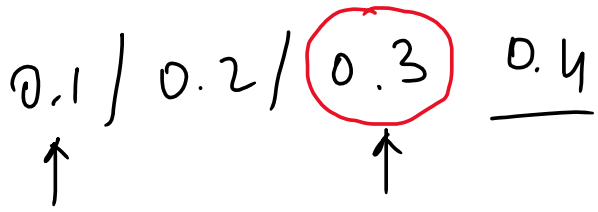
Curriculum Structure

29 January 2025 08:36



My Focus

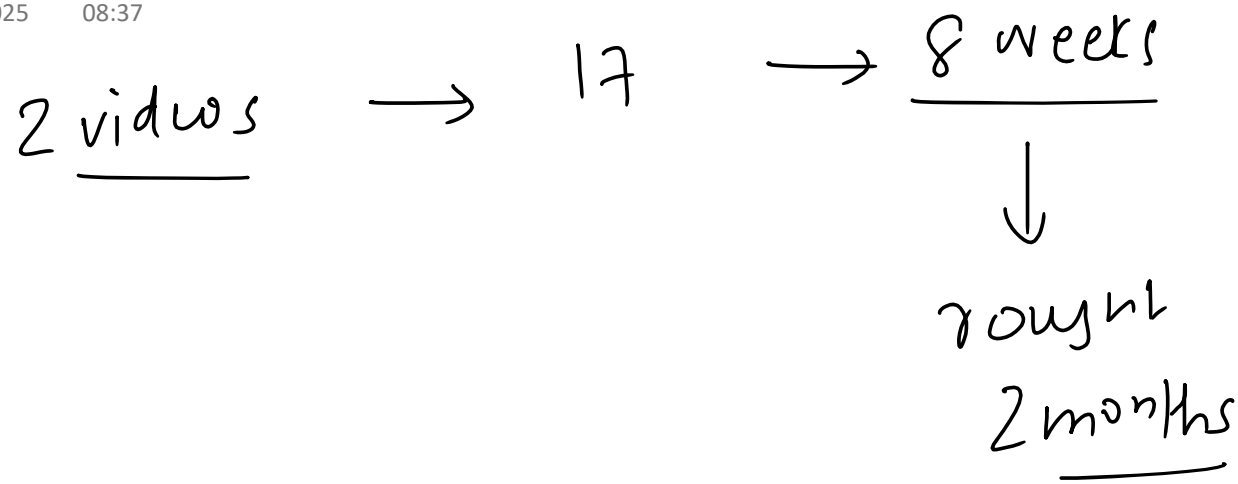
29 January 2025 08:35



1. Updated information
2. Clarity
3. Conceptual understanding
4. The 80 percent approach

Timeline

29 January 2025 08:37



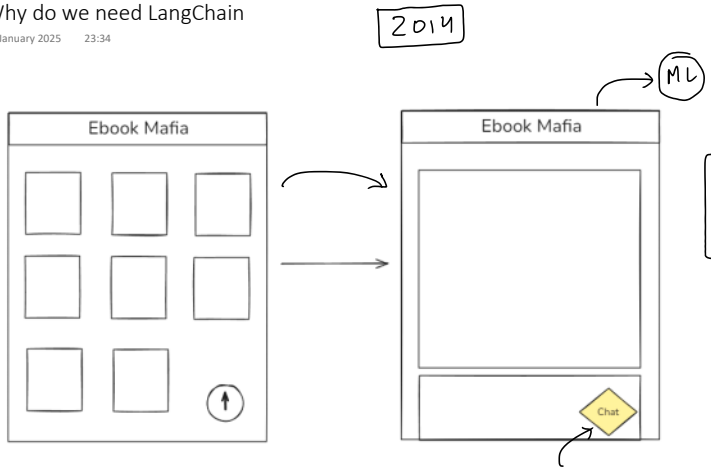
What is LangChain

07 January 2025 23:14

LangChain is an open-source framework for developing applications powered by large language models (LLMs).

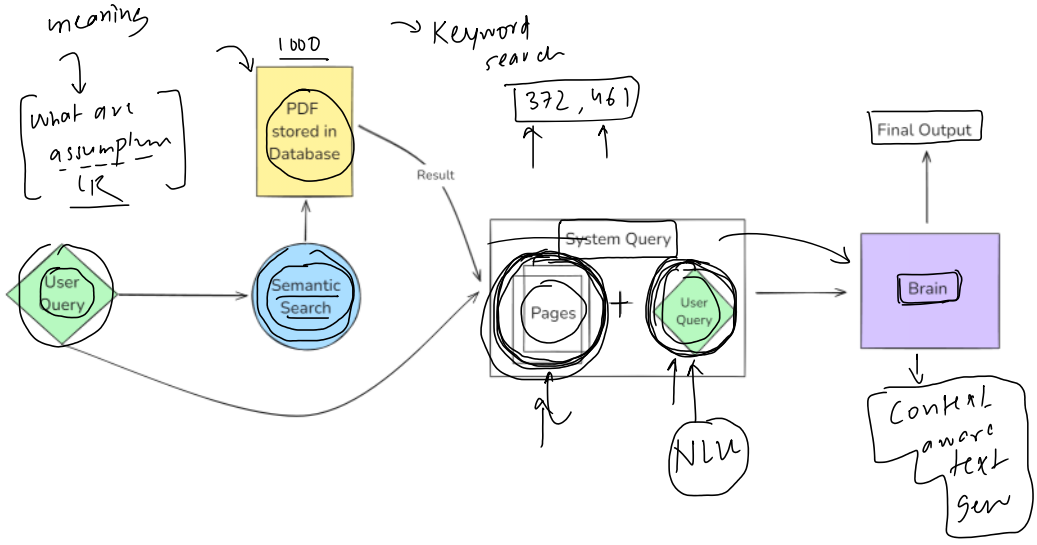
Why do we need LangChain

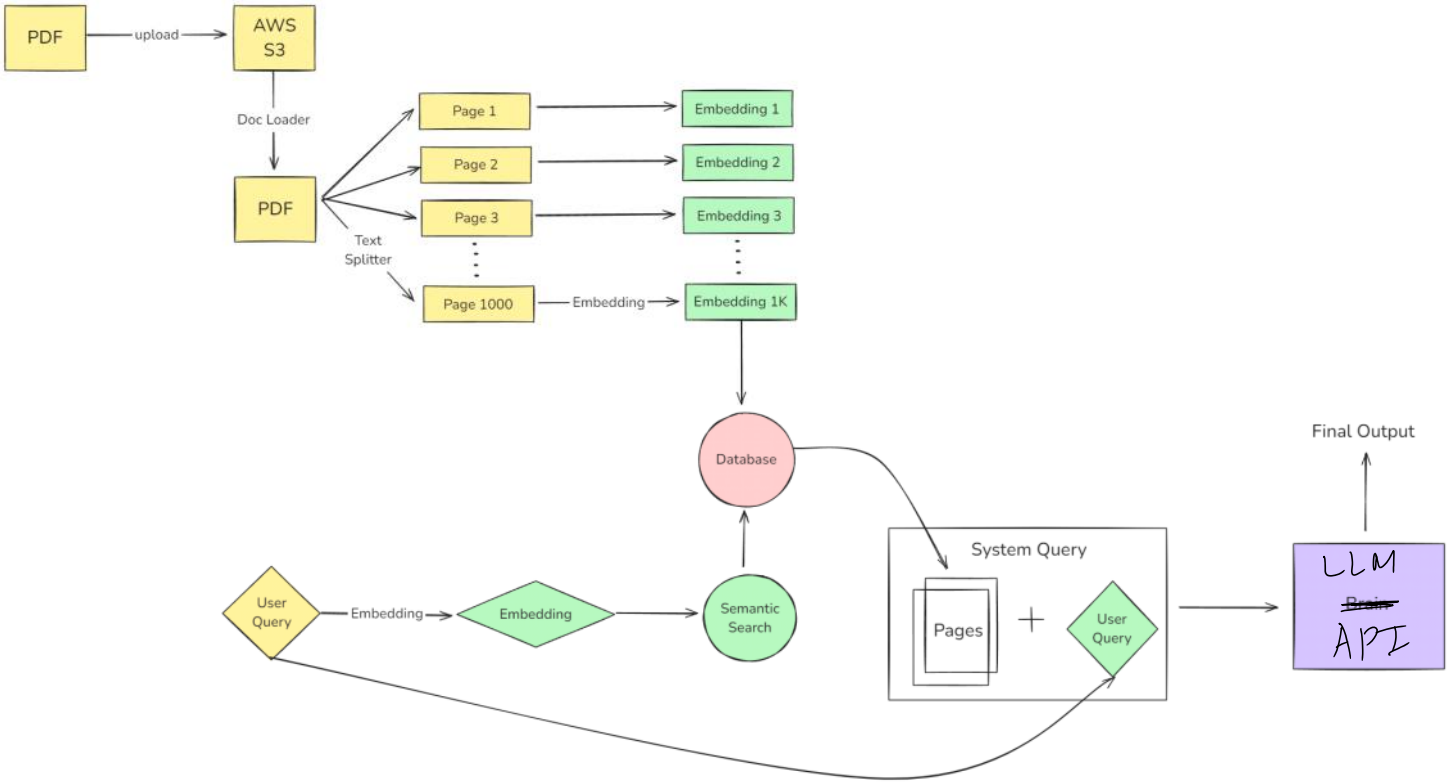
21 January 2025 23:34



Example Queries

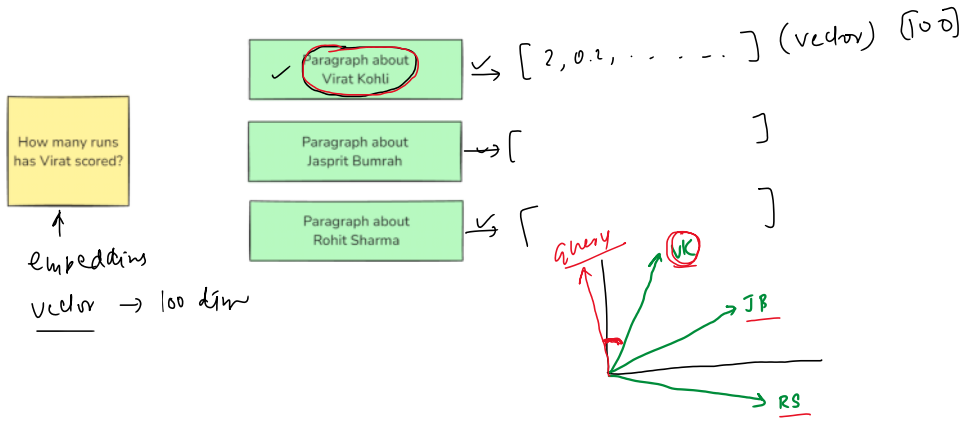
1. Explain page number 5 as if I am a 5 year old
2. Generate a True False exercise on Linear Regression
3. Generate notes for Decision Trees





Semantic search

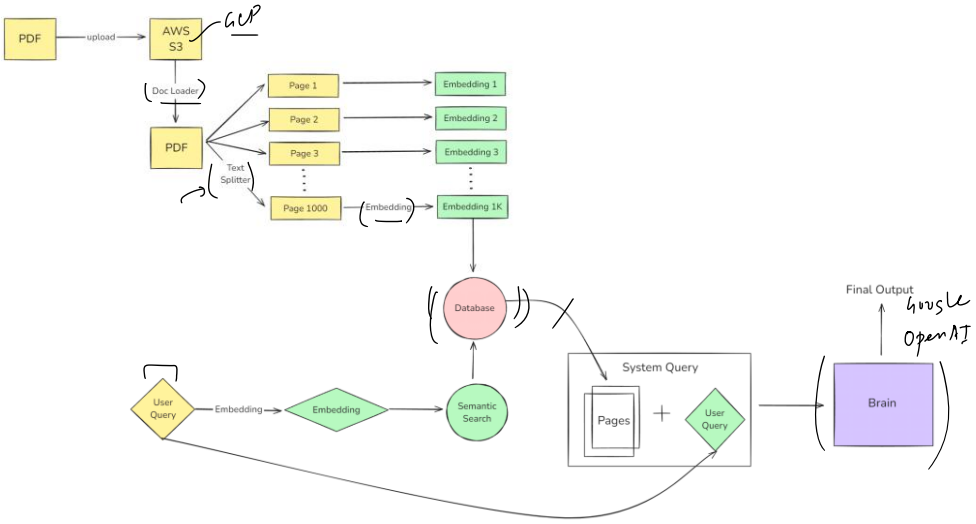
embedding → vector (set of nos.)



Benefits

21 January 2025 23:34

- Concept of chains
- Model Agnostic Development
- Complete ecosystem
- Memory and state handling



LLM
API

What can you build?

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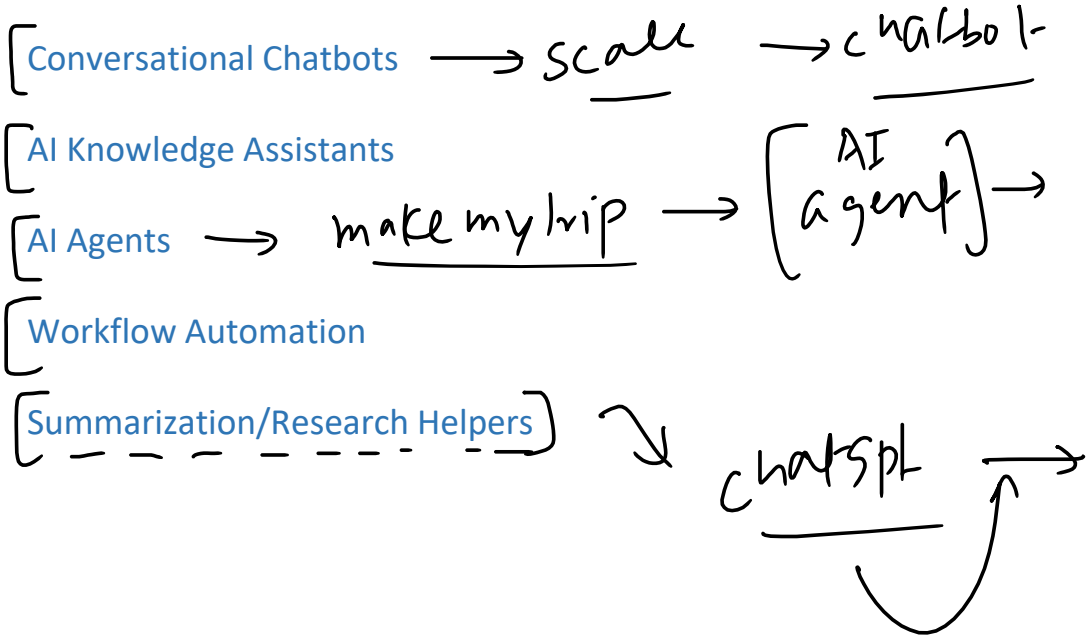


Alternatives

21 January 2025 23:34

LlamaIndex

Haystack

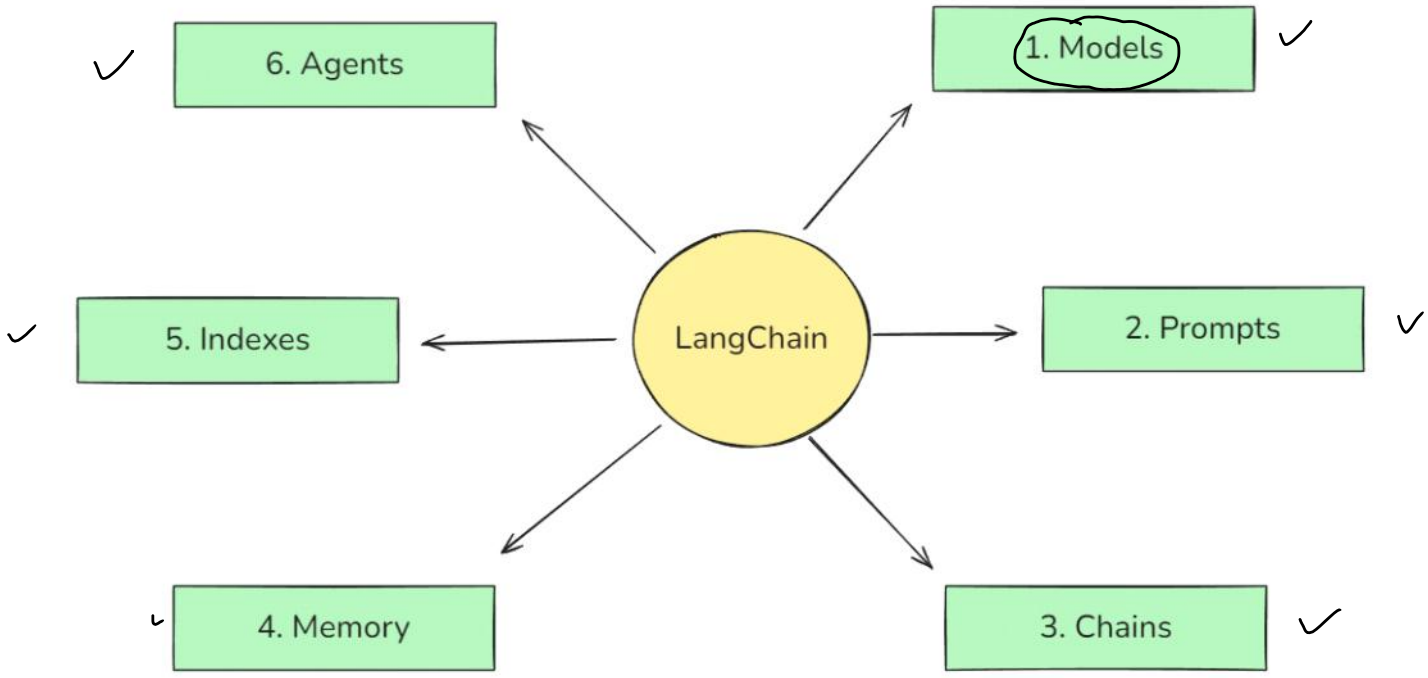
Recap

23 January 2025 10:29

LangChain is an open-source framework for developing applications powered by large language models (LLMs).

LangChain Components

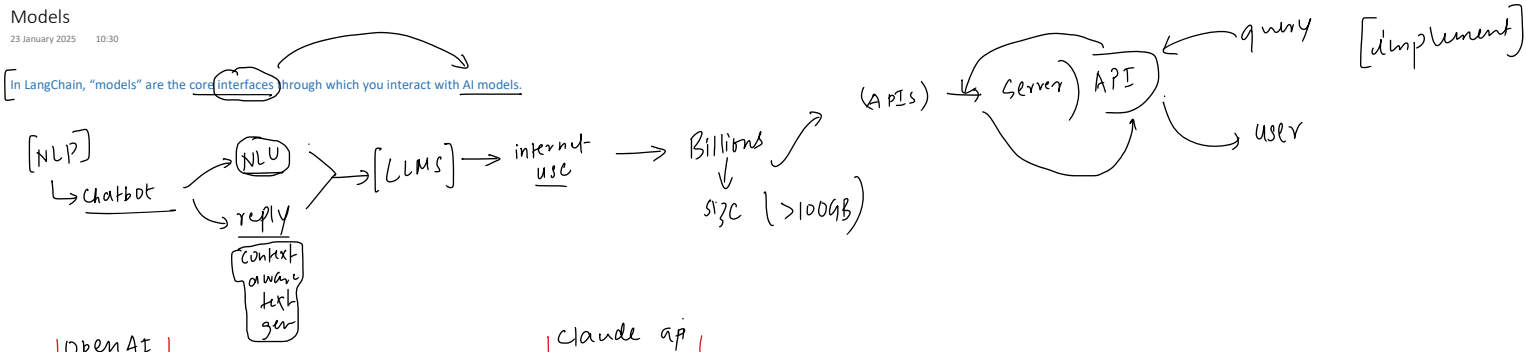
23 January 2025 10:30



Models

23 January 2025 10:30

In LangChain, "models" are the core interfaces through which you interact with AI models.



OpenAI

```python
Create a human-like response to a prompt

from openai import OpenAI
client = OpenAI()

completion = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {
            "role": "user",
            "content": "Write a haiku about recursion in programming."
        }
    ]
)

print(completion.choices[0].message)
```

Claude api

```python
claude_quickstart.py

import anthropic

client = anthropic.Anthropic()

message = client.messages.create(
    model="claude-3-5-sonnet-20241022",
    max_tokens=1000,
    temperature=0,
    system="You are a world-class poet. Respond only with short poems.",
    messages=[
        {
            "role": "user",
            "content": [
                {
                    "type": "text",
                    "text": "Why is the ocean salty?"
                }
            ]
        }
    ]
)
print(message.content)
```

Langchain

interface → API

OpenAI

```python
from langchain_openai import ChatOpenAI
from dotenv import load_dotenv

load_dotenv()

model = ChatOpenAI(model='gpt-4', temperature=0)

result = model.invoke("Now divide the result by 1.5")

print(result.content)
```

Langchain

Claude

```python
from langchain_anthropic import ChatAnthropic
from dotenv import load_dotenv

load_dotenv()

model = ChatAnthropic(model='claude-3-opus-20240229')

result = model.invoke("Hi who are you")

print(result.content)
```

Langchain

[language model] → LLMs

text → text

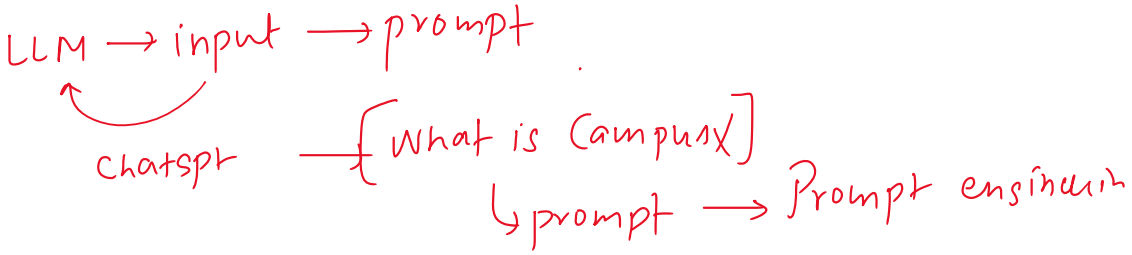
[Embedding model]

text → vector (output)

+ semantic search

Prompts

23 January 2025 10:30



1. Dynamic & Reusable Prompts

```
from langchain_core.prompts import PromptTemplate

prompt = PromptTemplate.from_template('Summarize {topic} in {emotion} tone')

print(prompt.format(topic='Cricket', length='fun'))
```

2. Role-Based Prompts

```
# Define the ChatPromptTemplate using from_template
chat_prompt = ChatPromptTemplate.from_template([
    ("system", "Hi you are a experienced {profession}"),
    ("user", "Tell me about {topic}"),
])

# Format the prompt with the variable
formatted_messages = chat_prompt.format_messages(profession="Doctor", topic="Viral Fever")
```

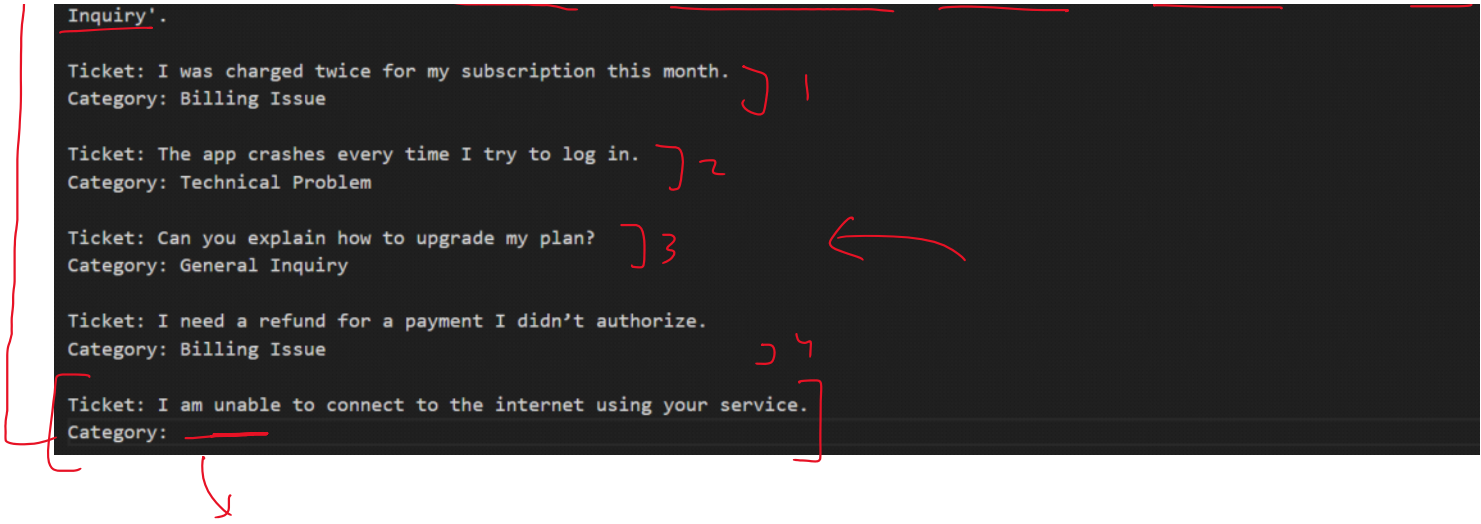
3. Few Shot Prompting

```
examples = [
    {"input": "I was charged twice for my subscription this month.", "output": "Billing Issue"},
    {"input": "The app crashes every time I try to log in.", "output": "Technical Problem"},
    {"input": "Can you explain how to upgrade my plan?", "output": "General Inquiry"},
    {"input": "I need a refund for a payment I didn't authorize.", "output": "Billing Issue"},
]
```

```
# Step 2: Create an example template
example_template = """
Ticket: {input}
Category: {output}
"""
```

```
# Step 3: Build the few-shot prompt template
few_shot_prompt = FewShotPromptTemplate(
    examples=examples,
    example_prompt=PromptTemplate(input_variables=["input", "output"], template=example_template),
    prefix="Classify the following customer support tickets into one of the categories: 'Billing Issue', 'Technical Problem', or 'General Inquiry'.\n\n",
    suffix="\nTicket: {user_input}\nCategory:",
    input_variables=["user_input"],
)
```

```
Classify the following customer support tickets into one of the categories: 'Billing Issue', 'Technical Problem', or 'General Inquiry'.

Ticket: I was charged twice for my subscription this month.
Category: Billing Issue
```


Inquiry'.
Ticket: I was charged twice for my subscription this month.
Category: Billing Issue
Ticket: The app crashes every time I try to log in.
Category: Technical Problem
Ticket: Can you explain how to upgrade my plan?
Category: General Inquiry
Ticket: I need a refund for a payment I didn't authorize.
Category: Billing Issue
Ticket: I am unable to connect to the internet using your service.
Category:

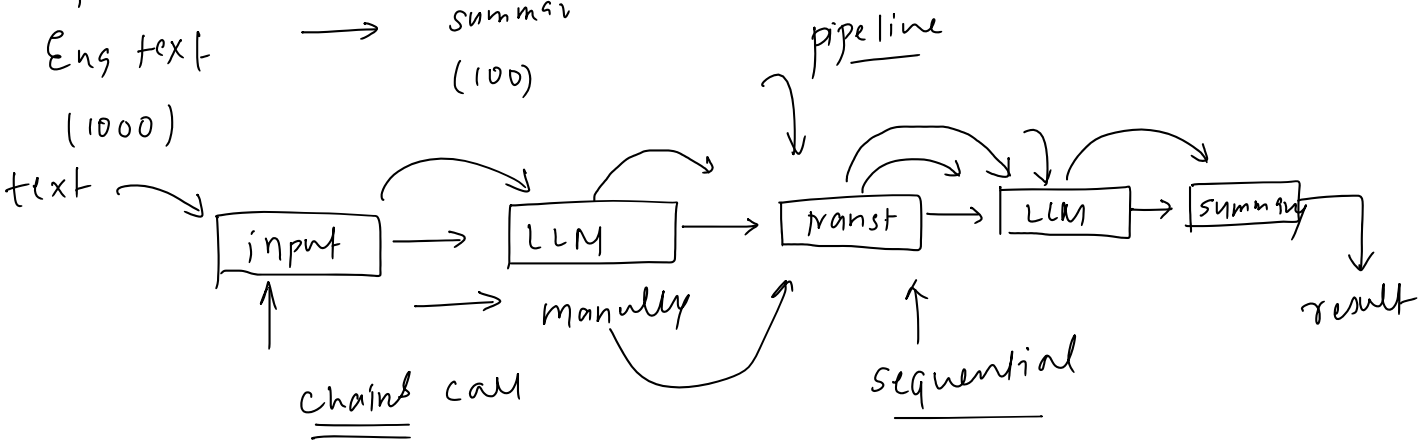
Chains →

23 January 2025 10:31

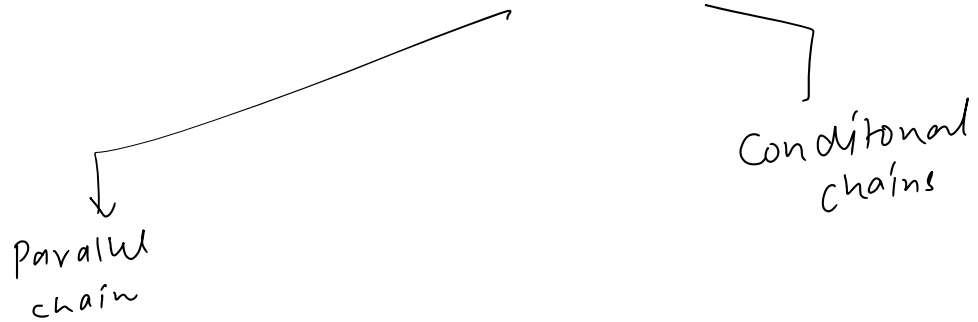
Pipelines → LLM + Pipeline

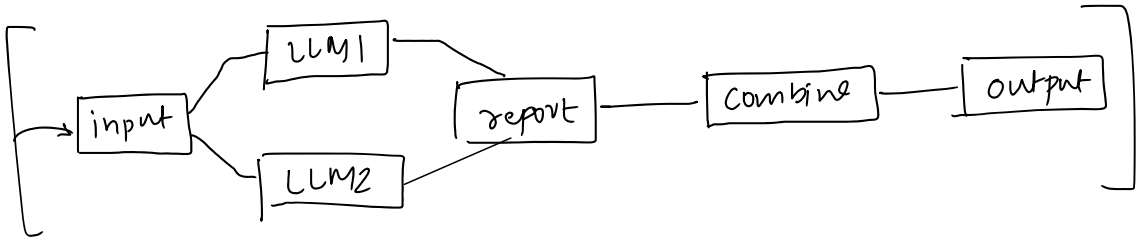
input

Eng text (1000) → Hindi summary (100)



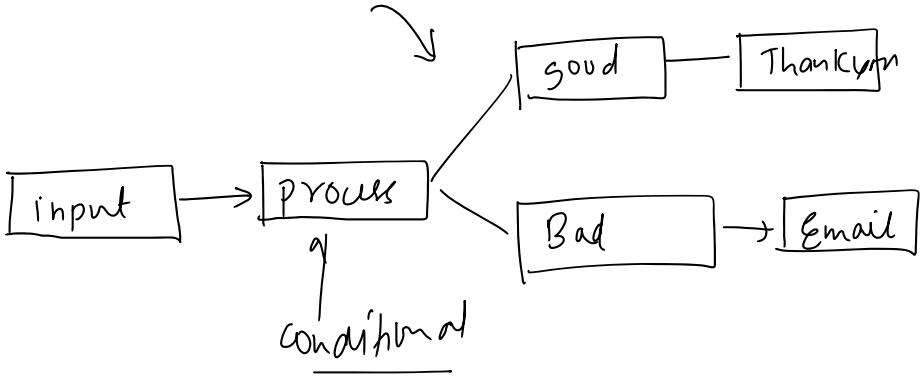
Complex pipeline





AI Agent

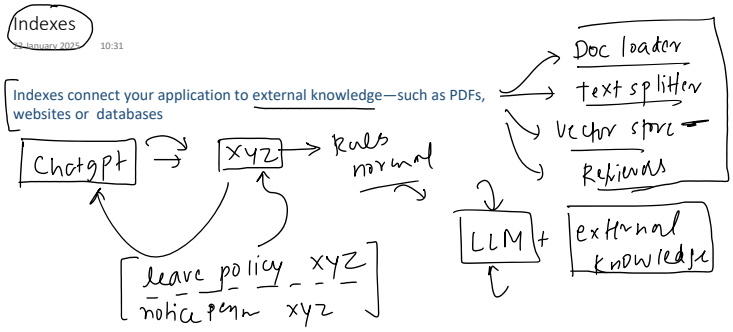
feedback



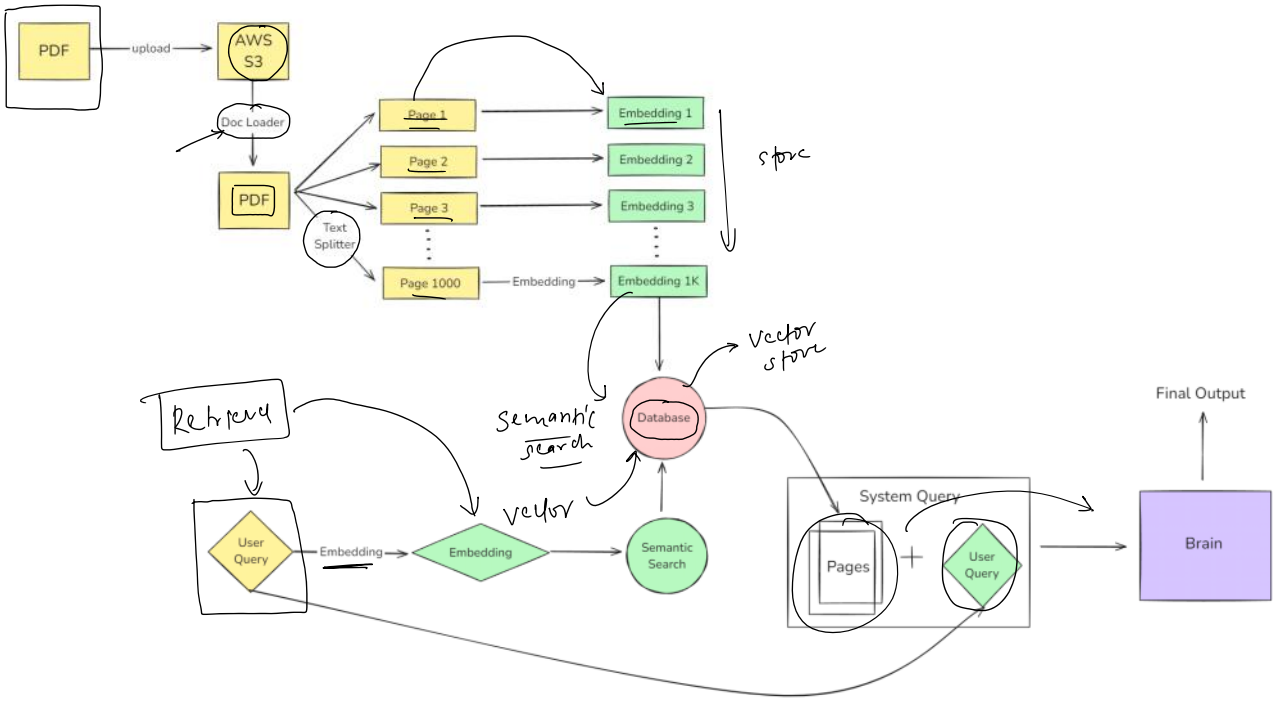
Indexes

23 January 2025 10:31

Indexes connect your application to external knowledge—such as PDFs, websites or databases



Rulebook 1000 pages



LLM
API

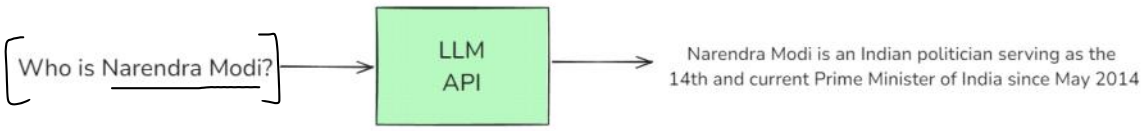
Memory

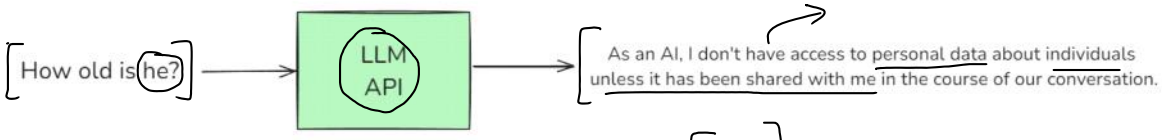
23 January 2025 10:31

LLM API calls are stateless

Chatbot

GPT

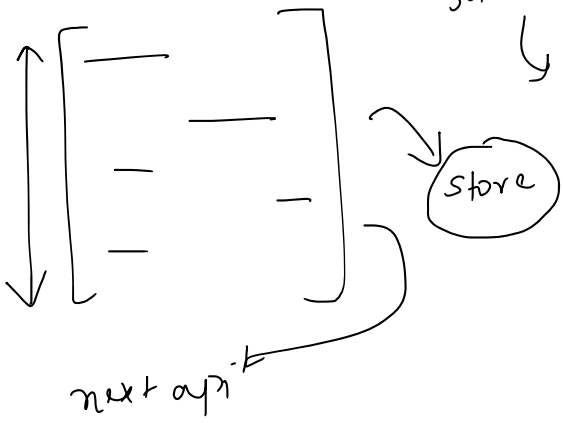




last n intrs [100]

summary → api call

- ConversationBufferMemory: Stores a transcript of recent messages. Great for short chats but can grow large quickly.
- ConversationBufferWindowMemory: Only keeps the last N interactions to avoid excessive token usage.
- Summarizer-Based Memory: Periodically summarizes older chat segments to keep a condensed memory footprint.
- Custom Memory: For advanced use cases, you can store specialized state (e.g., the user's preferences or key facts about them) in a custom memory class.



Agents

23 January 2025 10:31

AI Agents

- LLMs (NLU + Textgen) → chatbot

Chatbot → AI agent → Chatbot with super

AI agent

travel website

- Shimla
- ~~Manali~~

→ 24th Jan

- API
  - Indigo
  - Book the flight

AI agent-

- [Reasoning capability]
- TOOLS

Tools → Calculator, Weather API

User → AI agent → Can you multiply today's temp of Delhi with 3

- Chain of thought

→ [Delhi temp]

→ 25°C

→ 25, 3, *

75 → (75)

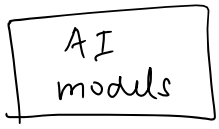
Recap

10 February 2025 09:21

+ Models

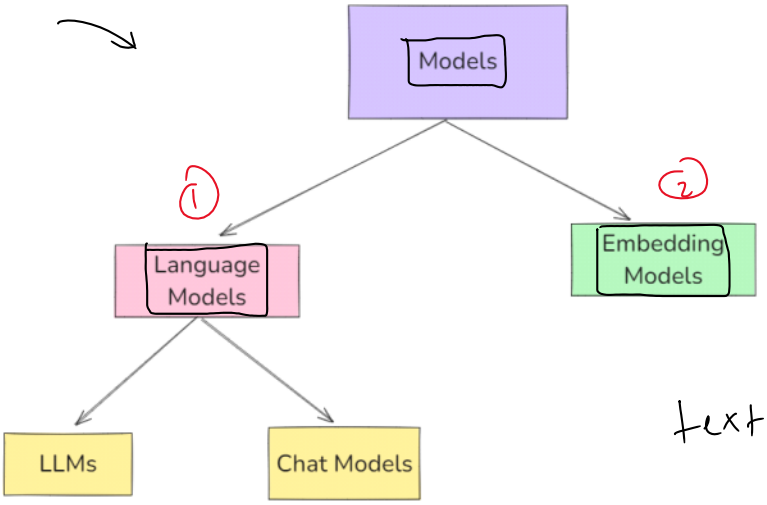
What are Models

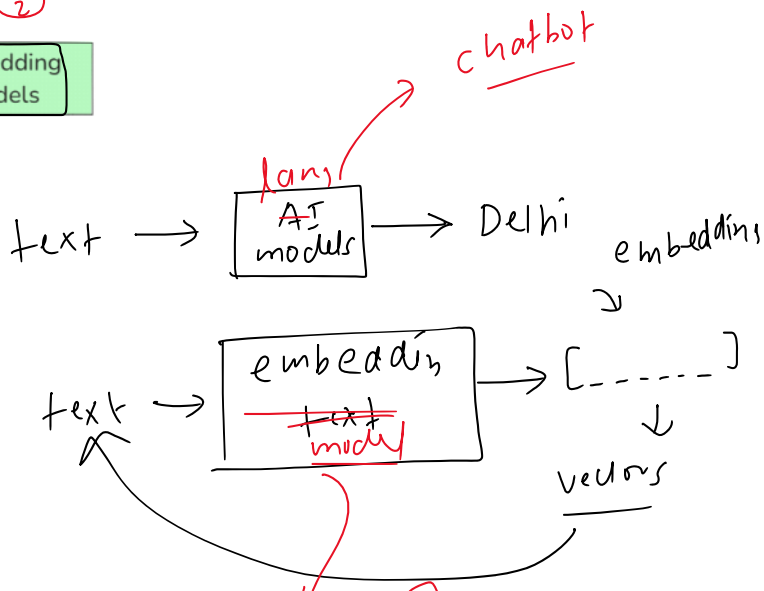
07 January 2025 23:15

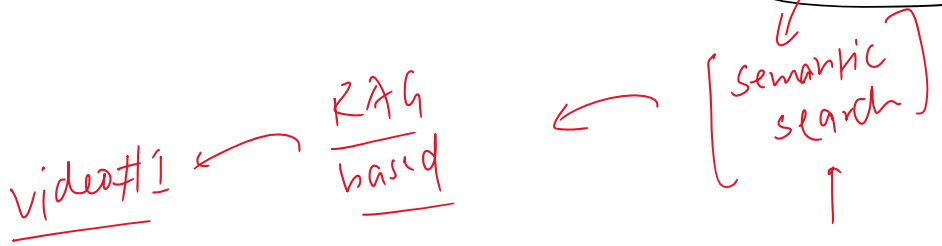


The Model Component in LangChain is a crucial part of the framework, designed to facilitate interactions with various language models and embedding models.

It abstracts the complexity of working directly with different LLMs, chat models, and embedding models, providing a uniform interface to communicate with them. This makes it easier to build applications that rely on AI-generated text, text embeddings for similarity search, and retrieval-augmented generation (RAG).

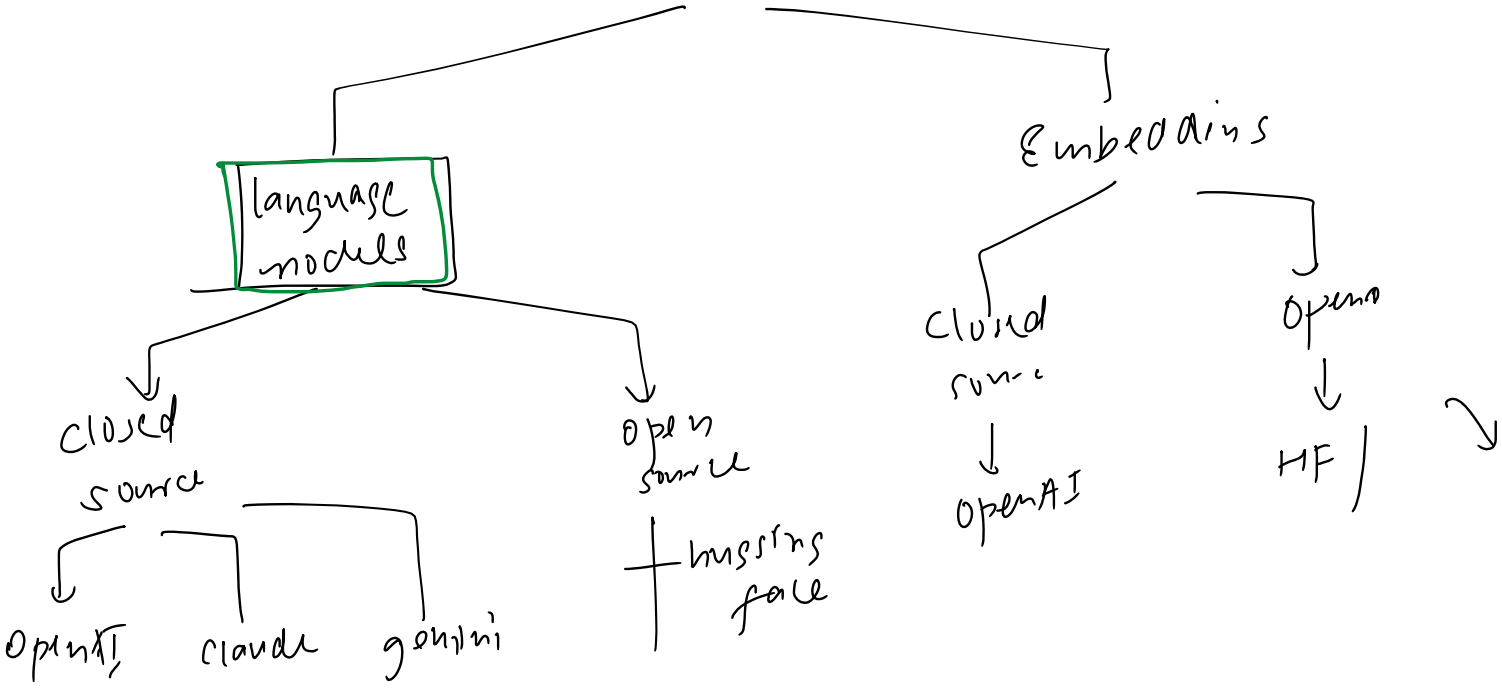






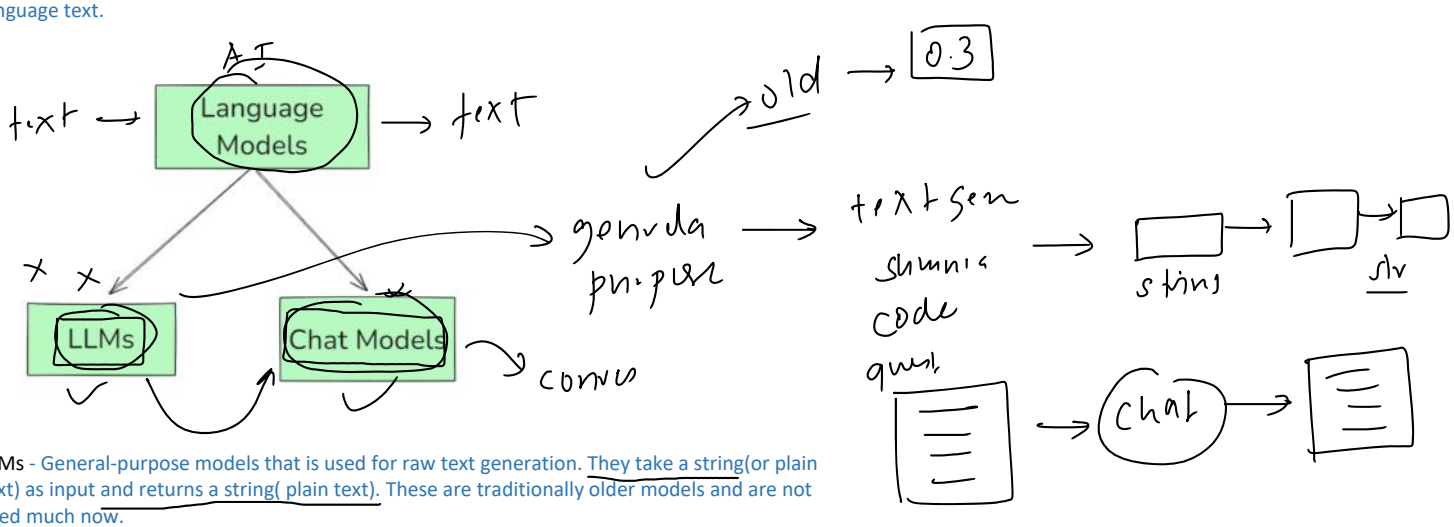
Plan of Action

10 February 2025 09:21



Language Models

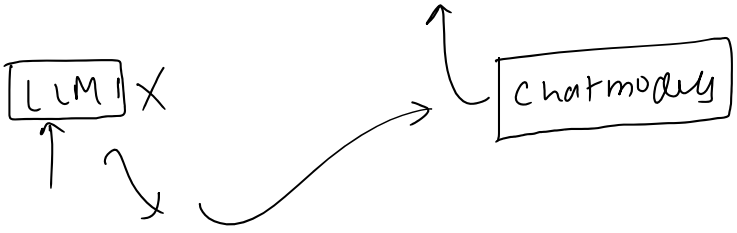
Language Models are AI systems designed to process, generate, and understand natural language text.



LLMs - General-purpose models that is used for raw text generation. They take a string(or plain text) as input and returns a string(plain text). These are traditionally older models and are not used much now.

Chat Models - Language models that are specialized for conversational tasks. They take a sequence of messages as inputs and return chat messages as outputs (as opposed to using plain text). These are traditionally newer models and used more in comparison to the LLMs.

Feature	LLMs (Base Models)	Chat Models (Instruction-Tuned)
Purpose	Free-form text generation	Optimized for multi-turn conversations
Training Data	General text corpora (books, articles)	Fine-tuned on chat datasets (dialogues, user-assistant conversations)
Memory & Context	No built-in memory	Supports structured conversation history
Role Awareness	No understanding of "user" and "assistant" roles	Understands "system", "user", and "assistant" roles
Example Models	GPT-3, Llama-2-7B, Mistral-7B, OPT-1.3B	GPT-4, GPT-3.5-turbo, Llama-2-Chat, Mistral-Instruct, Claude
Use Cases	Text generation, summarization, translation, creative writing, code generation	Conversational AI, chatbots, virtual assistants, customer support, AI tutors

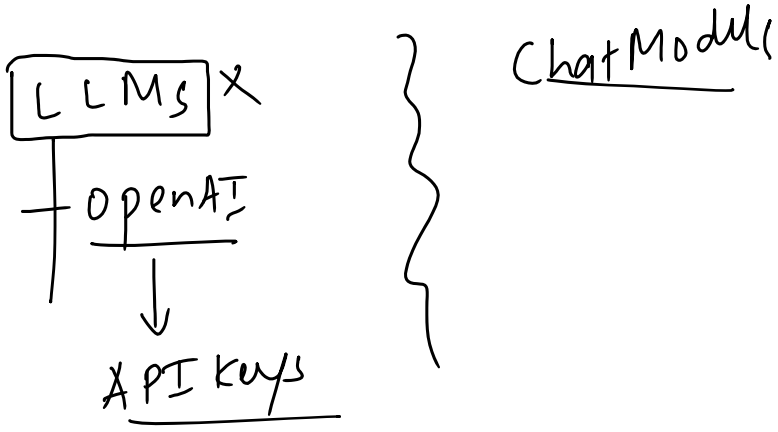


Setup

10 February 2025 09:21

Demo

1. OpenAI
2. Anthropic
3. Google
4. HuggingFace



`temperature` is a parameter that **controls the randomness of a language model's output**. It affects how **creative or deterministic** the responses are.

- **Lower values** (`0.0 - 0.3`) → More **deterministic** and predictable.
- **Higher values** (`0.7 - 1.5`) → More **random**, creative, and diverse.

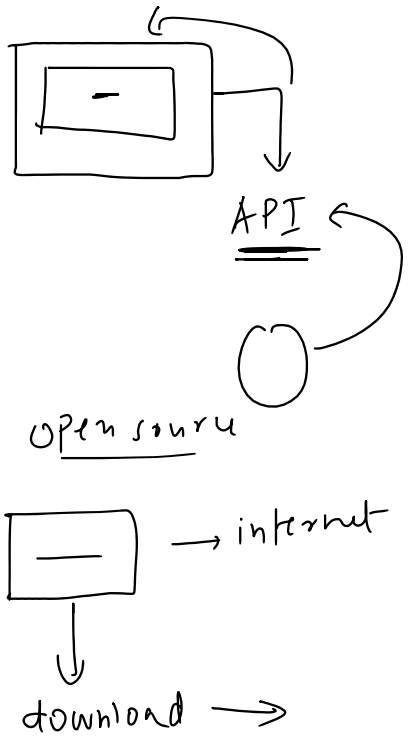
Use Case	Recommended Temperature
Factual answers (math, code, facts)	`0.0 - 0.3`
Balanced response (general QA, explanations)	`0.5 - 0.7`
Creative writing, storytelling, jokes	`0.9 - 1.2`
Maximum randomness (wild ideas, brainstorming)	`1.5+`

Open Source Models

11 February 2025 08:58

Open-source language models are freely available AI models that can be downloaded, modified, fine-tuned, and deployed without restrictions from a central provider. Unlike closed-source models such as OpenAI's GPT-4, Anthropic's Claude, or Google's Gemini, open-source models allow full control and customization.

Feature	Open-Source Models	Closed-Source Models
Cost	Free to use (no API costs)	Paid API usage (e.g., OpenAI charges per token)
Control	Can modify, fine-tune, and deploy anywhere	Locked to provider's infrastructure
Data Privacy	Runs locally (no data sent to external servers)	Sends queries to provider's servers
Customization	Can fine-tune on specific datasets	No access to fine-tuning in most cases
Deployment	Can be deployed on **on-premise** servers or **cloud**	Must use vendor's API



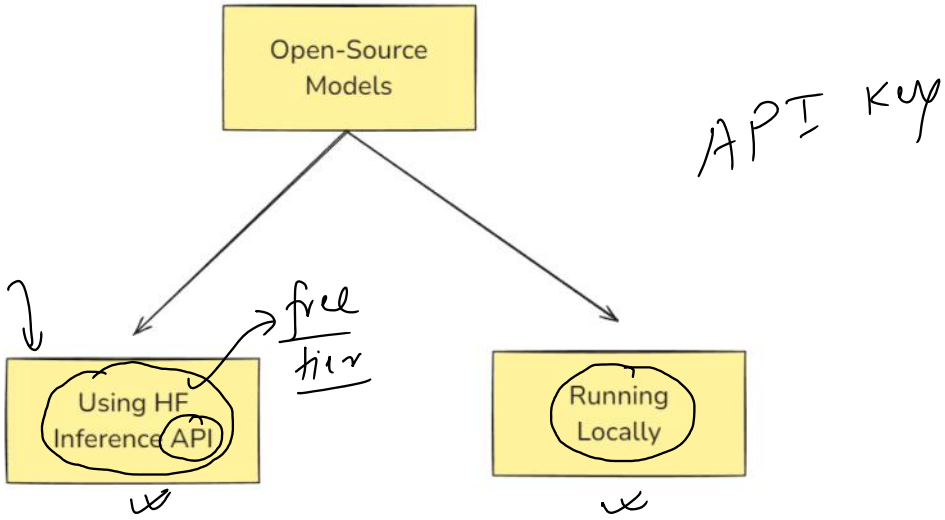
Some Famous Open Source Models

Model	Developer	Parameters	Best Use Case
LLaMA-2-7B/13B/70B	Meta AI	7B - 70B	General-purpose text generation
Mixtral-8x7B	Mistral AI	8x7B (MoE)	Efficient & fast responses
Mistral-7B	Mistral AI	7B	Best small-scale model (outperforms LLaMA-2-13B)
Falcon-7B/40B	TII UAE	7B - 40B	High-speed inference
BLOOM-176B	BigScience	176B	Multilingual text generation
GPT-J-6B	EleutherAI	6B	Lightweight and efficient
GPT-NeoX-20B	EleutherAI	20B	Large-scale applications
StableLM	Stability AI	3B - 7B	Compact models for chatbots

Where to find them?

HuggingFace - The largest repository of open-source LLMs

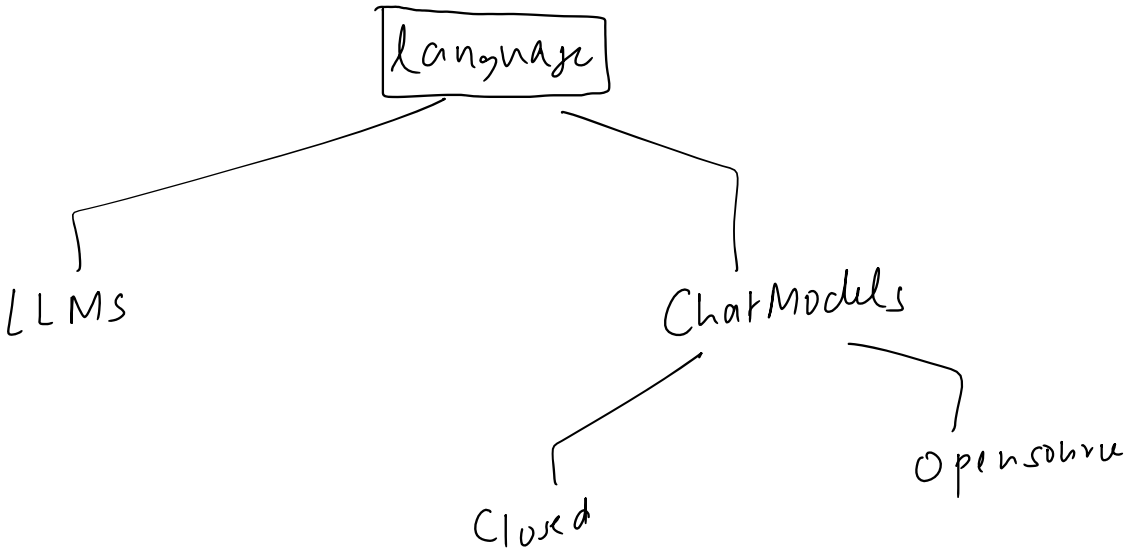
Ways to use Open-source Models

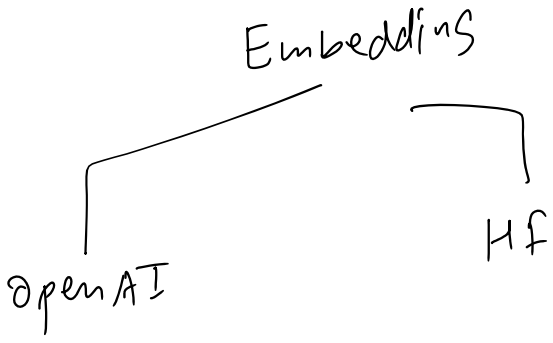


Disadvantages

Disadvantage	Details
High Hardware	Running large models (e.g., LLaMA-2-70B) requires expensive GPUs.

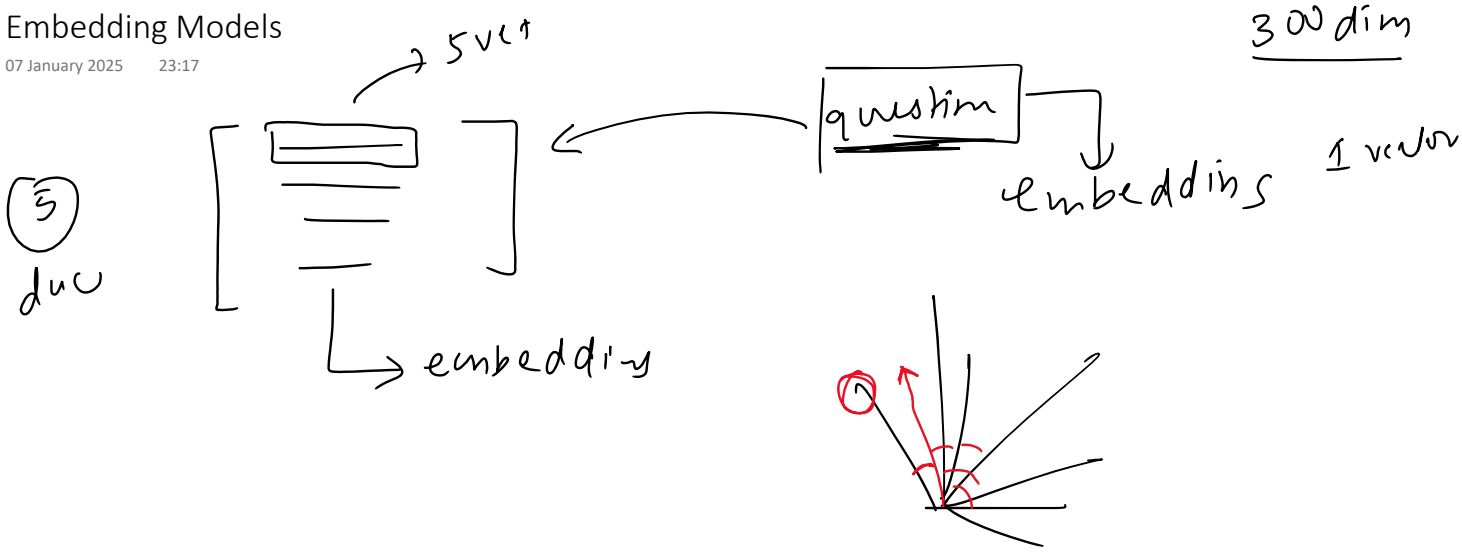
Disadvantage	Details
High Hardware Requirements	Running large models (e.g., LLaMA-2-70B) requires expensive GPUs.
Setup Complexity	Requires installation of dependencies like **PyTorch**, **CUDA**, **transformers**.
Lack of RLHF	Most open-source models don't have **fine-tuning with human feedback**, making them weaker in instruction-following.
Limited Multimodal Abilities	Open models don't support **images, audio, or video** like GPT-4V.





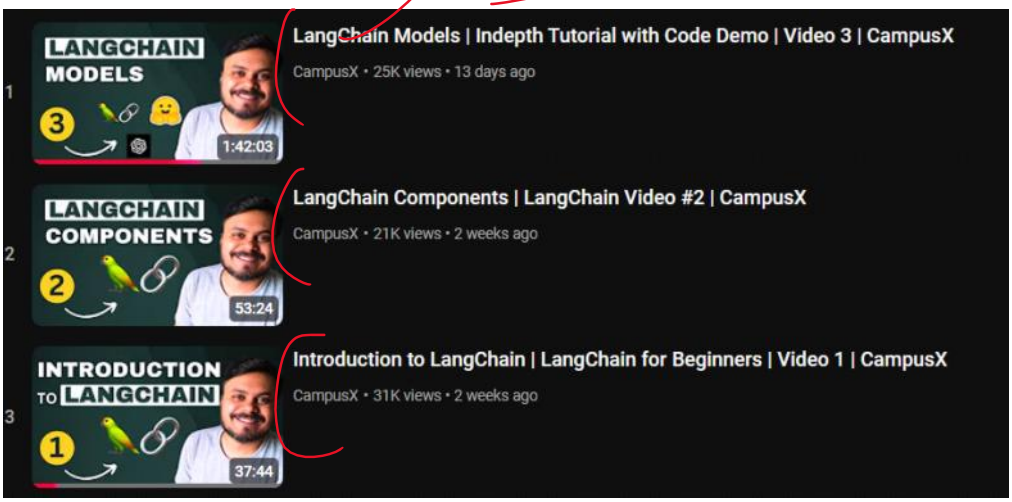
Embedding Models

07 January 2025 23:17



Recap

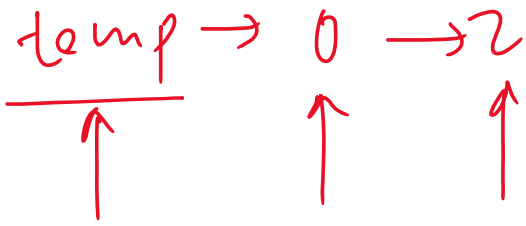
24 February 2025 09:42



④ → Prompts

A mistake from my side!

24 February 2025 09:42

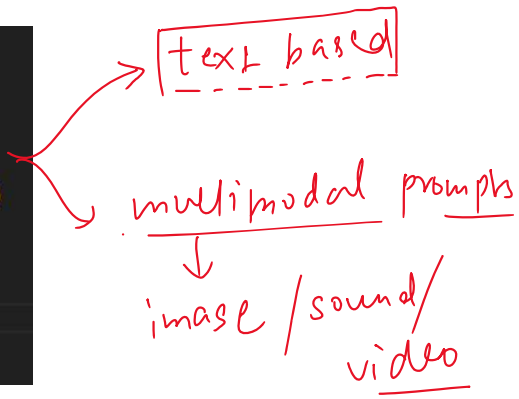


What are Prompts

10 January 2025 08:37

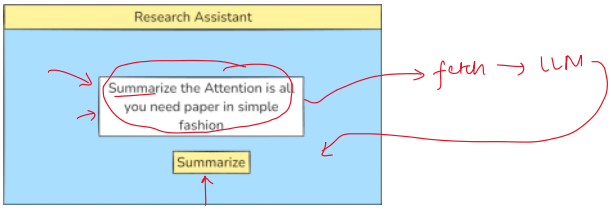
Prompts are the input instructions or queries given to a model to guide its output.

```python
from langchain_openai import ChatOpenAI
from dotenv import load_dotenv

load_dotenv()

model = ChatOpenAI(model='gpt-4', temperature=1.5, max_completion_tokens=10)

result = model.invoke("Write a 5 line poem on cricket")

print(result.content)
```



Static vs Dynamic Prompts

14 February 2025 00:01




```
paper_input = st.selectbox( "Select Research Paper Name", ["Select...", "Attention Is All You Need",
"BERT: Pre-training of Deep Bidirectional Transformers", "GPT-3: Language Models are Few-Shot
Learners", "Diffusion Models Beat GANs on Image Synthesis"] )

style_input = st.selectbox( "Select Explanation Style", ["Beginner-Friendly", "Technical", "Code-
Oriented", "Mathematical"] )

length_input = st.selectbox( "Select Explanation Length", ["Short (1-2 paragraphs)", "Medium (3-5
paragraphs)", "Long (detailed explanation)"] )
```

A **PromptTemplate** in LangChain is a structured way to create prompts dynamically by inserting variables into a predefined template. Instead of hardcoding prompts, PromptTemplate allows you to define placeholders that can be filled in at runtime with different inputs.

This makes it reusable, flexible, and easy to manage, especially when working with dynamic user inputs or automated workflows.

Prompt Template

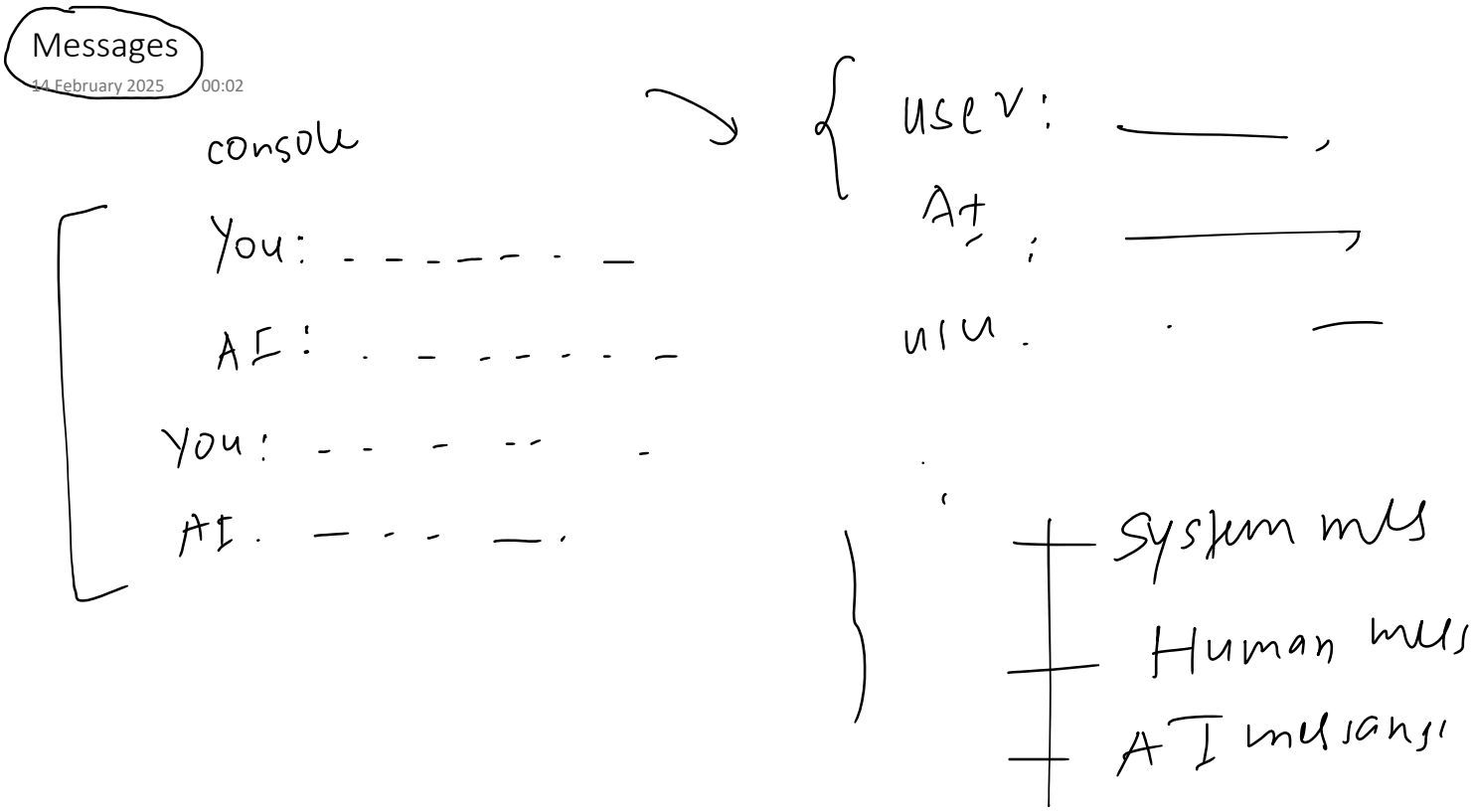
10 January 2025 08:37

A **PromptTemplate** in LangChain is a structured way to create prompts dynamically by inserting variables into a predefined template. Instead of hardcoding prompts, PromptTemplate allows you to define placeholders that can be filled in at runtime with different inputs.

This makes it reusable, flexible, and easy to manage, especially when working with dynamic user inputs or automated workflows.

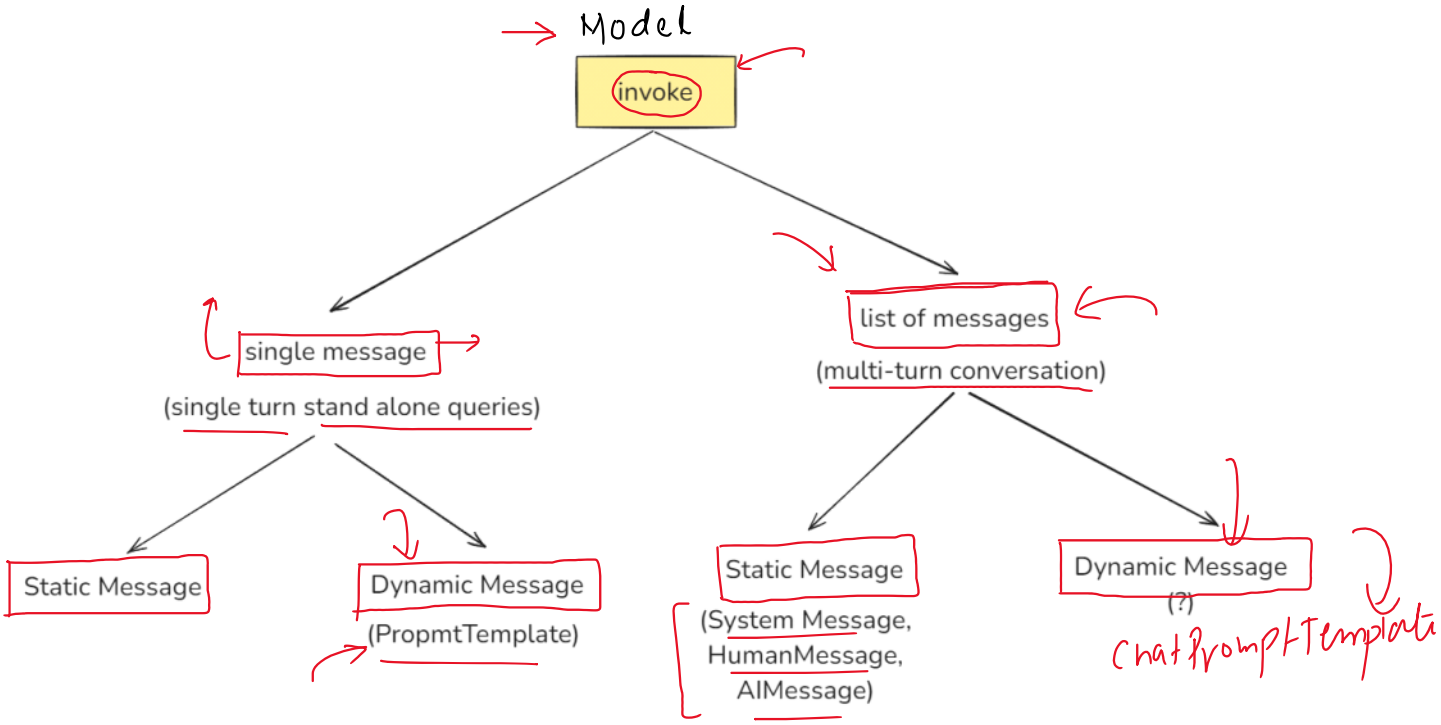
Why use PromptTemplate over f strings?

1. Default validation
2. Reusable
3. LangChain Ecosystem


Messages
14 February 2025 00:02
console
You:
AI:
You:
AI.
{ user:
At
system msg
Human msg
AI msg

Chat Prompt Templates

14 February 2025 00:02



System Message → dynamic

[You are a helpful {domain} expert

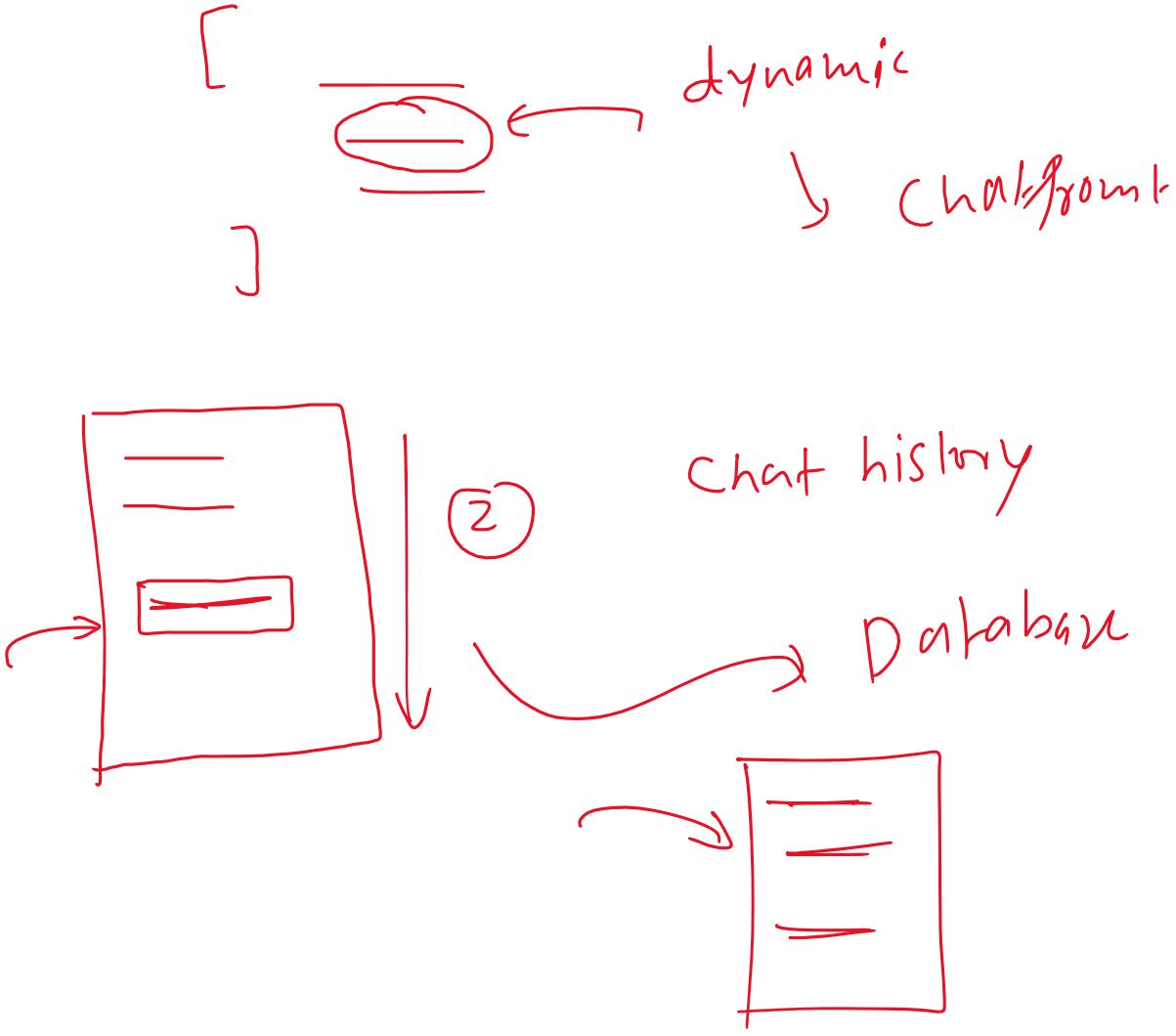
Human me

Explain about {topic}

Message Placeholder

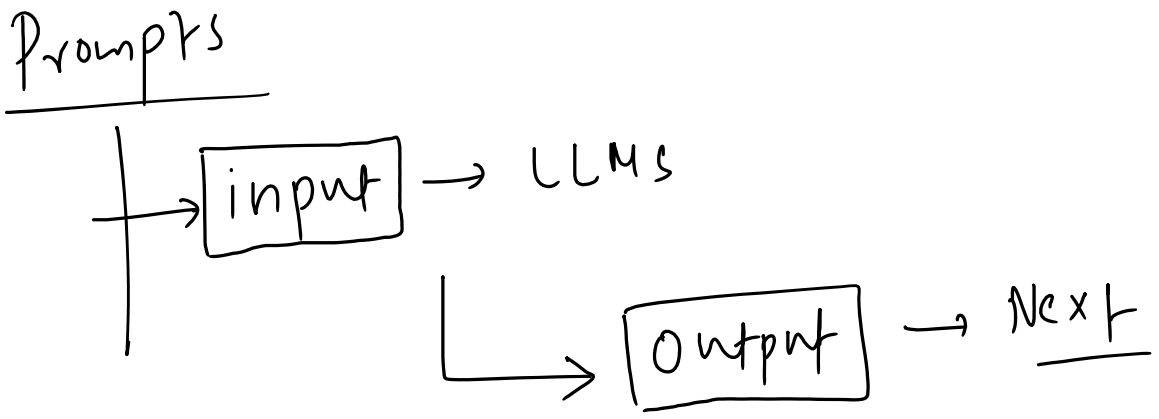
14 February 2025 00:09

A MessagesPlaceholder in LangChain is a special placeholder used inside a ChatPromptTemplate to dynamically insert chat history or a list of messages at runtime.



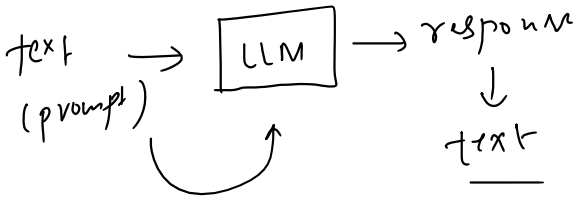
Recap

28 February 2025 00:12



Structured Output

28 February 2025 00:13



In LangChain, structured output refers to the practice of having language models return responses in a well-defined data format (for example, JSON), rather than free-form text. This makes the model output easier to parse and work with programmatically.

[Prompt] - *Can you create a one-day travel itinerary for Paris?*

[LLM's Unstructured Response]

Here's a suggested itinerary: *Morning: Visit the Eiffel Tower.*

Afternoon: Walk through the Louvre Museum.

Evening: Enjoy dinner at a Seine riverside café.

[JSON enforced output]

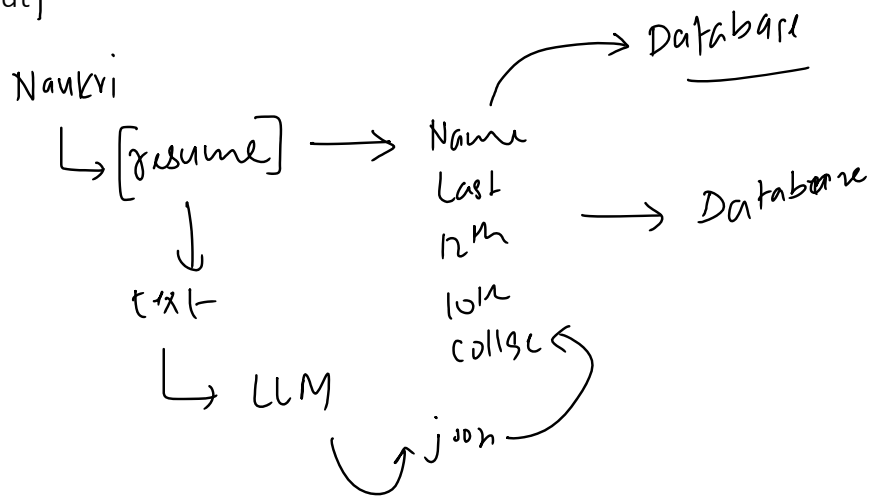
```
[
  {"time": "Morning", "activity": "Visit the Eiffel Tower"},
  {"time": "Afternoon", "activity": "Walk through the Louvre Museum"},
  {"time": "Evening", "activity": "Enjoy dinner at a Seine riverside café"}
]
```

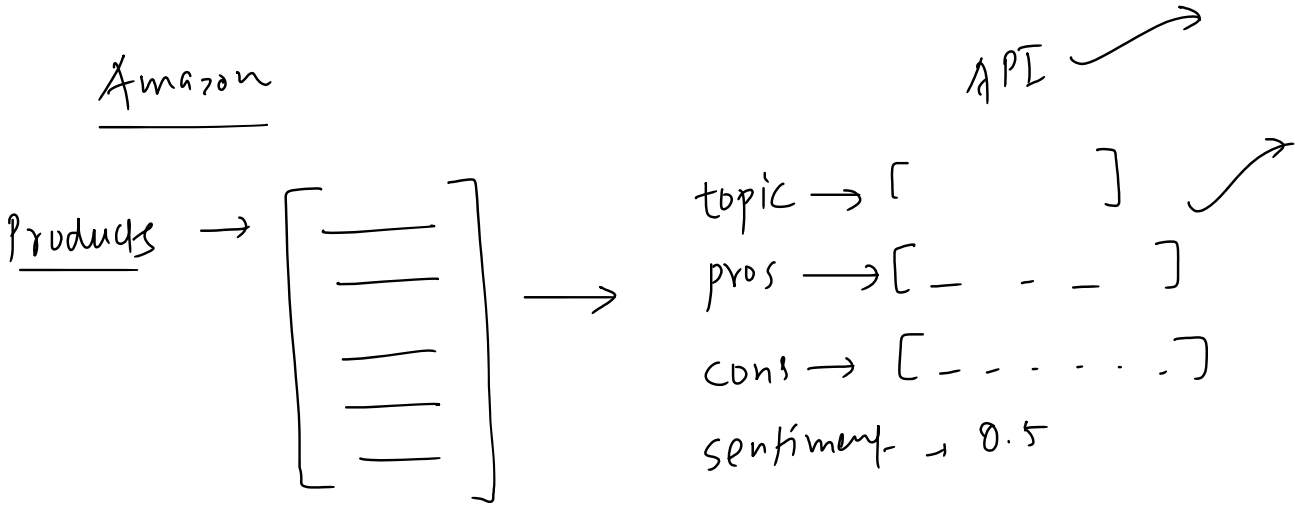
Source - https://www.linkedin.com/pulse/structured-outputs-from-llms-langchain-output-parsers-vijay-chaudhary-wgjqc

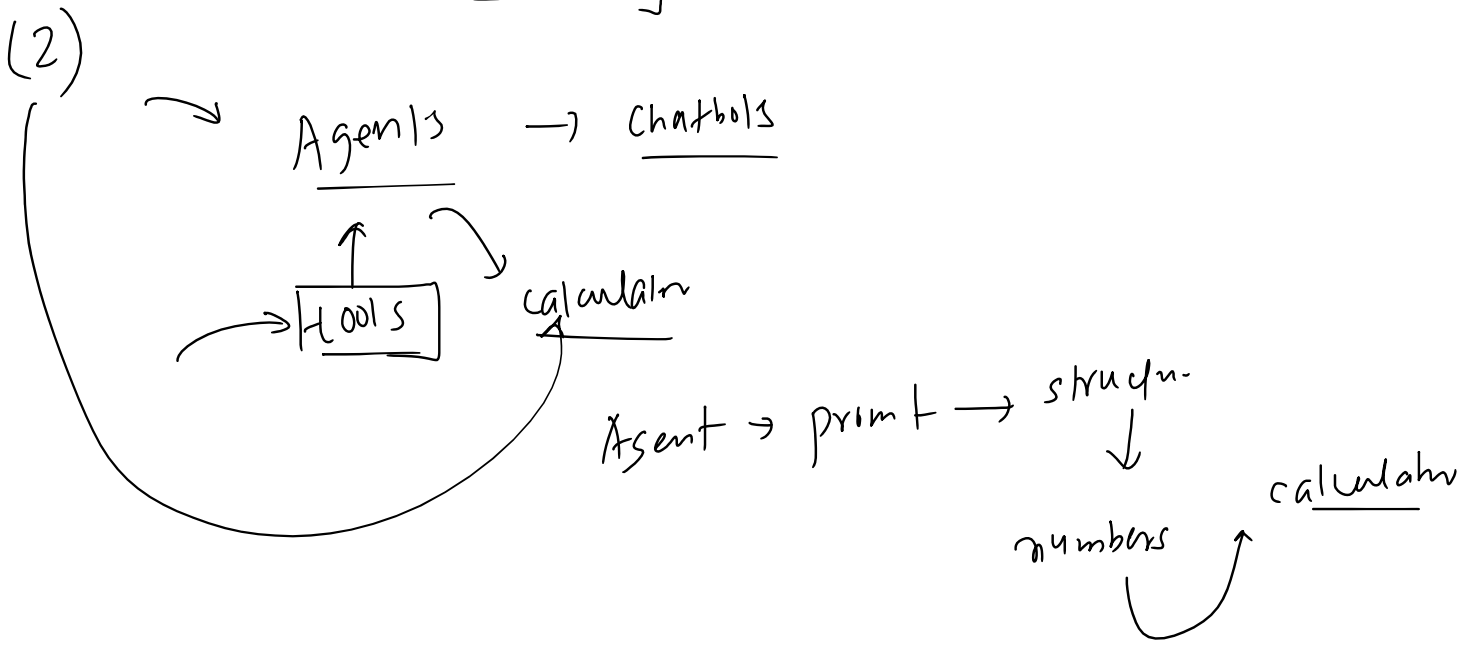
Why do we need Structured Output

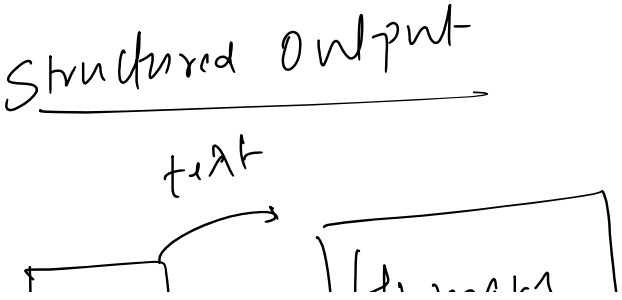
28 February 2025 00:13

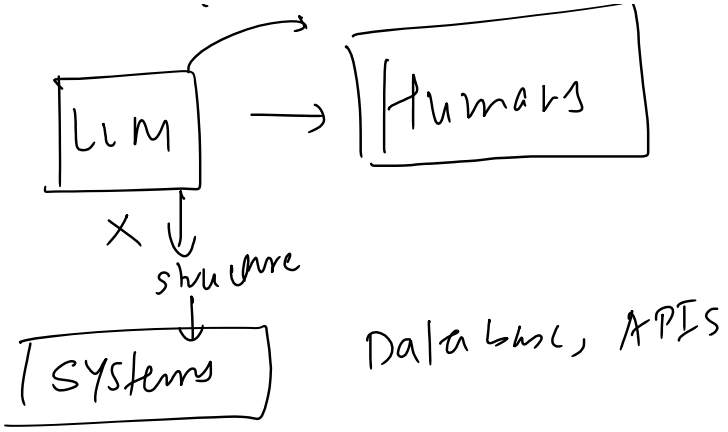
Data Extraction
API building
Agents





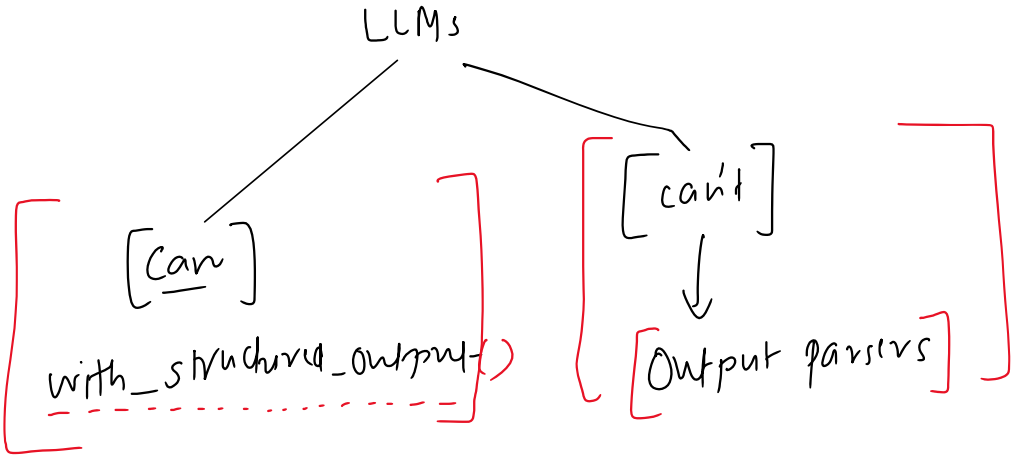





LLM
Humans
structure
Systems
Databases, APIs

Ways to get Structured Output

28 February 2025 00:14

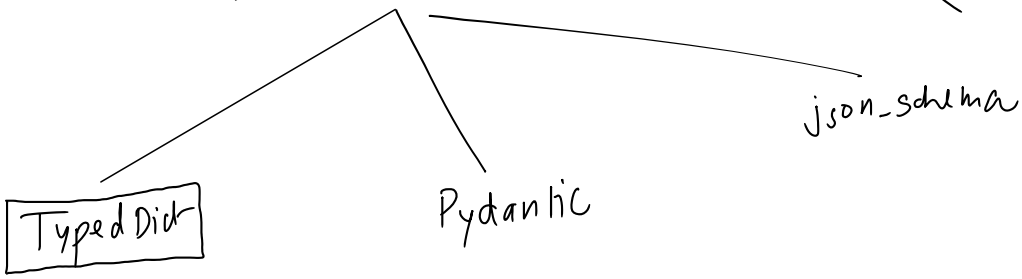


with_structured_output) → data-format

28 February 2025 00:15

(model_invoke)

You are an AI assistant that extracts structured insights from text. Given a product review, extract: - Summary: A brief overview of the main points. - Sentiment: Overall tone of the review (positive, neutral, negative). Return the response in JSON format.



TypedDict

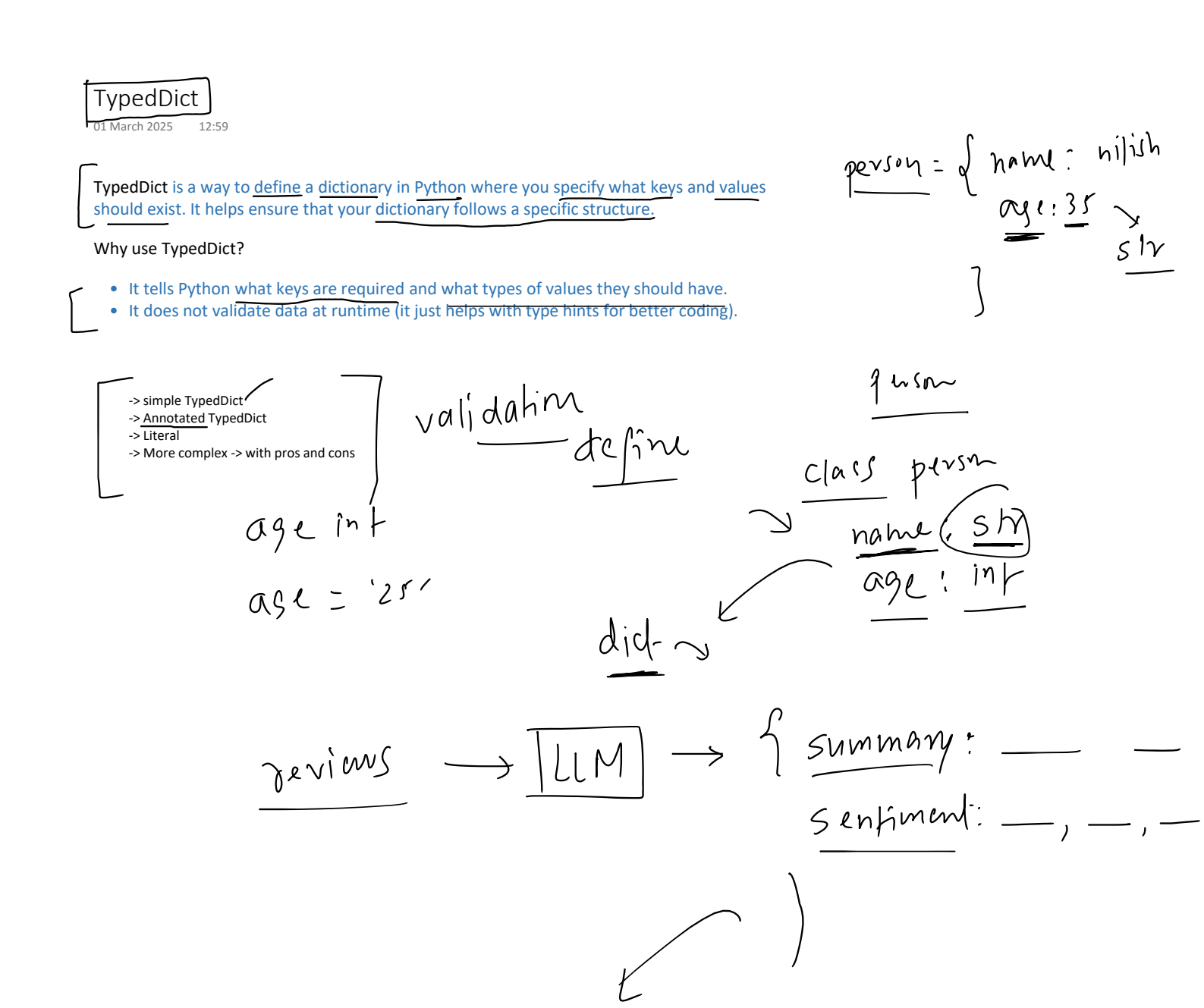
01 March 2025 12:59

TypedDict is a way to define a dictionary in Python where you specify what keys and values should exist. It helps ensure that your dictionary follows a specific structure.

Why use TypedDict?

- It tells Python what keys are required and what types of values they should have.
- It does not validate data at runtime (it just helps with type hints for better coding).

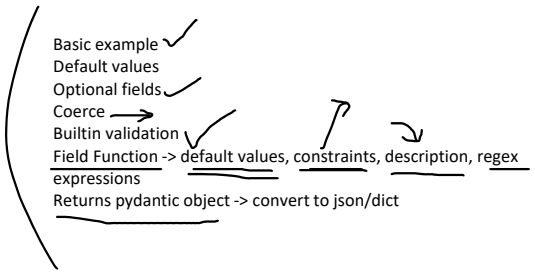
-> simple TypedDict
-> Annotated TypedDict
-> Literal
-> More complex -> with pros and cons



Pydantic

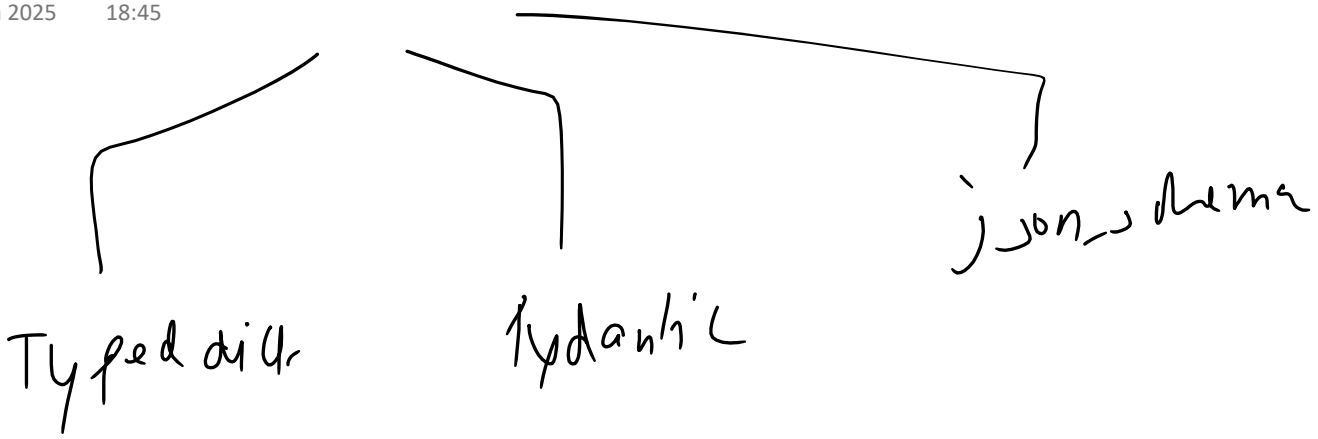
01 March 2025 12:59

Pydantic is a data validation and data parsing library for Python. It ensures that the data you work with is correct, structured, and type-safe.



Json

01 March 2025 18:45



When to use what?

01 March 2025 12:59

✅ Use `TypedDict` if:

- You **only need type hints** (basic structure enforcement).
- You **don't need validation** (e.g., checking numbers are positive).
- You **trust the LLM** to return correct data.

✅ Use `Pydantic` if:

- You need **data validation** (e.g., sentiment must be `"positive"`, `"neutral"`, or `"negative"`).
- You need **default values** if the LLM **misses fields**.
- You want **automatic type conversion** (e.g., `"100"` → `100`).

✅ Use `JSON Schema` if:

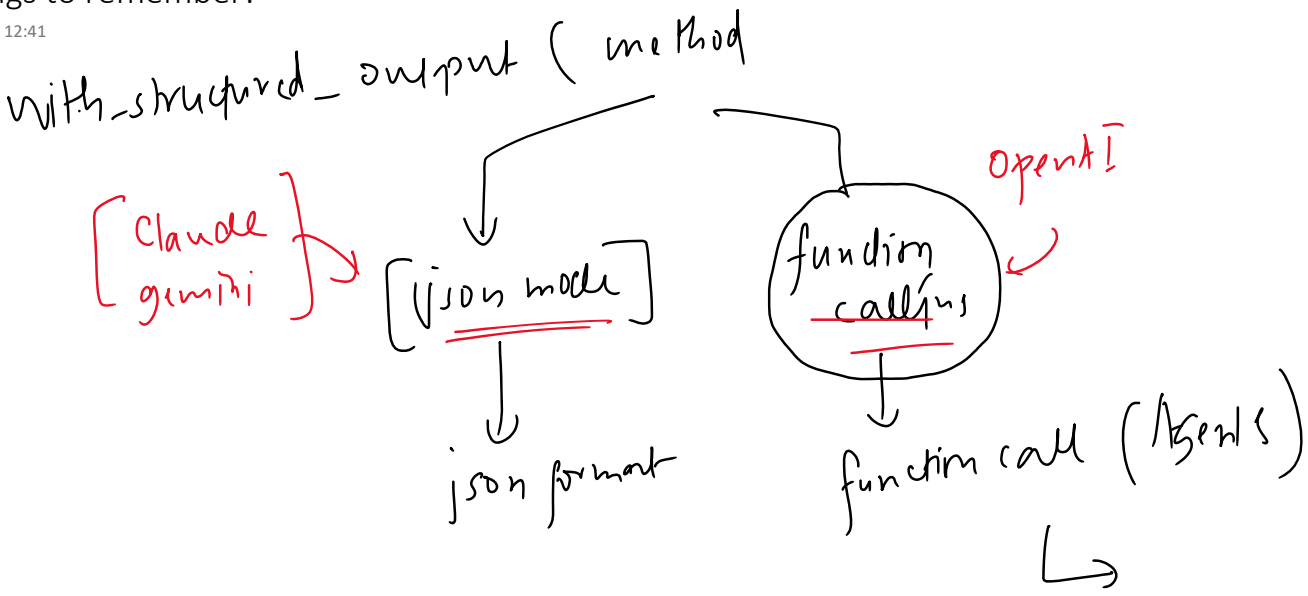
- You **don't want to ~~import extra Python libraries~~** (Pydantic).
- You need **validation but don't need Python objects**.
- You want to define structure **in a standard JSON format**.

🚀 When to Use What?

Feature	TypedDict ✅	Pydantic 🚀	JSON Schema 🌍
Basic structure	✅	✅	✅
Type enforcement	✅	✅	✅
Data validation	❌	✅	✅
Default values	❌	✅	❌
Automatic conversion	❌	✅	❌
Cross-language compatibility	❌	❌	✅

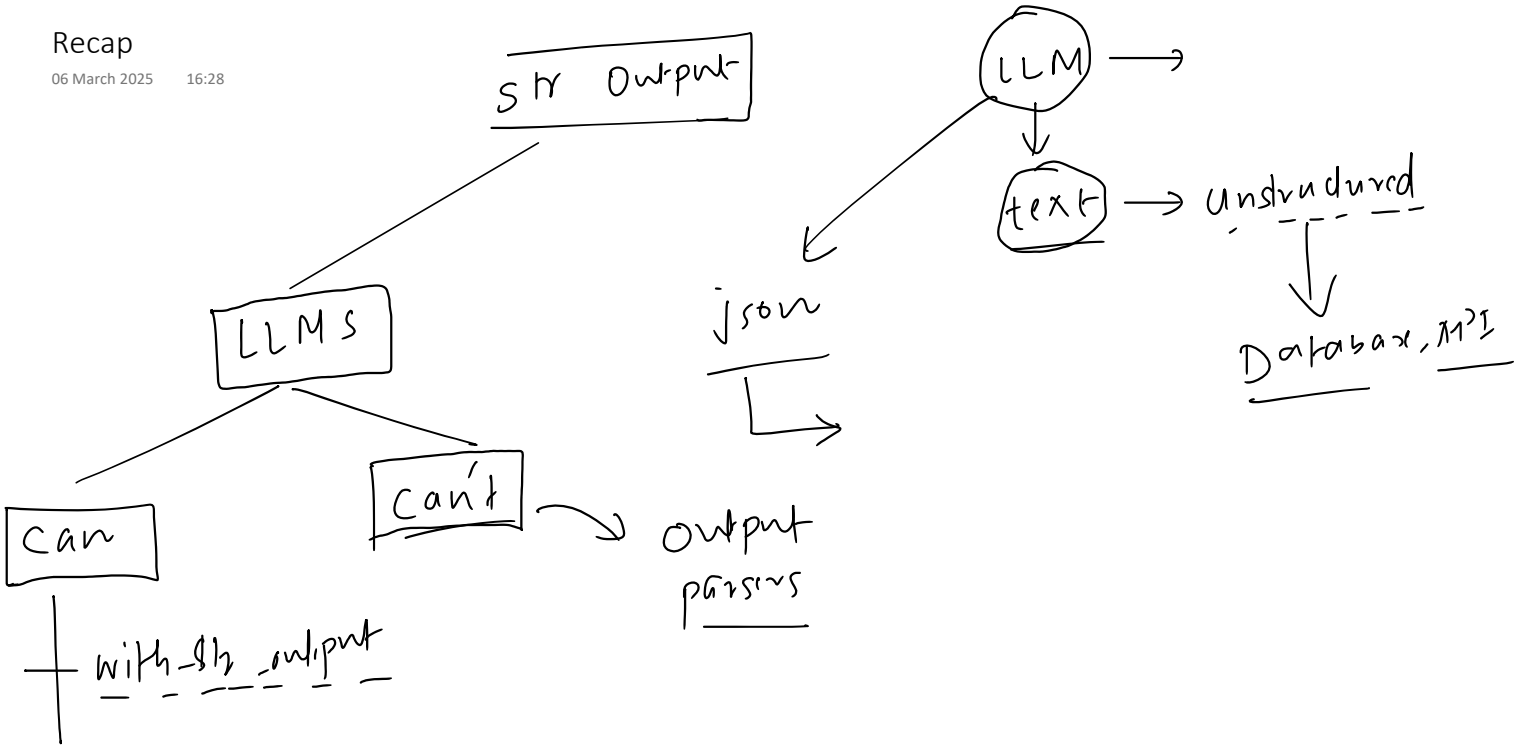
A few things to remember!

03 March 2025 12:41



Recap

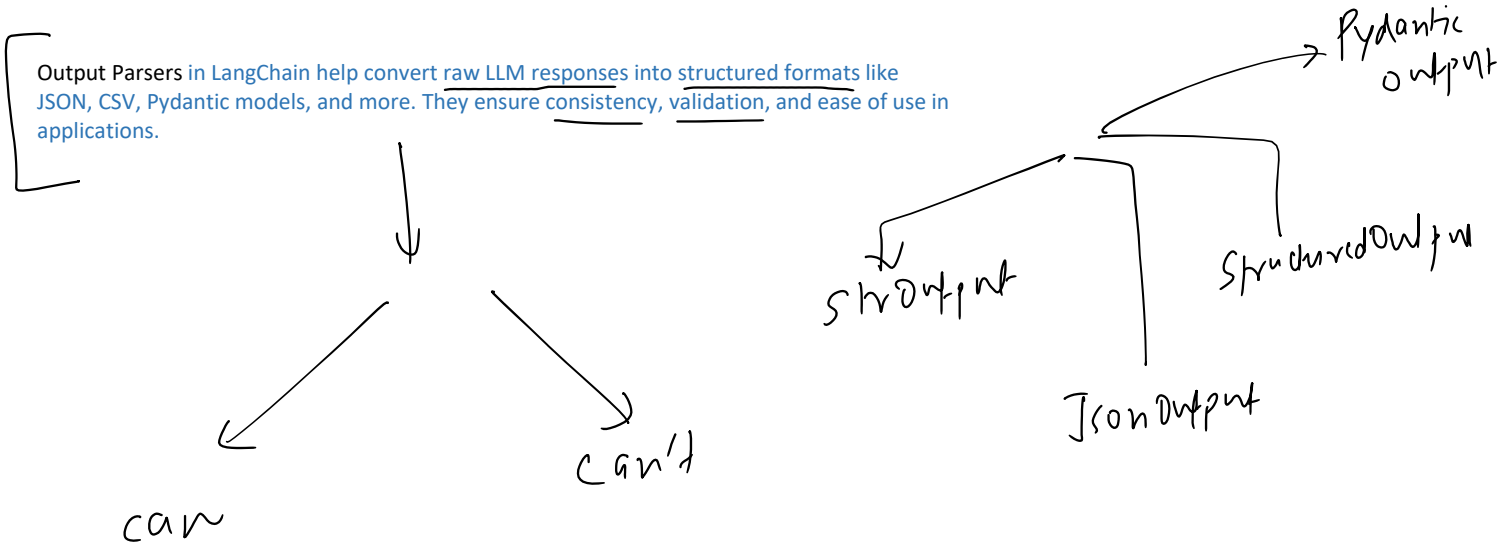
06 March 2025 16:28



Output Parsers

06 March 2025 16:29

Output Parsers in LangChain help convert raw LLM responses into structured formats like JSON, CSV, Pydantic models, and more. They ensure consistency, validation, and ease of use in applications.



StrOutputParser

06 March 2025 16:29

The StrOutputParser is the simplest output parser in LangChain. It is used to parse the output of a Language Model (LLM) and return it as a plain string.

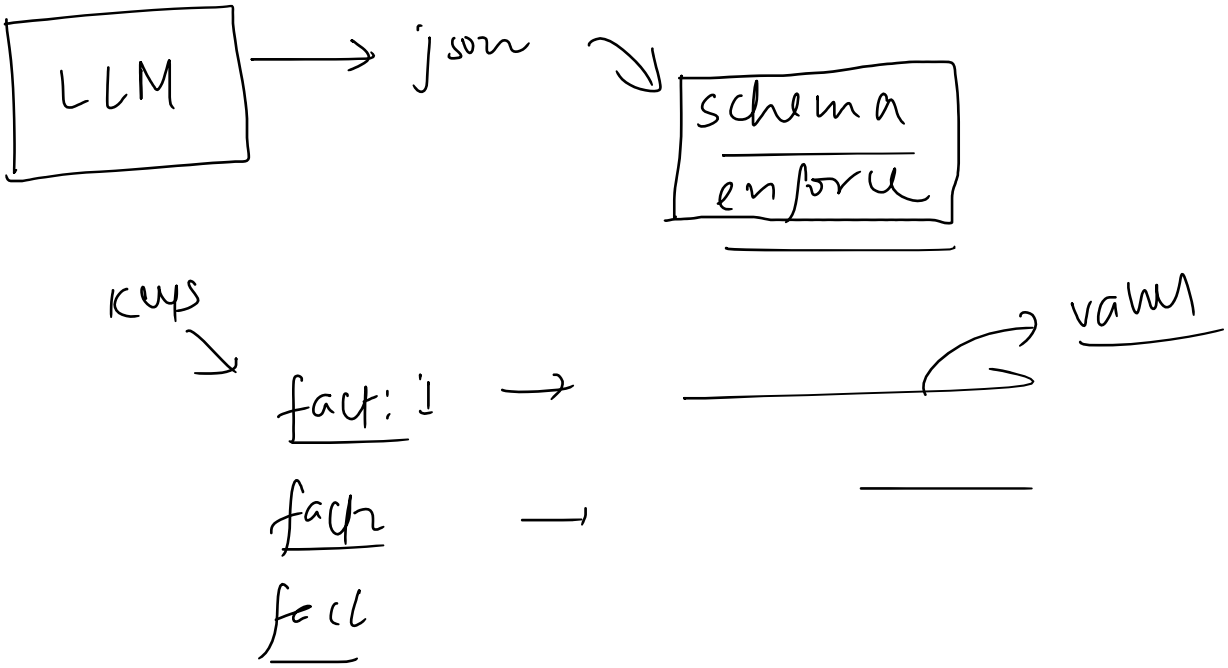
result.content

```
content='A black hole is a region in space where gravity is so strong that nothing, not e
ven light, can escape its pull. It is formed when a massive star collapses upon itself.'
additional_kwargs={'refusal': None} response_metadata={'token_usage': {'completion_tokens
': 37, 'prompt_tokens': 15, 'total_tokens': 52, 'completion_tokens_details': {'accepted_p
rediction_tokens': 0, 'audio_tokens': 0, 'reasoning_tokens': 0, 'rejected_prediction_toke
ns': 0}, 'prompt_tokens_details': {'audio_tokens': 0, 'cached_tokens': 0}}, 'model_name':
 'gpt-3.5-turbo-0125', 'system_fingerprint': None, 'finish_reason': 'stop', 'logprobs': N
one} id='run-a7b90203-58f8-47c5-a01b-01184b6aec14-0' usage_metadata={'input_tokens': 15,
'output_tokens': 37, 'total_tokens': 52, 'input_token_details': {'audio': 0, 'cache_read'
: 0}, 'output_token_details': {'audio': 0, 'reasoning': 0}}
```

topic → LLM → [Detailed report] → LLM → 5 line summary

JSONOutputParser

06 March 2025 16:29

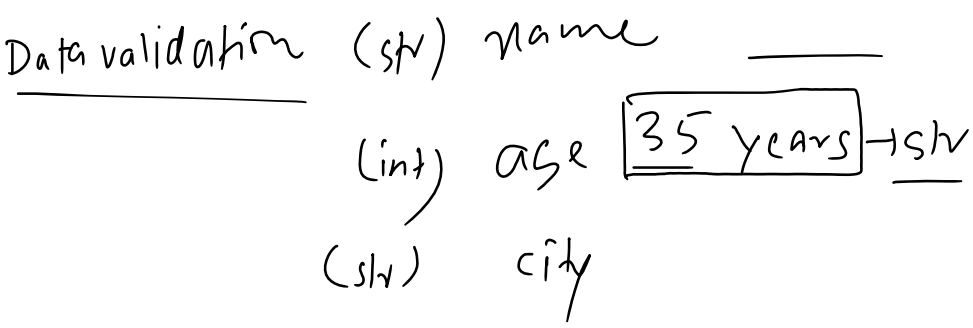


StructuredOutputParser

06 March 2025 16:29

StructuredOutputParser is an output parser in LangChain that helps extract structured JSON data from LLM responses based on predefined field schemas.

It works by defining a list of fields (ResponseSchema) that the model should return, ensuring the output follows a structured format.



PydanticOutputParser

06 March 2025 16:30

🔹 What is `PydanticOutputParser` in LangChain?

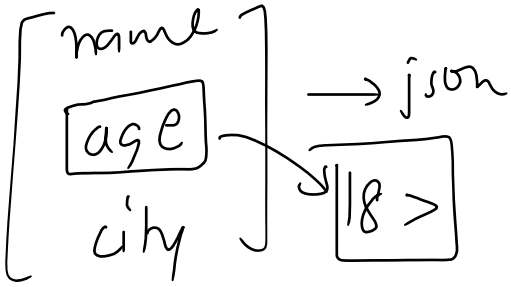
`PydanticOutputParser` is a **structured output parser** in LangChain that uses **Pydantic models** to enforce schema validation when processing LLM responses.

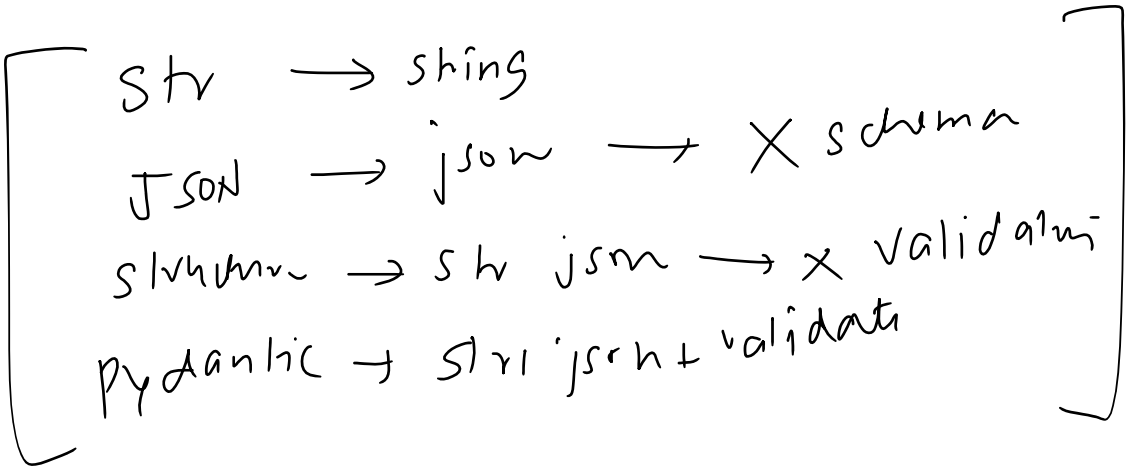
→ schema

[pydantic object]

🚀 Why Use `PydanticOutputParser`?

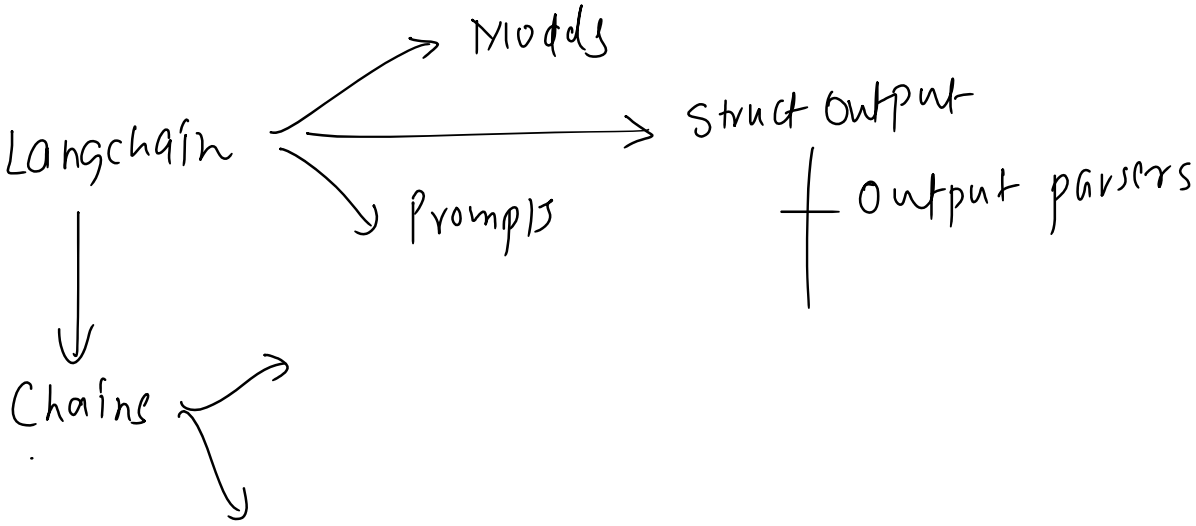
- ✅ **Strict Schema Enforcement** → Ensures that LLM responses follow a well-defined structure.
- ✅ **Type Safety** → Automatically converts LLM outputs into Python objects.
- ✅ **Easy Validation** → Uses Pydantic's built-in validation to catch incorrect or missing data.
- ✅ **Seamless Integration** → Works well with other LangChain components.





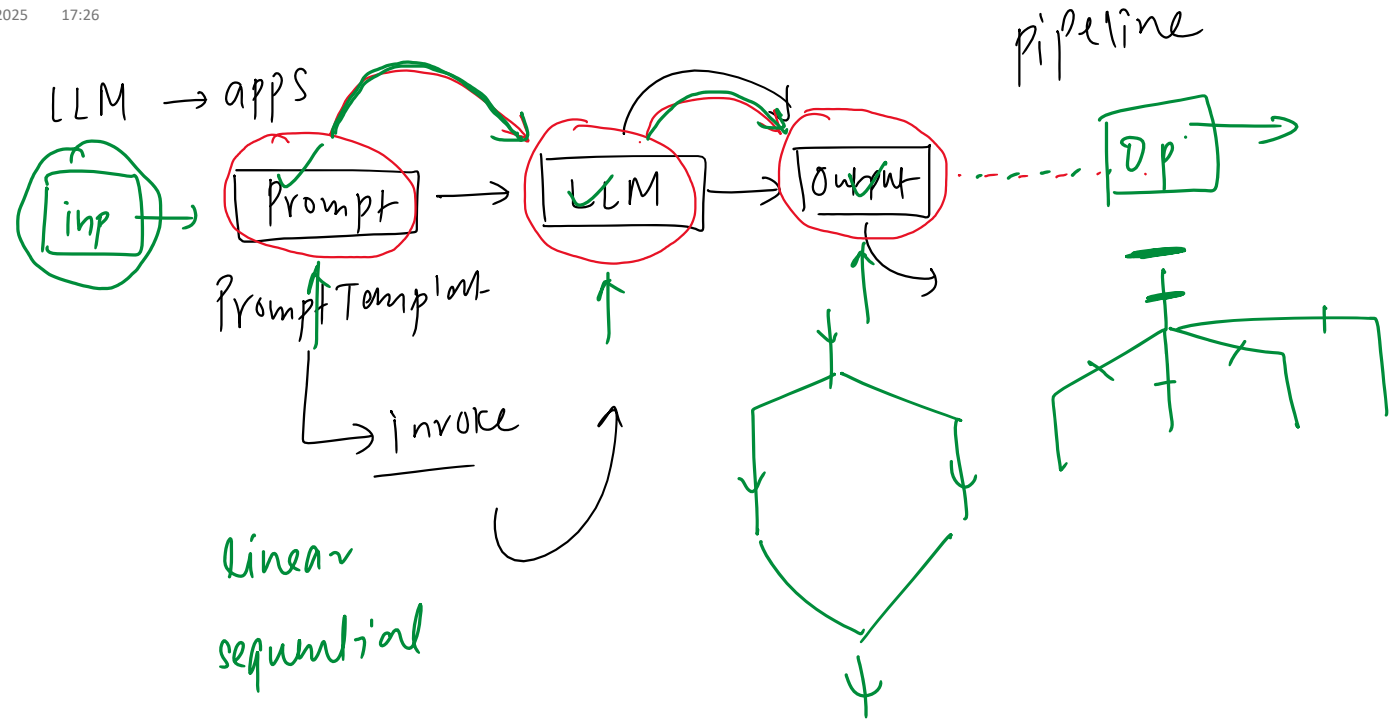
Recap

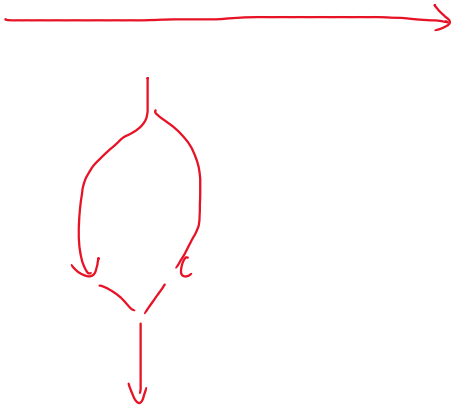
11 March 2025 17:26



What & Why

11 March 2025 17:26





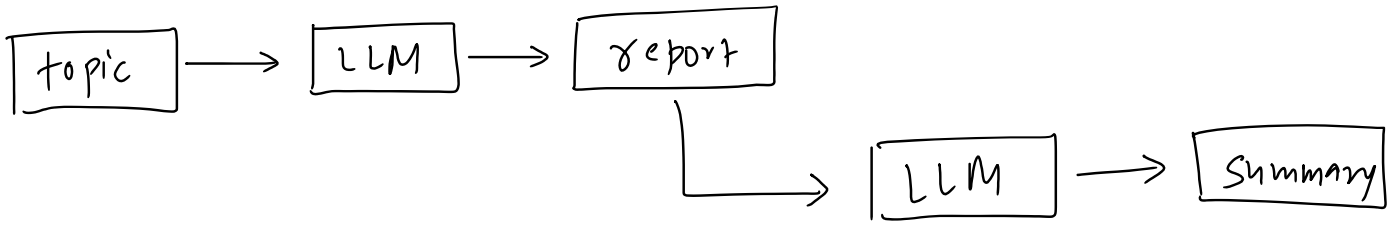
simple chains

Simple Chain

11 March 2025 17:26

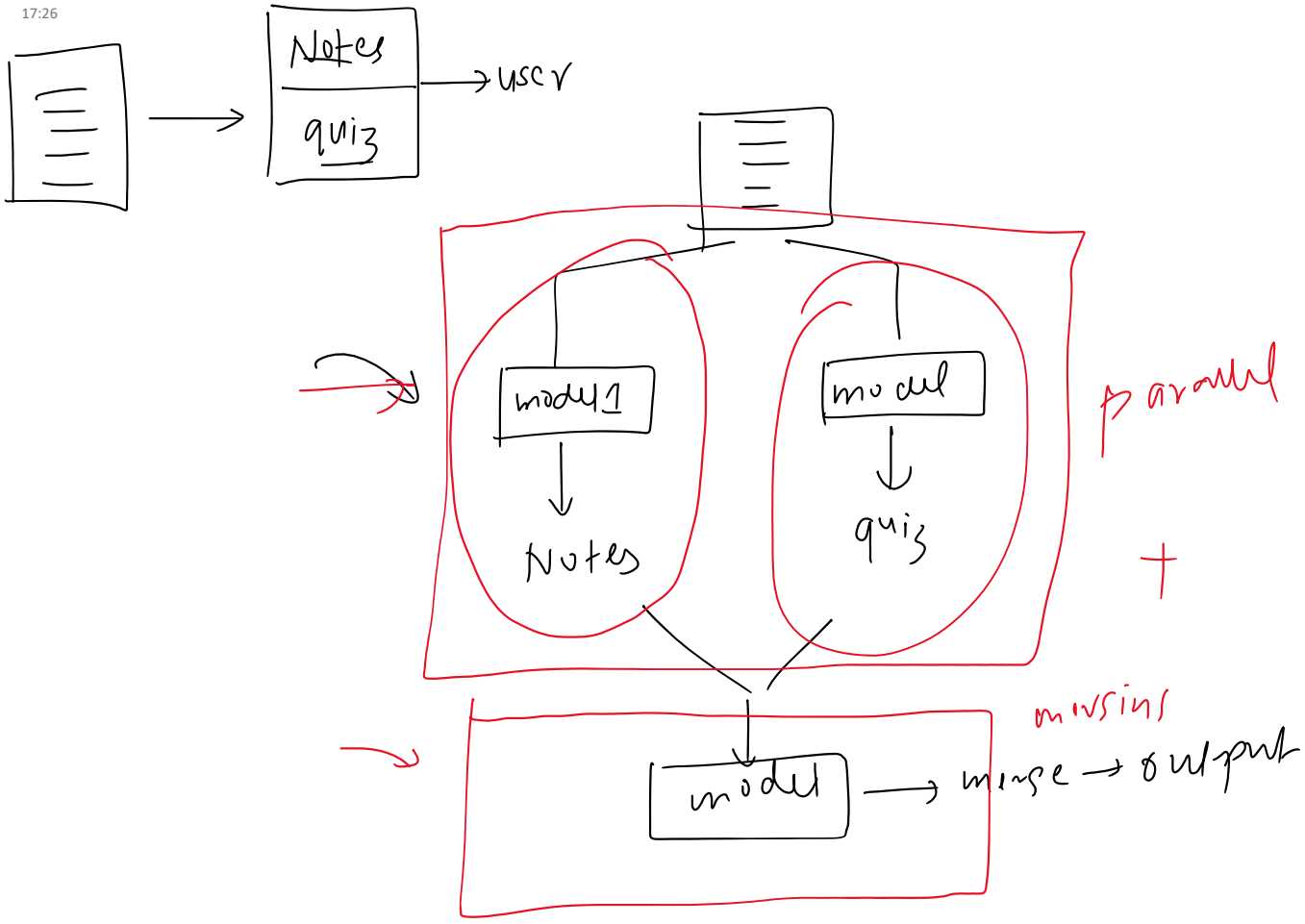
Sequential Chain

11 March 2025 17:26



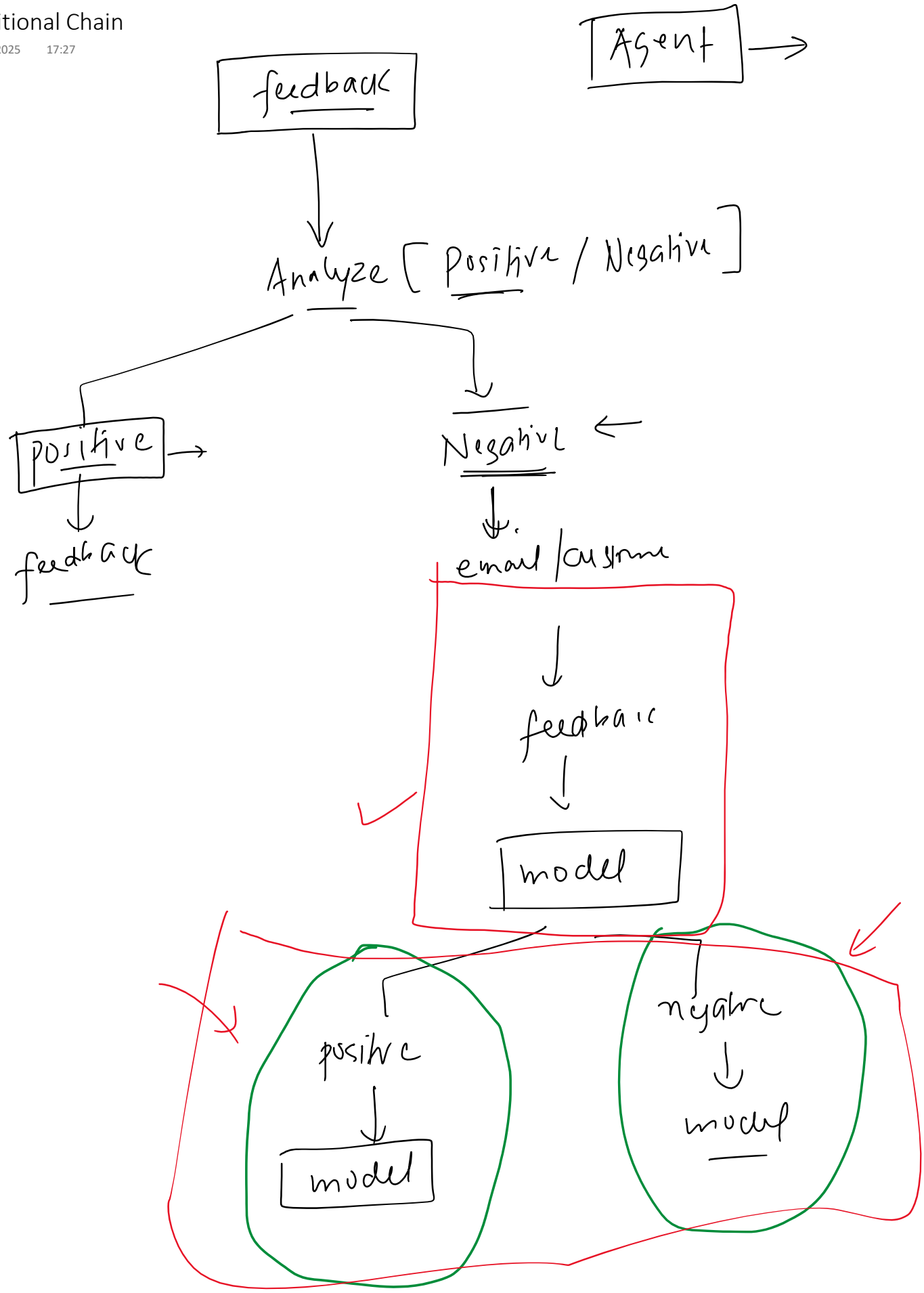
Parallel Chain

11 March 2025 17:26



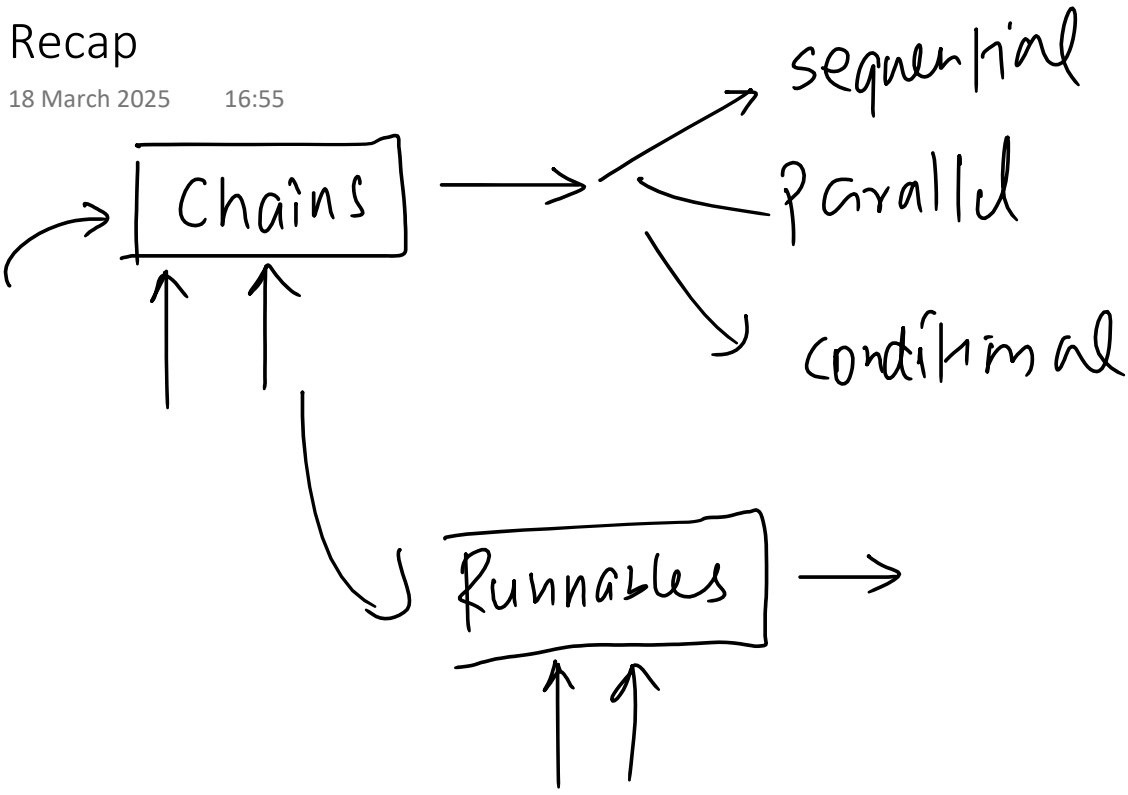
Conditional Chain

11 March 2025 17:27



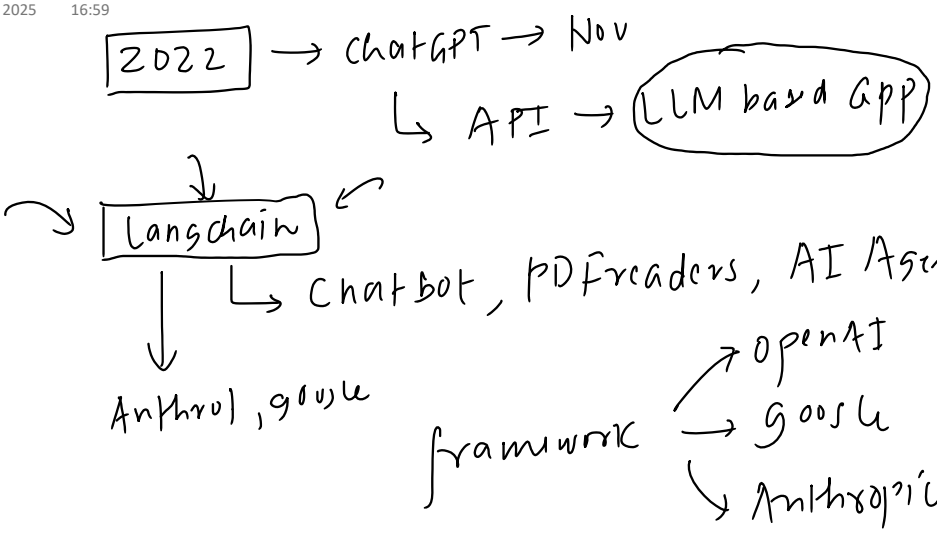
Recap

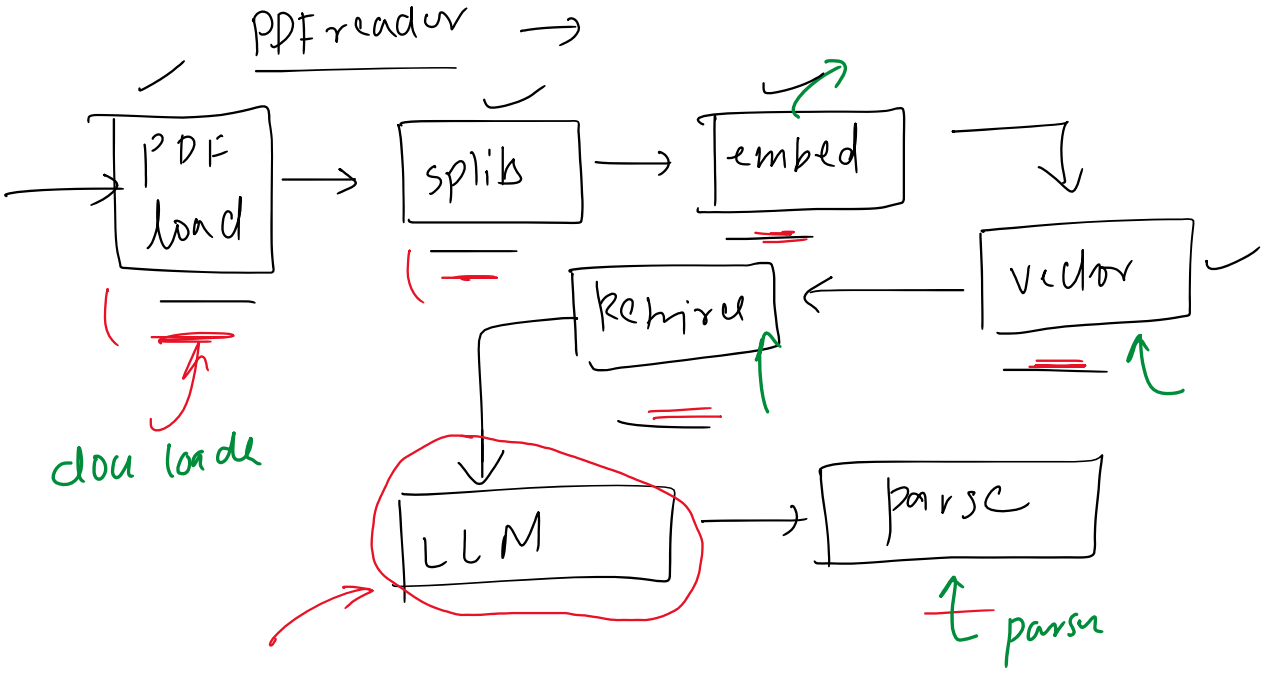
18 March 2025 16:55

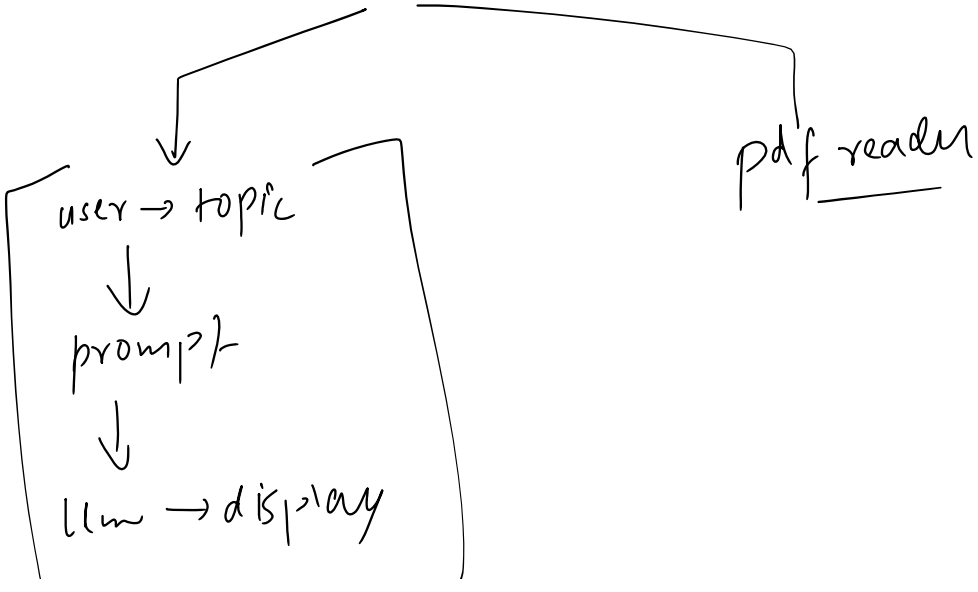


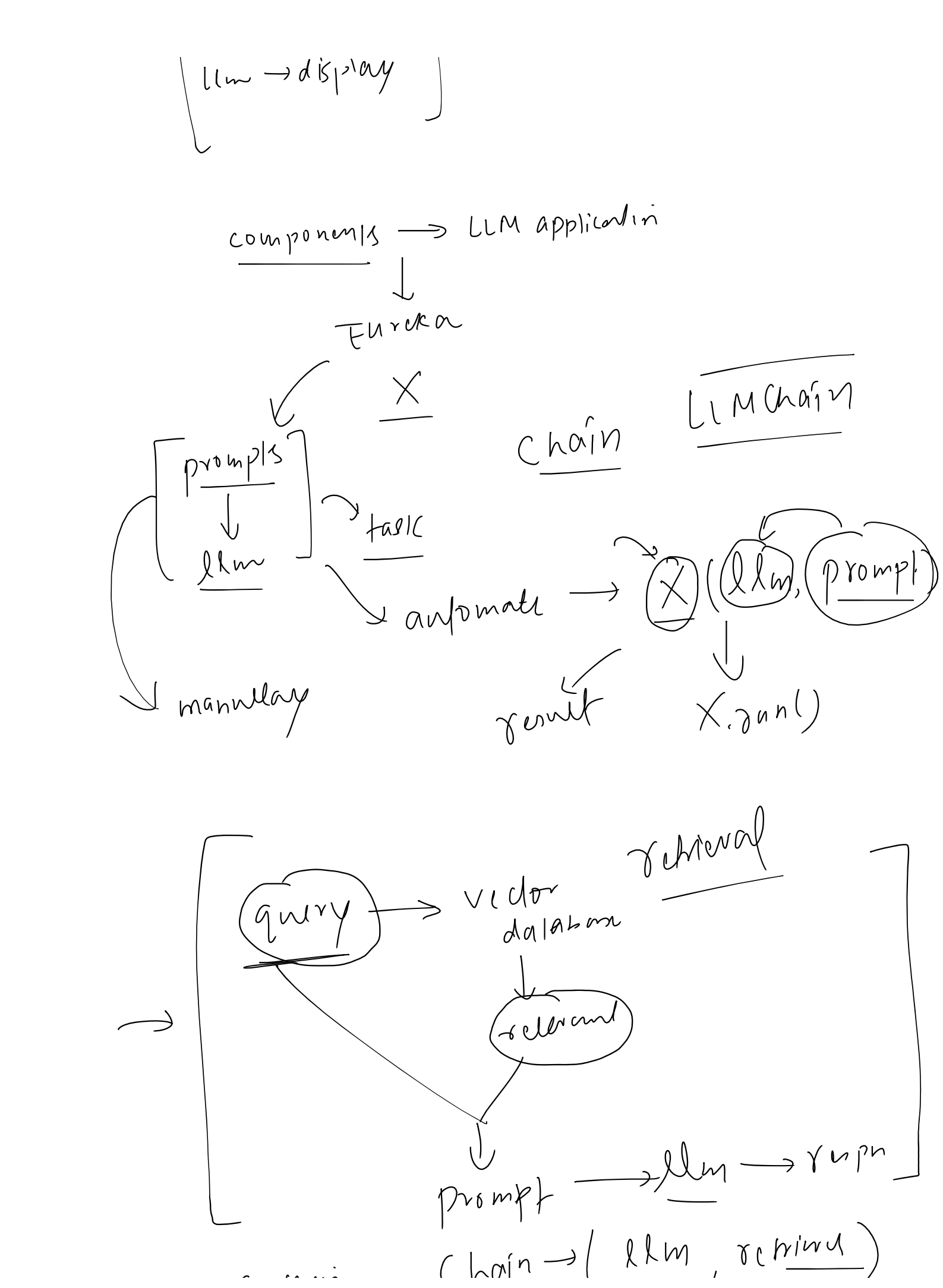
The Why

18 March 2025 16:59








llm → display
components → LLM application
Eureka
X
LLMChain
Chain
prompts
llm
task
automate
manually
X(llm, prompt)
result
X.run()
retrieval
query
vector database
relevant
prompt
llm → resp
chain → (llm , retrieval)

RetrieveQAChain Chain → (llm , retriver)

LLMChain

LLMchain ↔ LLMChain

Chain Name	Description
LLMChain	Basic chain that calls an LLM with a prompt template.
SequentialChain	Chains multiple LLM calls in a specific sequence.
SimpleSequentialChain	A simplified version of SequentialChain for easier use.
ConversationalRetrievalChain	Handles conversational Q&A with memory and retrieval.
RetrievalQA	Fetches relevant documents and uses an LLM for question-answering.
RouterChain	Directs user queries to different chains based on intent.
MultiPromptChain	Uses different prompts for different user intents dynamically.
HydeChain (Hypothetical Document Embeddings)	Generates hypothetical answers to improve document retrieval.
AgentExecutorChain	Orchestrates different tools and actions dynamically using an agent.
SQLDatabaseChain	Connects to SQL databases and answers natural language queries.

Chains
↓
use cases
↓
resive task
↓
chain

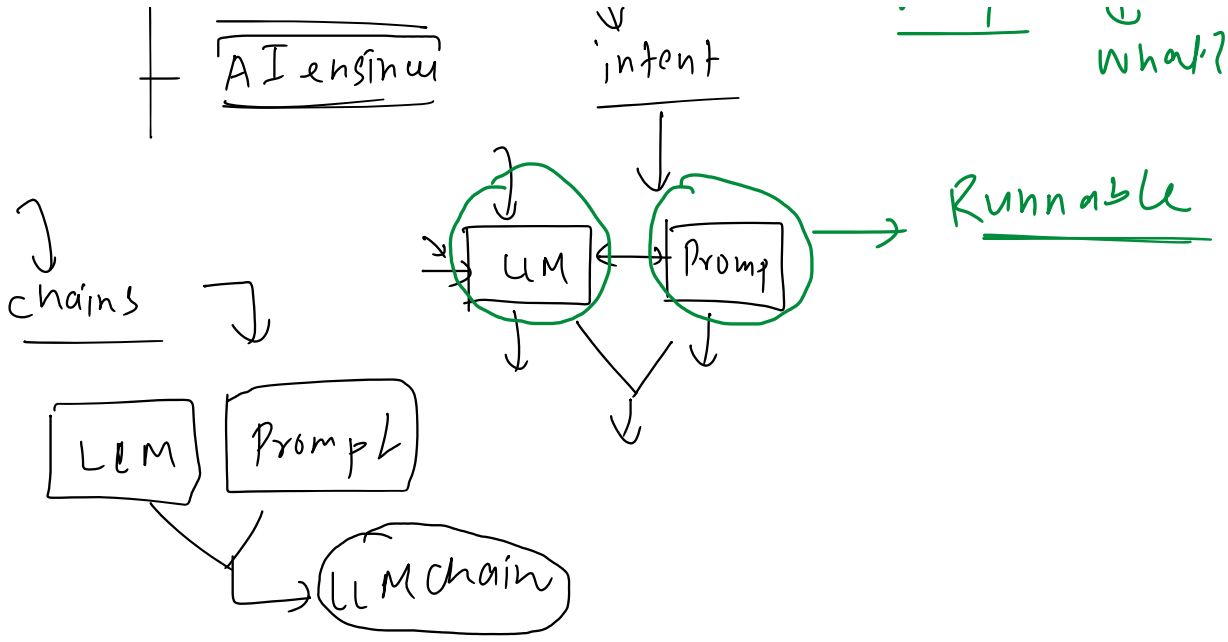
→ topic → joke → explanation

Problem

too many chains
- codebase
- AI engine

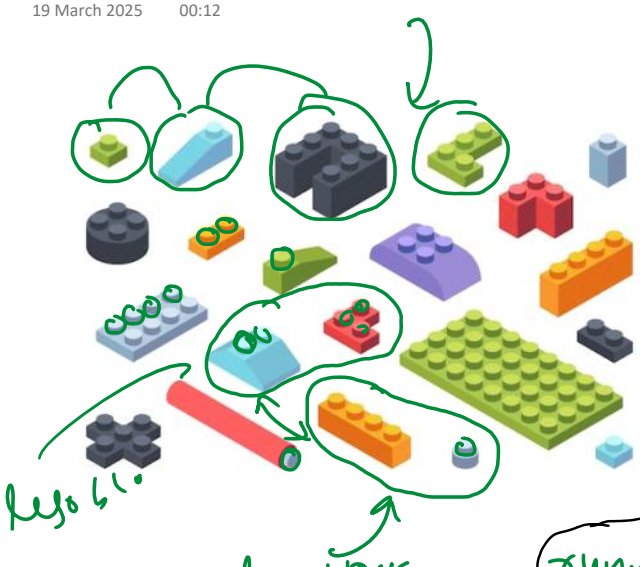
↓ intent

why? ↓ what?


AI engine
intent
what?
Runnable
LLM
Prompt
chains
LLM
Prompt
LLMchain

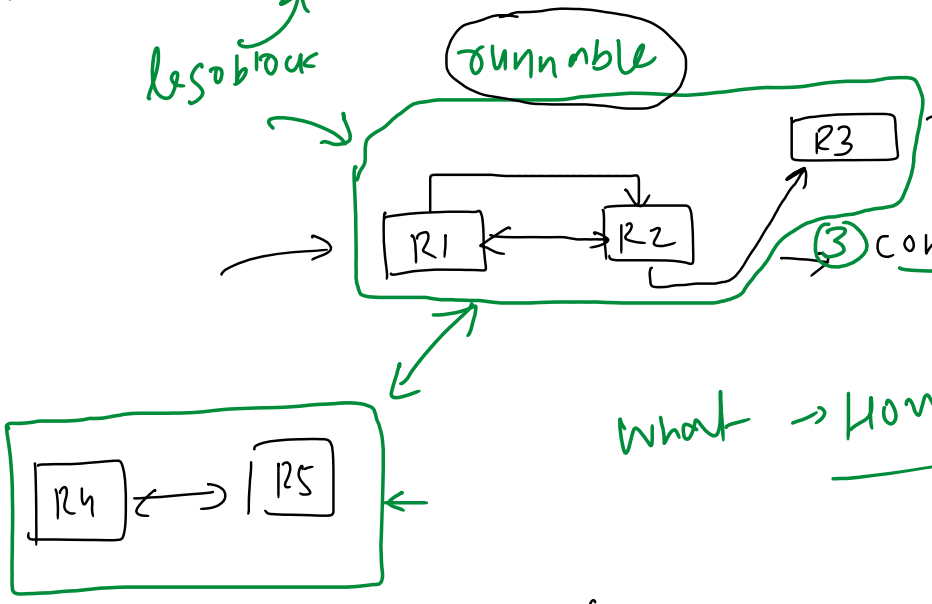
The What

19 March 2025 00:12



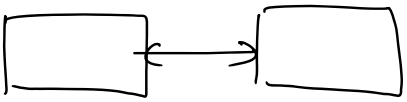
1. unit of work
   - input
   - process
   - output

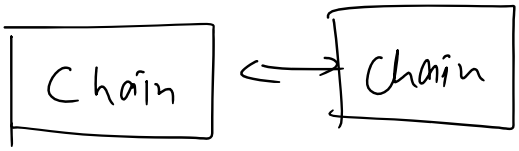
2. common interface
   - invoke()
   - batch()
   - stream()

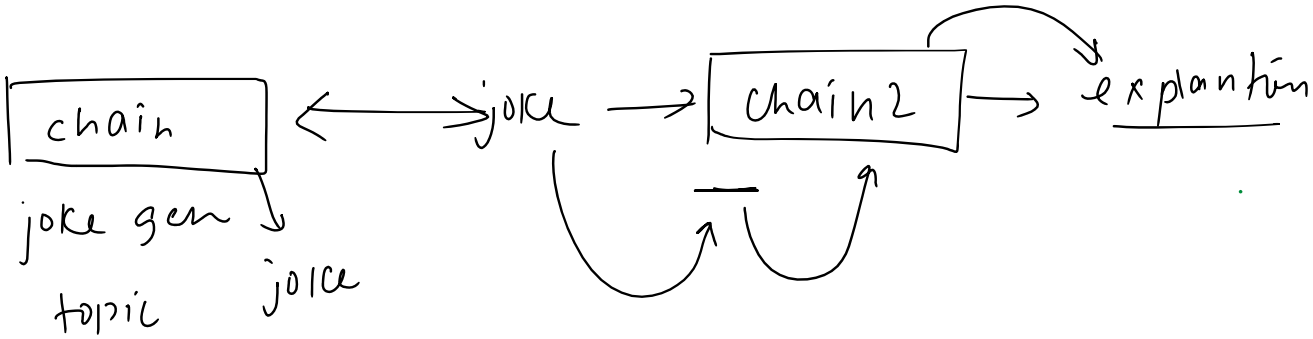


what → How →

chains

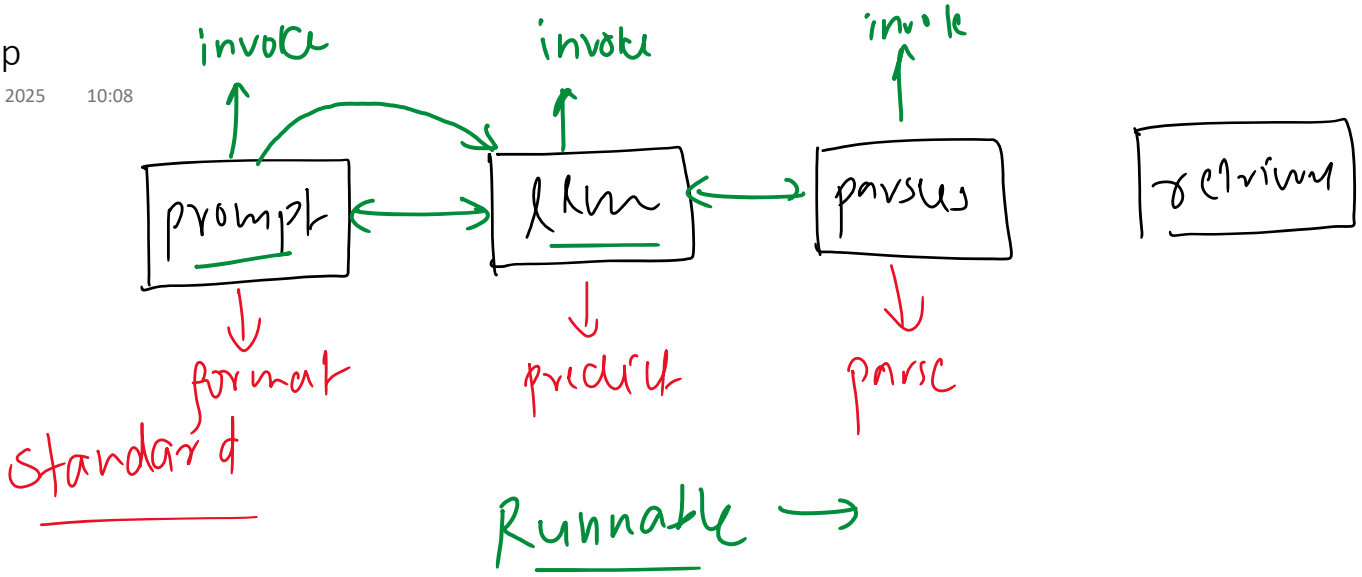






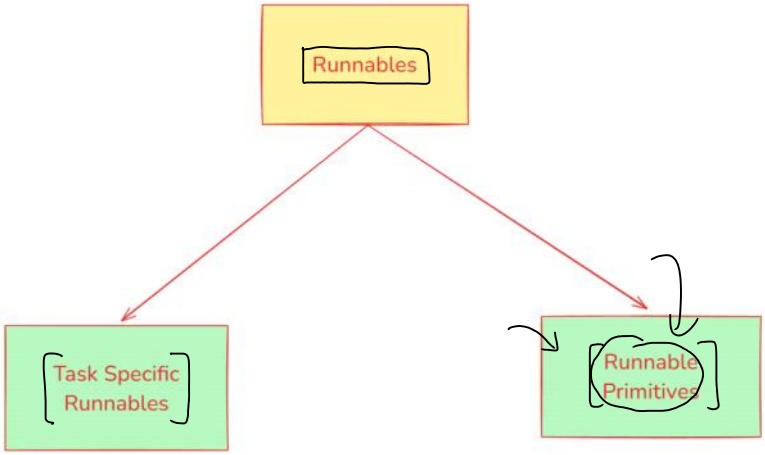
Recap

20 March 2025 10:08



Plan of Action

20 March 2025 11:02



✅ **Definition**: These are **core LangChain components** that have been converted into Runnables so they can be used in pipelines.

✅ **Purpose**: Perform **task-specific** operations like LLM calls, prompting, retrieval, etc.

✅ **Examples**:

- `ChatOpenAI` → Runs an LLM model.
- `PromptTemplate` → Formats prompts dynamically.
- `Retriever` → Retrieves relevant documents.

✅ **Definition**: These are **fundamental building blocks** for structuring execution logic in AI workflows.

✅ **Purpose**: They **help orchestrate execution** by defining how different Runnables interact (sequentially, in parallel, conditionally, etc.).

✅ **Examples**:

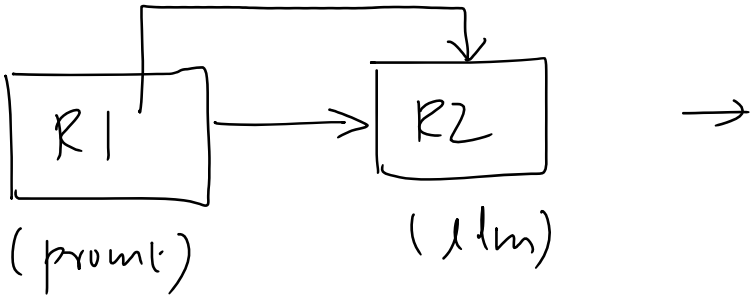
- `RunnableSequence` → Runs steps **in order** (`|` operator).
- `RunnableParallel` → Runs multiple steps **simultaneously**.
- `RunnableMap` → Maps the same input across multiple functions.
- `RunnableBranch` → Implements **conditional execution** (if-else logic).
- `RunnableLambda` → Wraps **custom Python functions** into Runnables.
- `RunnablePassthrough` → Just forwards input as output (acts as a placeholder).

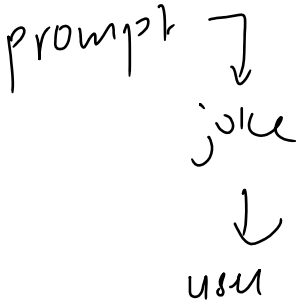
1. RunnableSequence

20 March 2025 12:10

RunnableSequence is a sequential chain of runnables in LangChain that executes each step one after another, passing the output of one step as the input to the next.

It is useful when you need to compose multiple runnables together in a structured workflow.



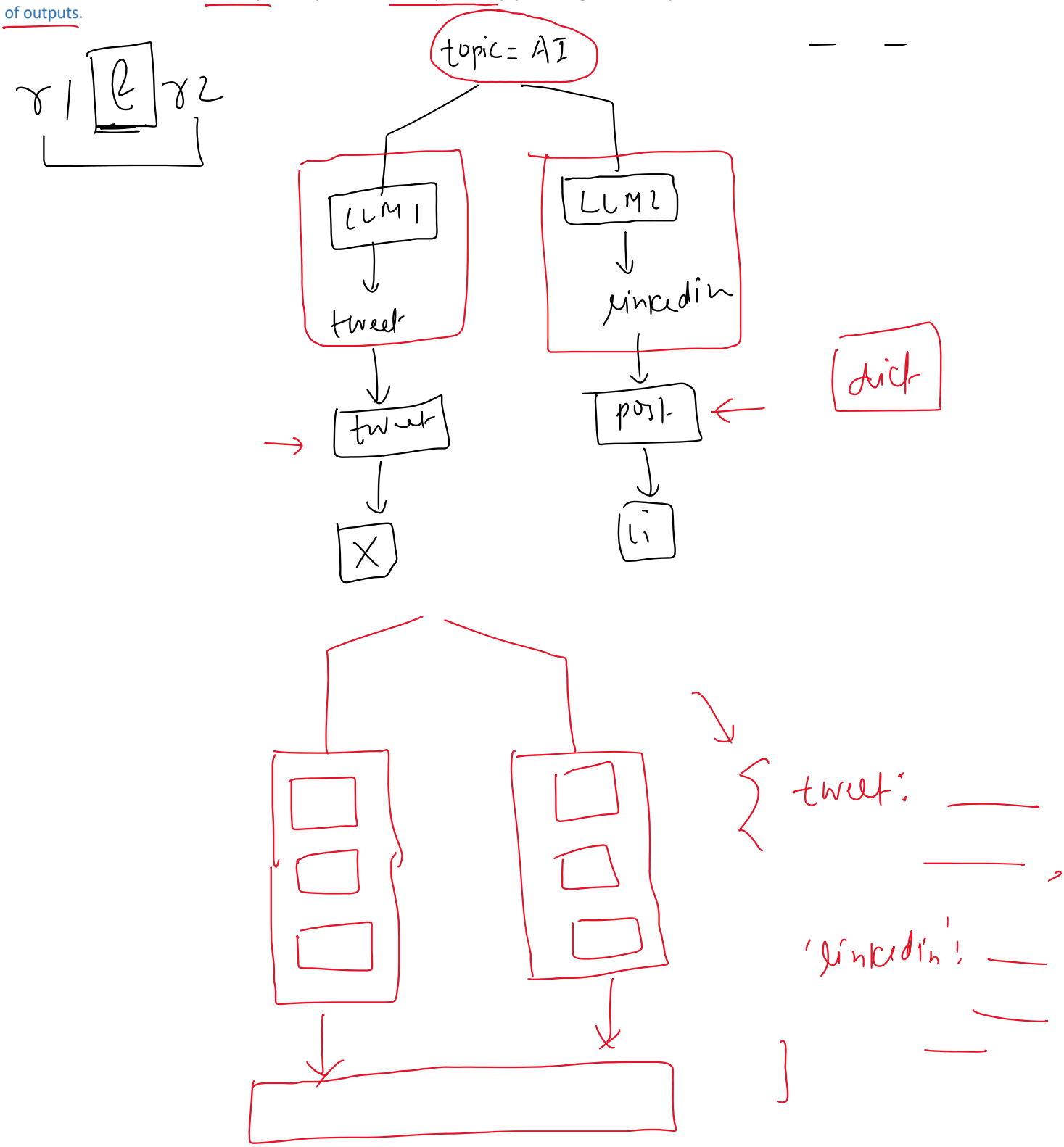


2. RunnableParallel

20 March 2025 18:33

RunnableParallel is a runnable primitive that allows multiple runnables to execute in parallel.

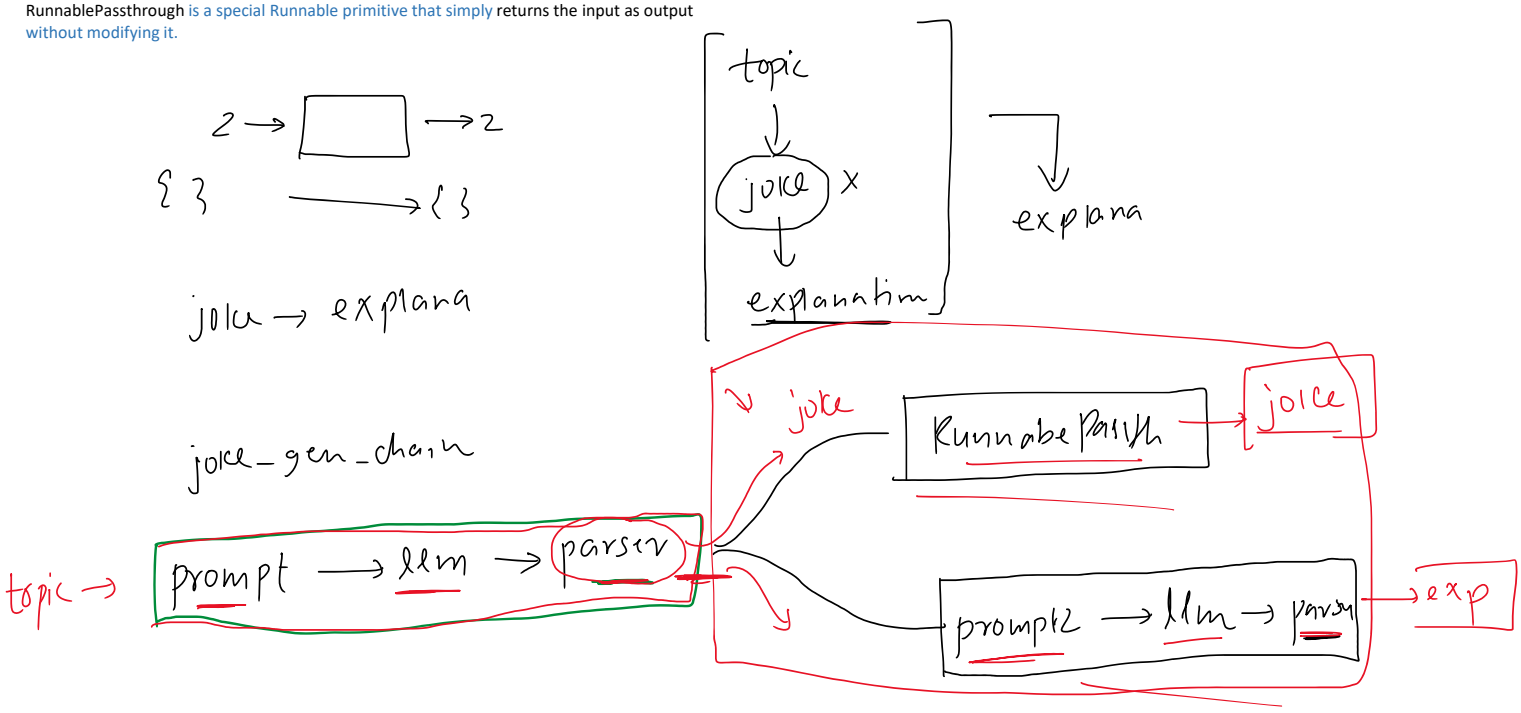
Each runnable receives the same input and processes it independently, producing a dictionary of outputs.



3. RunnablePassthrough

20 March 2025 22:34

RunnablePassthrough is a special Runnable primitive that simply returns the input as output without modifying it.

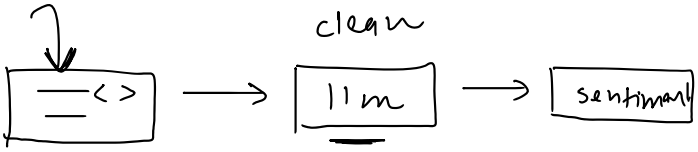


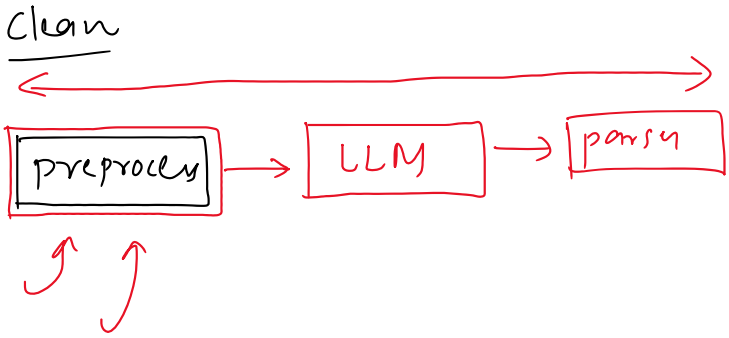
4. RunnableLambda

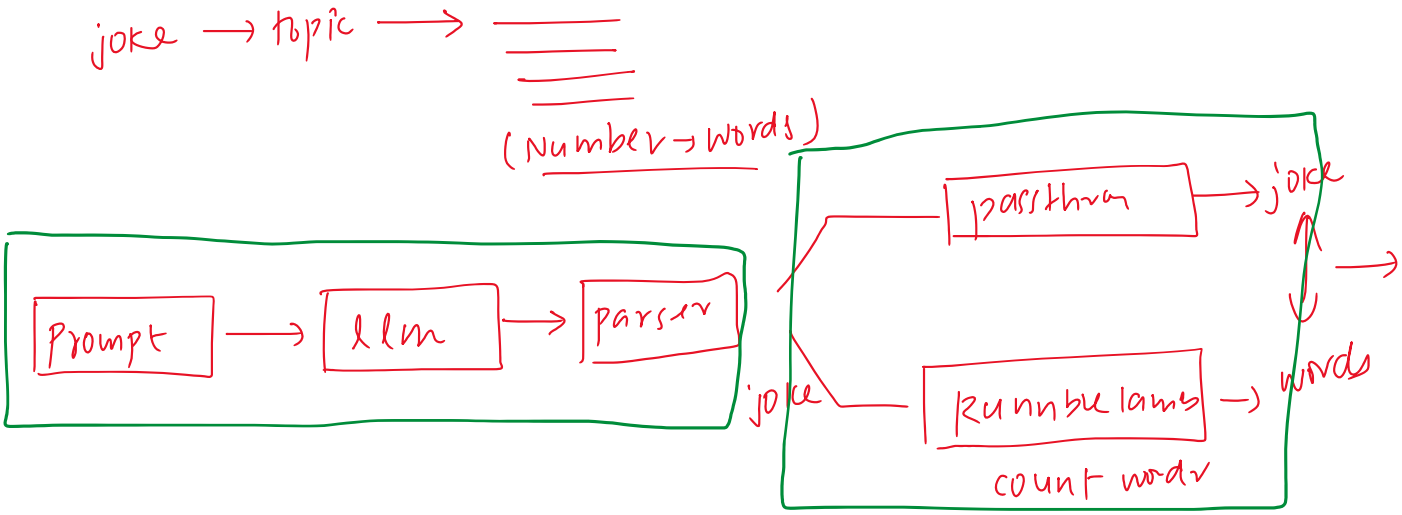
20 March 2025 23:18

RunnableLambda is a runnable primitive that allows you to apply custom Python functions within an AI pipeline.

It acts as a middleware between different AI components, enabling preprocessing, transformation, API calls, filtering, and post-processing in a LangChain workflow.





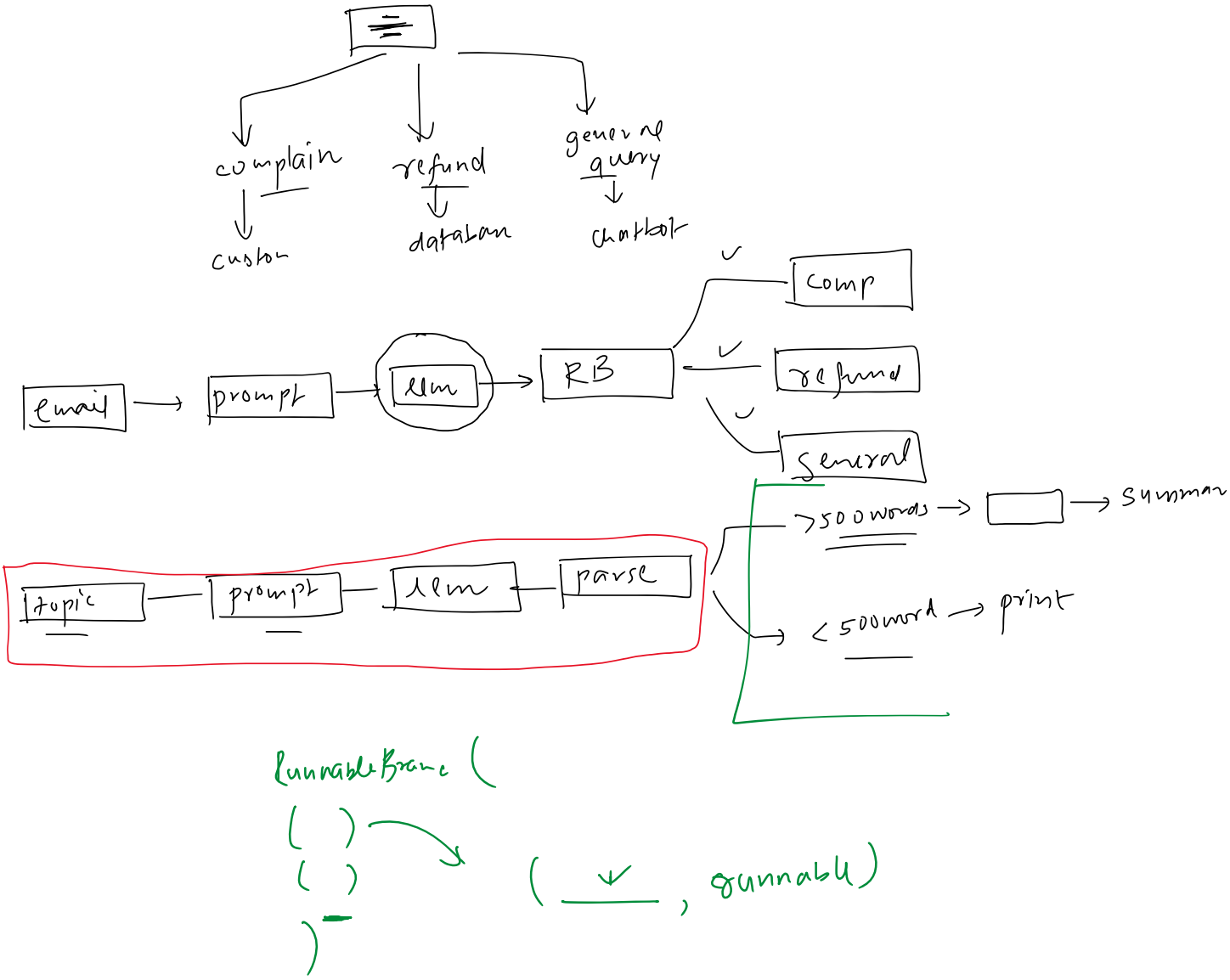


5. RunnableBranch → conditional chains

21 March 2025 08:02

RunnableBranch is a control flow component in LangChain that allows you to conditionally route input data to different chains or runnables based on custom logic.

It functions like an if/elif/else block for chains — where you define a set of condition functions, each associated with a runnable (e.g., LLM call, prompt chain, or tool). The first matching condition is executed. If no condition matches, a default runnable is used (if provided).



RunnableBranch (

( )

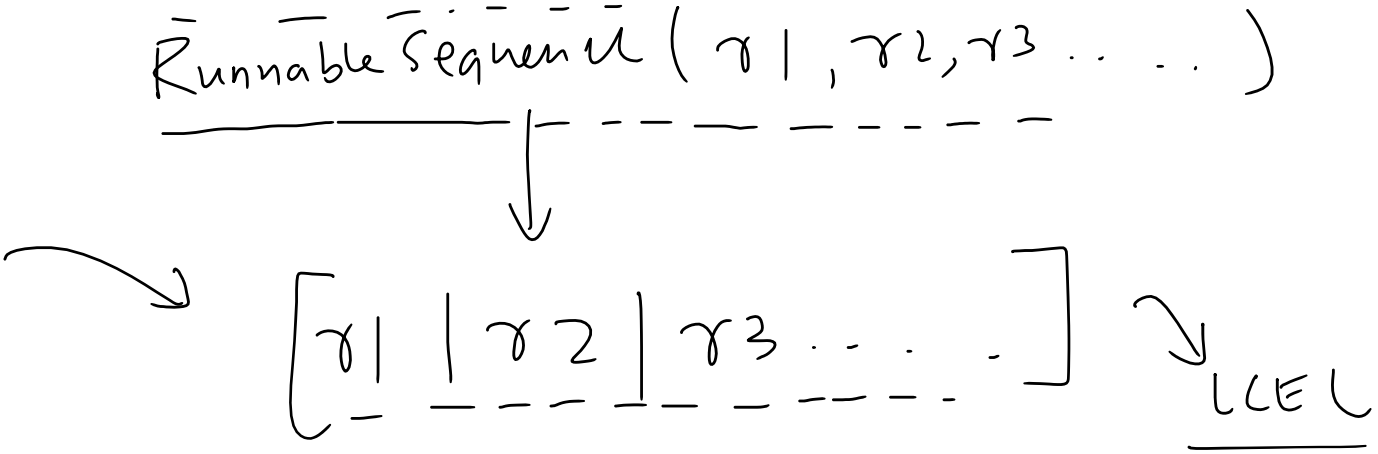
( )

)

(, runnable)

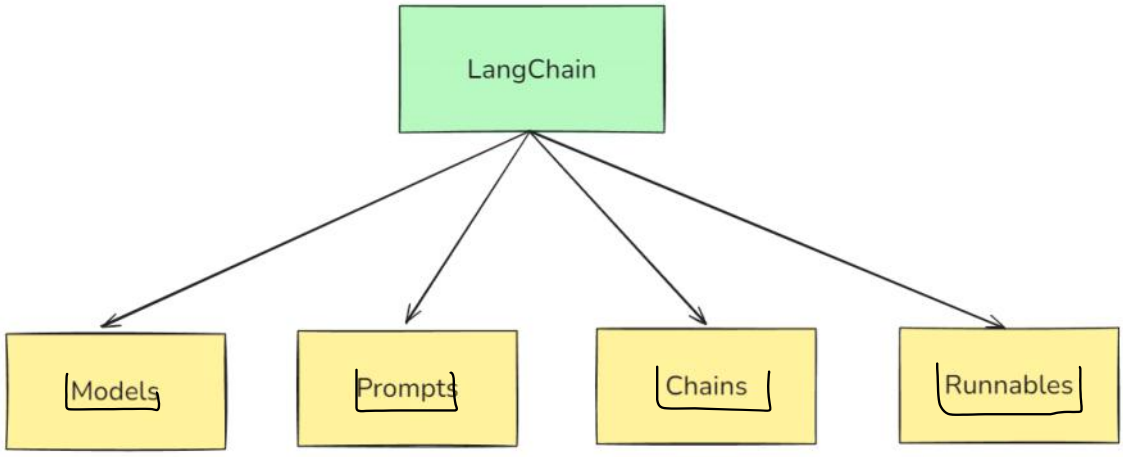
LCEL

21 March 2025 08:39



Plan of Action

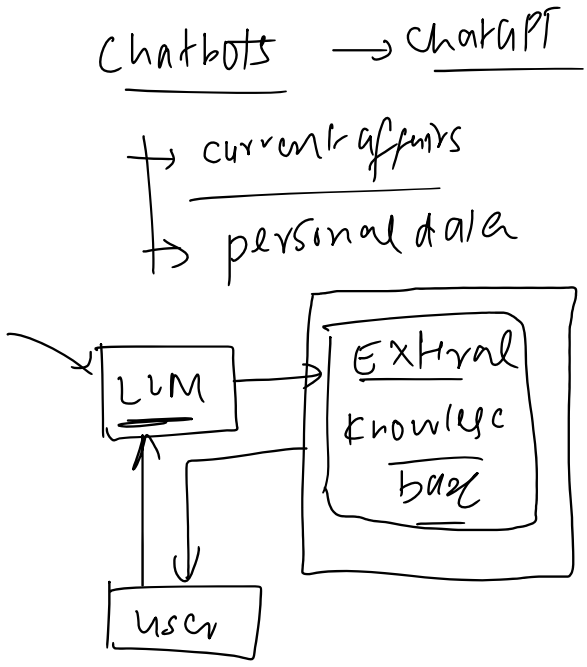
27 March 2025 10:59

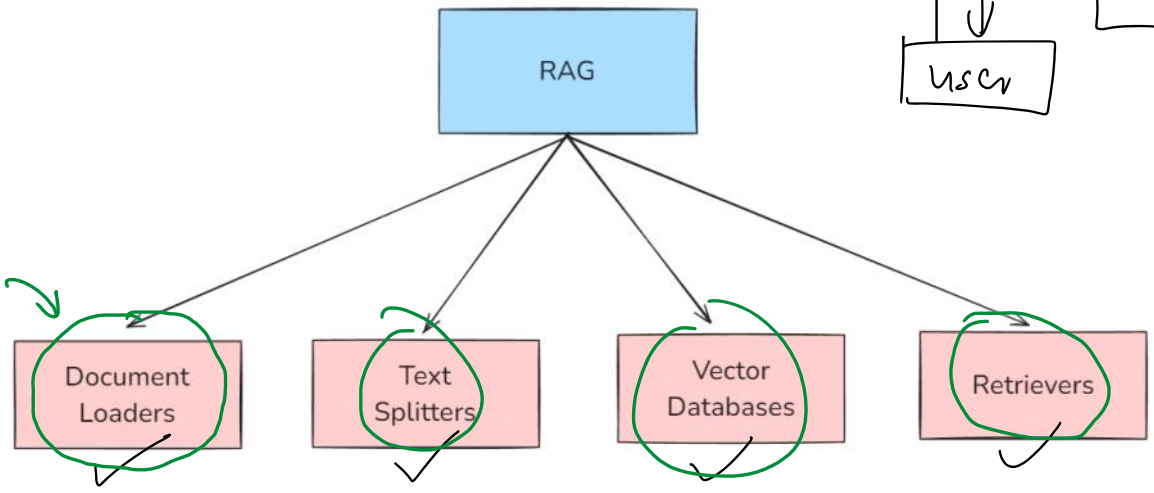


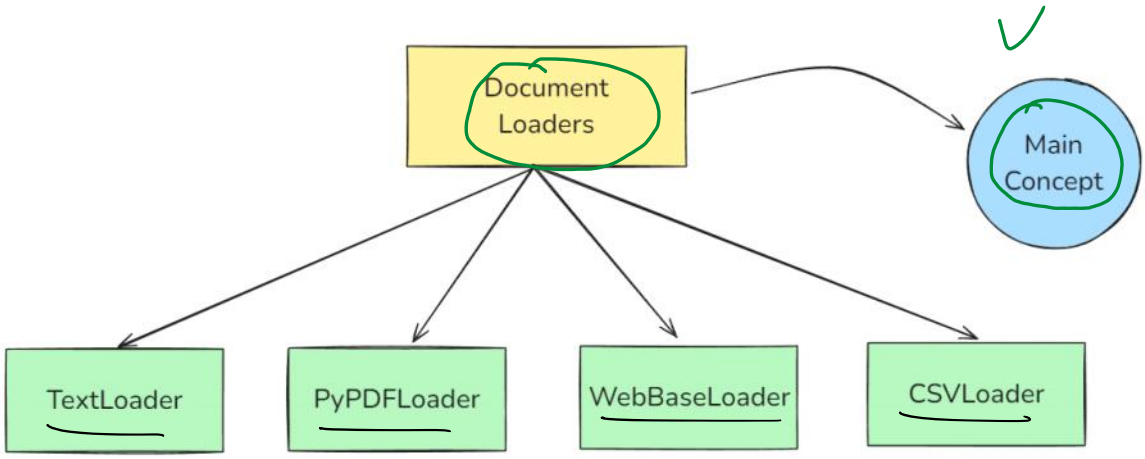
RAG is a technique that combines information retrieval with language generation, where a model retrieves relevant documents from a knowledge base and then uses them as context to generate accurate and grounded responses.

Benefits of using RAG

1. Use of up-to-date information
2. Better privacy
3. No limit of document size





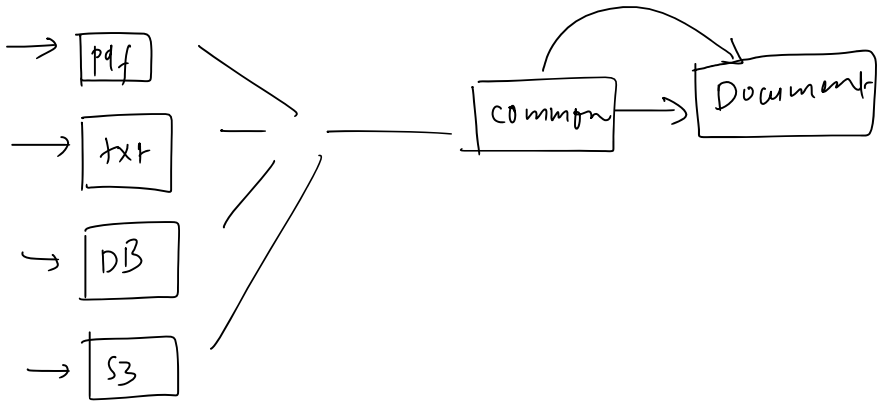

Document Loaders
Main Concept
TextLoader
PyPDFLoader
WebBaseLoader
CSVLoader

Document Loaders in LangChain

27 March 2025 16:20

Document loaders are components in LangChain used to load data from various sources into a standardized format (usually as Document objects), which can then be used for chunking, embedding, retrieval, and generation.

```
Document(
    page_content="The actual text content",
    metadata={"source": "filename.pdf", ...}
)
```



TextLoader

27 March 2025 16:50

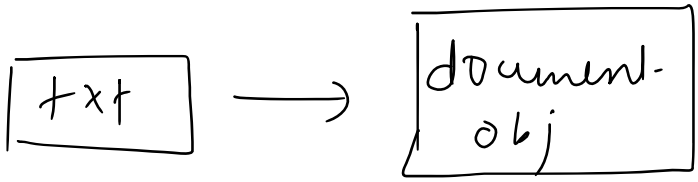
TextLoader is a simple and commonly used document loader in LangChain that reads plain text (.txt) files and converts them into LangChain Document objects.

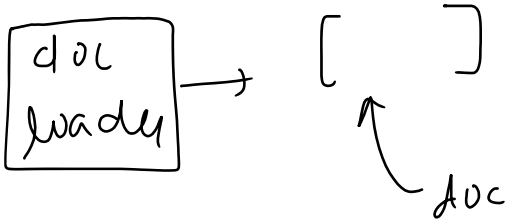
Use Case

- Ideal for loading chat logs, scraped text, transcripts, code snippets, or any plain text data into a LangChain pipeline.

Limitation

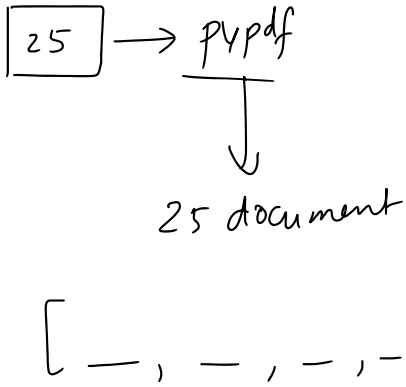
- Works only with .txt files





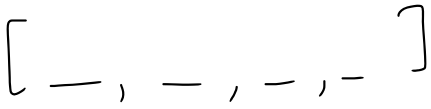
PyPDFLoader → text → scanned pdf

27 March 2025 17:50



PyPDFLoader is a document loader in LangChain used to load content from PDF files and convert each page into a Document object.

```
[
    Document(page_content="Text from page 1", metadata={"page": 0, "source": "file.pdf"}),
    Document(page_content="Text from page 2", metadata={"page": 1, "source": "file.pdf"}),
    ...
]
```



Limitations:

- It uses the PyPDF library under the hood — not great with scanned PDFs or complex layouts.

Use Case	Recommended Loader
Simple, clean PDFs	`PyPDFLoader`
PDFs with tables/columns	`PDFPlumberLoader`
Scanned/image PDFs	`UnstructuredPDFLoader` or `AmazonTextractPDFLoader`
Need layout and image data	`PyMuPDFLoader`
Want best structure extraction	`UnstructuredPDFLoader`

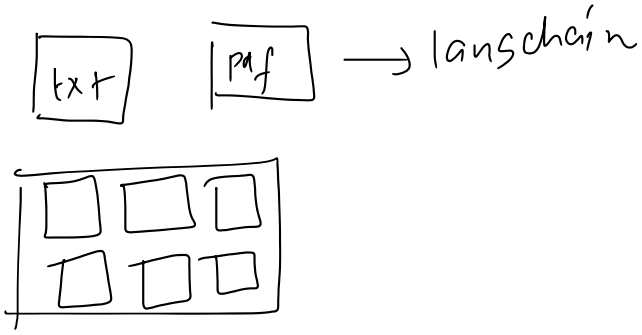
DirectoryLoader

27 March 2025 18:44

DirectoryLoader is a document loader that lets you load multiple documents from a directory (folder) of files.

Glob Pattern	What It Loads
`"**/*.txt"`	All `.txt` files in all subfolders
`"*.pdf"`	All `.pdf` files in the root directory
`"data/*.csv"`	All `.csv` files in the `data/` folder
`"**/*"`	All files (any type, all folders)

`**` = recursive search through subfolders



Load vs Lazy load → 500 pdf → generator of docs →

27 March 2025 23:51

eager loading

`load()`

- **Eager Loading** (loads everything at once).
- Returns: A **list of** `Document` **objects**.
- Loads all documents **immediately** into memory.
- Best when:
  - The number of documents is small.
  - You want everything loaded upfront.

`lazy_load()`

- **Lazy Loading** (loads on demand).
- Returns: A **generator** of `Document` objects.
- Documents are **not all loaded at once**; they're fetched one at a time as needed.
- Best when:
  - You're dealing with **large documents or lots of files**.
  - You want to **stream** processing (e.g., chunking, embedding) without using lots of memory.

WebBaseLoader

28 March 2025 00:34

WebBaseLoader is a document loader in LangChain used to load and extract text content from web pages (URLs).

It uses BeautifulSoup under the hood to parse HTML and extract visible text.

When to Use:

- For blogs, news articles, or public websites where the content is primarily text-based and static.

Limitations:

- Doesn't handle JavaScript-heavy pages well (use SeleniumURLLoader for that).
- Loads only static content (what's in the HTML, not what loads after the page renders).

CSVLoader

28 March 2025 01:48

CSVLoader is a document loader used to load CSV files into LangChain Document objects — one per row, by default.

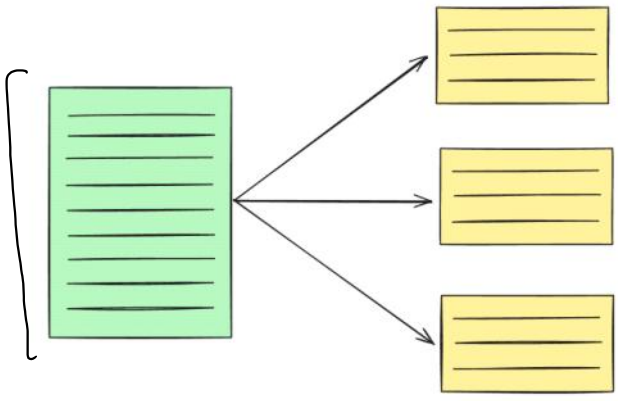
Other Document Loaders

28 March 2025 01:58

Text Splitting

01 April 2025 18:10

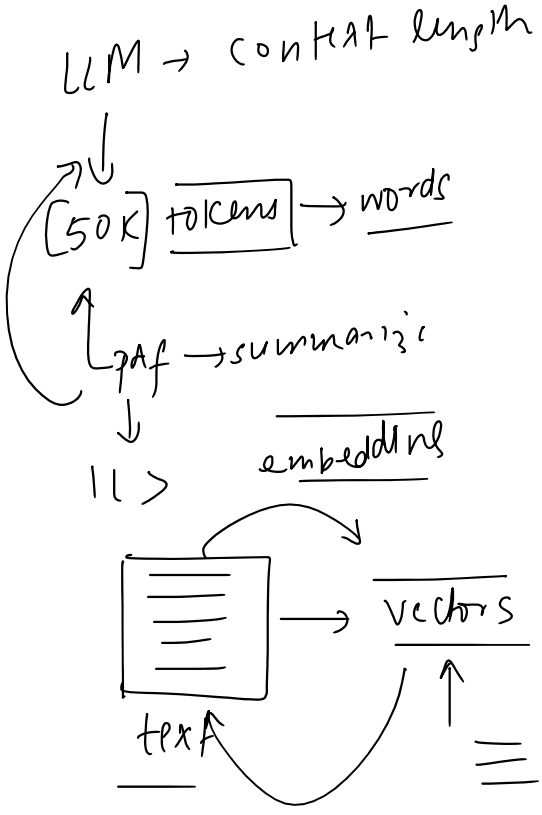
Text Splitting is the process of breaking large chunks of text (like articles, PDFs, HTML pages, or books) into smaller, manageable pieces (chunks) that an LLM can handle effectively.

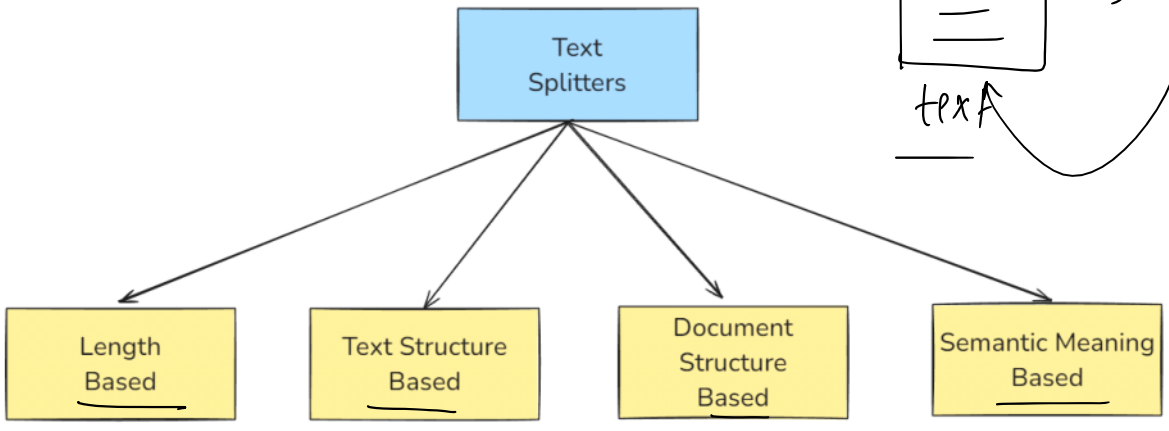


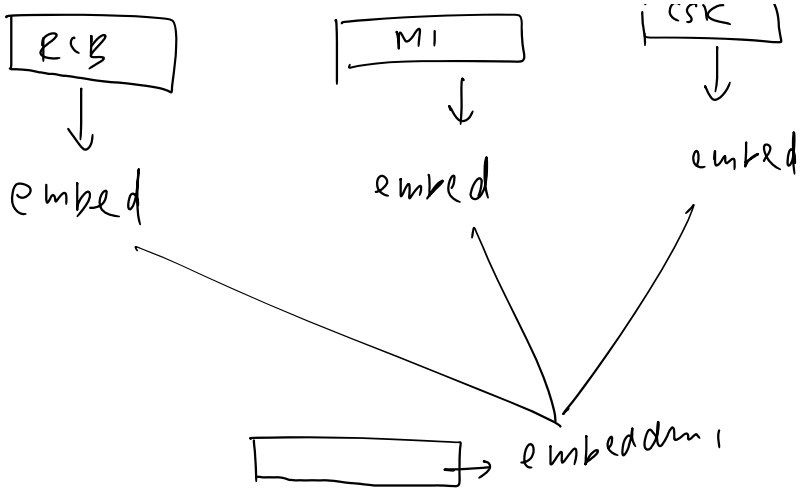
- Overcoming model limitations: Many embedding models and language models have maximum input size constraints. Splitting allows us to process documents that would otherwise exceed these limits.
- Downstream tasks - Text Splitting improves nearly every LLM powered task

Task	Why Splitting Helps
Embedding	Short chunks yield more accurate vectors
Semantic Search	Search results point to focused info, not noise
Summarization	Prevents hallucination and topic drift

- Optimizing computational resources: Working with smaller chunks of text can be more memory-efficient and allow for better parallelization of processing tasks.





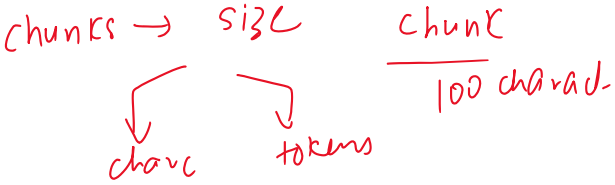

RCB
MI
CSK
embed
embed
embed
embeddings

1. Length Based Text Splitting

01 April 2025 18:10

Space exploration has led to incredible scientific discoveries. From landing on the Moon to exploring Mars, humanity continues to push the boundaries of what's possible beyond our planet.

These missions have not only expanded our knowledge of the universe but have also contributed to advancements in technology here on Earth. Satellite communications, GPS, and even certain medical imaging techniques trace their roots back to innovations driven by space programs.



Space exploration has led to incredible scientific discoveries. From landing on the Moon to explorin → c1

g Mars, humanity continues to push the boundaries of what's possible beyond our planet. These missi c2

ons have not only expanded our knowledge of the universe but have also contributed to advancements in c3

n technology here on Earth. Satellite communications, GPS, and even certain medical imaging techniqu c4

es trace their roots back to innovations driven by space programs. c5

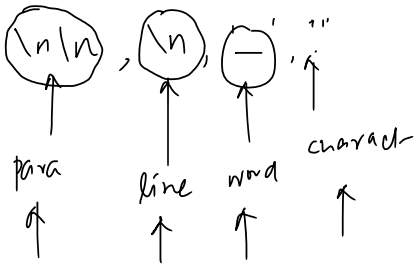
2. Text-Structured Based

01 April 2025 18:10

My name is Nitish
I am 35 years old

I live in Gurgaon
How are you

structure



chunks

Allowed

[chunk_size = 10]

chunksize = 10

[My name is Nit

My name is Nitish (17)
I am 35 years old (17)
I live in Gurgaon (17)
How are you (11)

My name is Nitish (17)/X
I am 35 years old (17)
(34)

I live in Gurgaon (17)
How are you (11)
(28) X

My name is Nitish (17)

My (2) name (4) is (2) Nitish (6)

My name (7) is (2) Nitish (6)

My name is (10) Nitish (6)

[My name is] [Nitish]

I am 35 years old (17)

I (1) am (2) 35 (2) years (5) old (3)

[I am 35] (7) [years old]

I live in Gurgaon (17)

I (1) live (4) in (2) gurgaon (7)

[I live in] [gurgaon]

How are you (11)

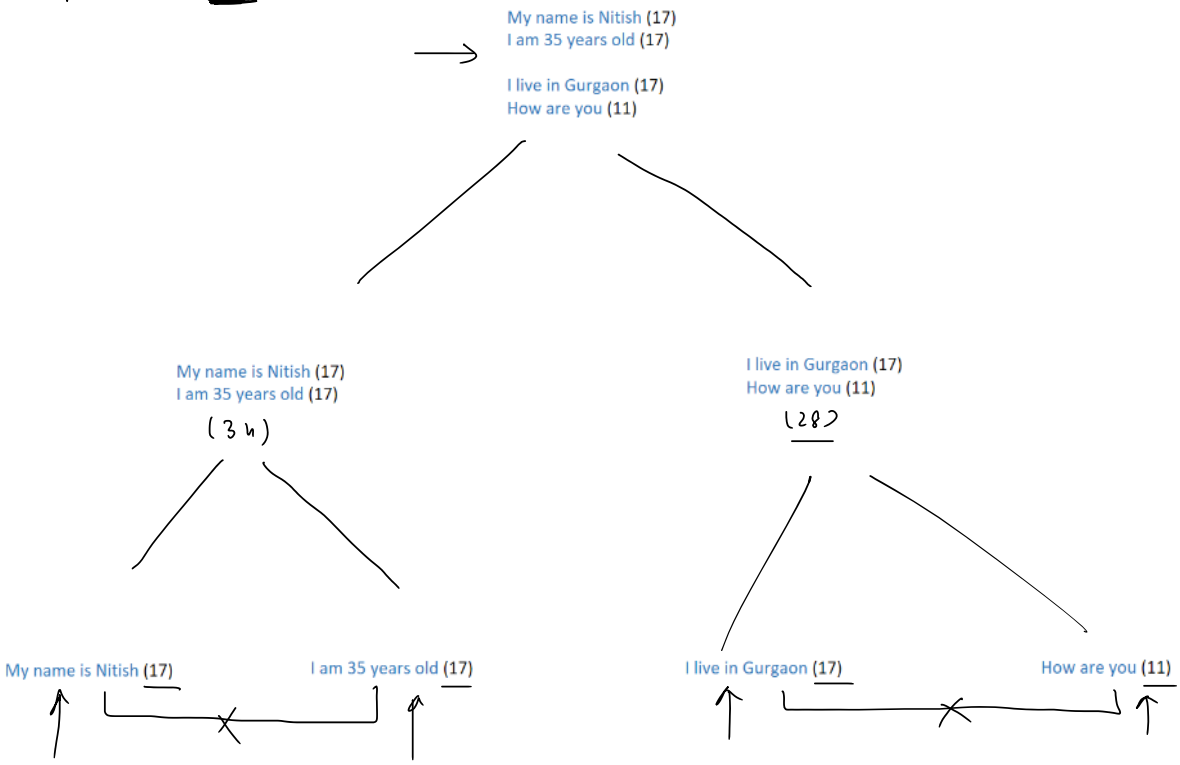
How (3) are (3) you (3)

[How are] [you

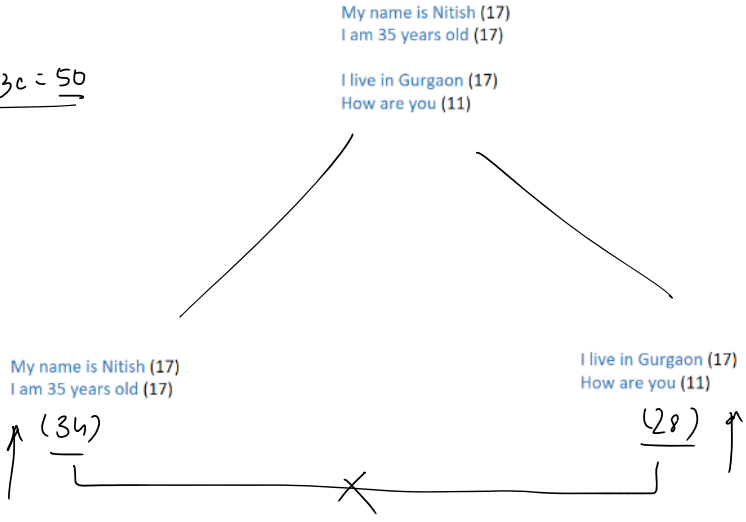
Chunk size = 25

My name is Nitish (17)
I am 35 years old (17)

Chunk size = 25



Chunk size = 50



3. Document-Structured Based

01 April 2025 18:11

markdown → LLM

```
# 📘 Project Name: Smart Student Tracker

A simple Python-based project to manage and track student data,

---

## 🔍 Features

- Add new students with relevant info
- View student details
- Check if a student is passing
- Easily extendable class-based design

---

## 🛠 Tech Stack

- Python 3.10+
- No external dependencies
```

code →

```python
class Student:
    def __init__(self, name, age, grade):
        self.name = name
        self.age = age
        self.grade = grade  # Grade is a float (like 8.5 or 9.2)

    def get_details(self):
        return f"Name: {self.name}, Age: {self.age}, Grade: {self.grade}"

    def is_passing(self):
        return self.grade >= 6.0


# Example usage
student1 = Student("Aarav", 20, 8.2)
print(student1.get_details())

if student1.is_passing():
    print("The student is passing.")
else:
    print("The student is not passing.")
```

```
# First, try to split along Markdown headings (starting with level 2)
"\n#{1,6} ",
# Note the alternative syntax for headings (below) is not handled here
# Heading level 2
# ---------------
# End of code block
"```\n",
# Horizontal lines
"\n\\*\\*\\*+\n",
"\n---+\n",
"\n___+\n",
# Note that this splitter doesn't handle horizontal lines defined
# by *three or more* of ***, ---, or ___, but this is not handled
"\n\n",
"\n",
" ",
"",
```

```
# First, try to split along class definitions
"\nclass ",
"\ndef ",
"\n\tdef ",
# Now split by the normal type of lines
"\n\n",
"\n",
" ",
"",
```

4. Semantic Meaning Based

01 April 2025 18:11

2 chunks

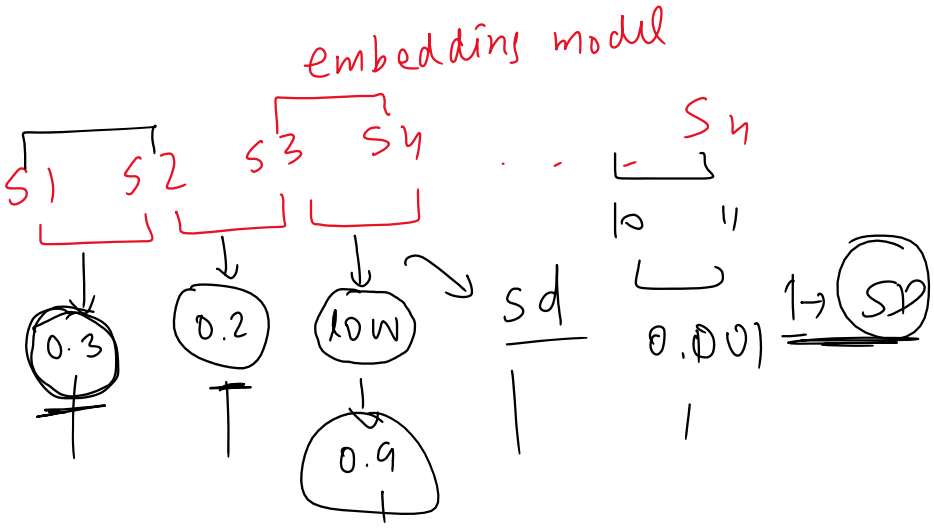
(len) (structure)

Farmers were working hard in the fields, preparing the soil and planting seeds for the next season. The sun was bright, and the air smelled of earth and fresh grass.

The Indian Premier League (IPL) is the biggest cricket league in the world. People all over the world watch the matches and cheer for their favourite teams.

→ 2 para ✗

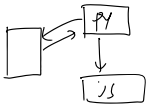
Terrorism is a big danger to peace and safety. It causes harm to people and creates fear in cities and villages. When such attacks happen, they leave behind pain and sadness. To fight terrorism, we need strong laws, alert security forces, and support from people who care about peace and safety.



Why Vector Stores?

05 April 2025 17:40

Movie id	Movie name	Director	Actor	Genre	Release Date	Outcome
M001	3 Idiots	Raju Hirani	Aamir Khan	Drama, Romance	2009	Super Hit
M002	Chennai Express	Rohit Shetty	Shah Rukh Khan	Romance, Comedy	2014	Super Hit
M003	Inception	C Nolan	L Di Caprio	Thriller, Sci-Fi	2009	Blockbuster
M004	Stree	Amar Kaushik	Rajkumar Rao	Horror, Comedy	2019	Hit



Keyword match

m1 m2

Movie id	Plot
M001	In the present day, Farhan receives a call from Chatur, saying that Ranchi is coming. Farhan is so excited that he fakes a heart attack to get off a flight and picks up Raju from his home (who forgets to wear his pants). Farhan and Raju meet Chatur at the water tower of their college ICE (Imperial College of Engineering), where Chatur informs them that he has found Rancho. Chatur taunts Farhan and Raju that now he has a mansion worth $3.5 million in the US, with a heated swimming pool, a maple wood floor living room and a Lamborghini for a car. Chatur reveals that Rancho is in Shimla...
M002	Rahul Mithaiwala (Shahrukh Khan) is a forty-year old bachelor who lives in Mumbai. His parents died in a car accident when he was eight years old and was brought up by grandparents. His grandfather has a sweet-selling chain store - Y.Y. Mithaiwala. Before his birth centenary celebration, two of Rahul's friends suggest a vacation in Goa which he accepts. On the eve of the celebration, his grandfather dies whilst watching a cricket match. His grandmother tells him that his grandfather desired to have his ashes divided into two parts - to be immersed in the Ganges River and Rameswaram respectively. She requests Rahul to go to Rameswaram and immerse them. Rahul reluctantly accepts her request but was also eager to attend the Goa trip...
M003	Dominick "Dom" Cobb (Leonardo DiCaprio) and business partner Arthur (Joseph Gordon-Levitt) are "extractors", people who perform corporate espionage using an experimental military technology to infiltrate the subconscious of their targets and extract information while experiencing shared dreaming. Their latest target is Japanese businessman Saito (Ken Watanabe). The extraction from Saito fails when sabotaged by a memory of Cobb's deceased wife Mal (Marion Cotillard). After Cobb's and Arthur's associate sells them out, Saito reveals that he was actually auditioning the team to perform the difficult act of "inception": planting an idea in a person's subconscious...
M004	In the peculiar town of Chanderi, India, the residents believe in the myth of an angry woman ghost, referred to as "Stree" (Hindi for woman) (Flora Saini), who stalks men during Durga Puja festival. This is explained by the sudden disappearance of these men, leaving their clothes behind. She is said to stalk the men of the town, whispering their names and causing disappearances if they look back at her. The whole town protects itself from Stree during the 4 nights of Durga Puja by writing "OO Stree, Kal Aana" on their walls. Additionally, men are advised to avoid going out alone after 10 PM during the festival and to move in groups for safety. This practice reflects a societal parallel to the precautions typically advised to women for their own protection...

512 dim

0.32	0.63	0.13	0.45	0.0	0.1	•	•	•	0.78	0.99

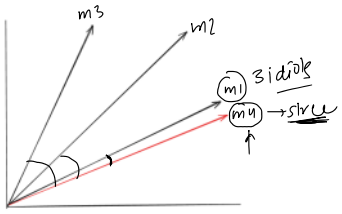
compare similarity score →

0.44	0.61	0.0	0.91	0.0	0.99	•	•	•	0.91	0.99

0.21	0.0	0.0	0.01	0.0	0.77	•	•	•	0.0	0.99

0.33	0.03	0.10	0.22	0.99	0.77	•	•	•	0.22	0.88

512 dim

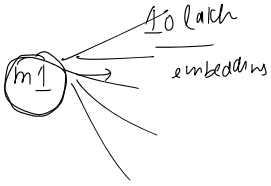


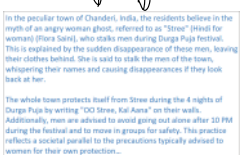
C1 → gen embedding vectors

C2 → storage

C3 → semantic search intelligence

↳ Vector store





vectors

0.33	0.03	0.10	0.22	0.99	0.77	•	•	•	0.22	0.88

What are Vector Stores

05 April 2025 17:38

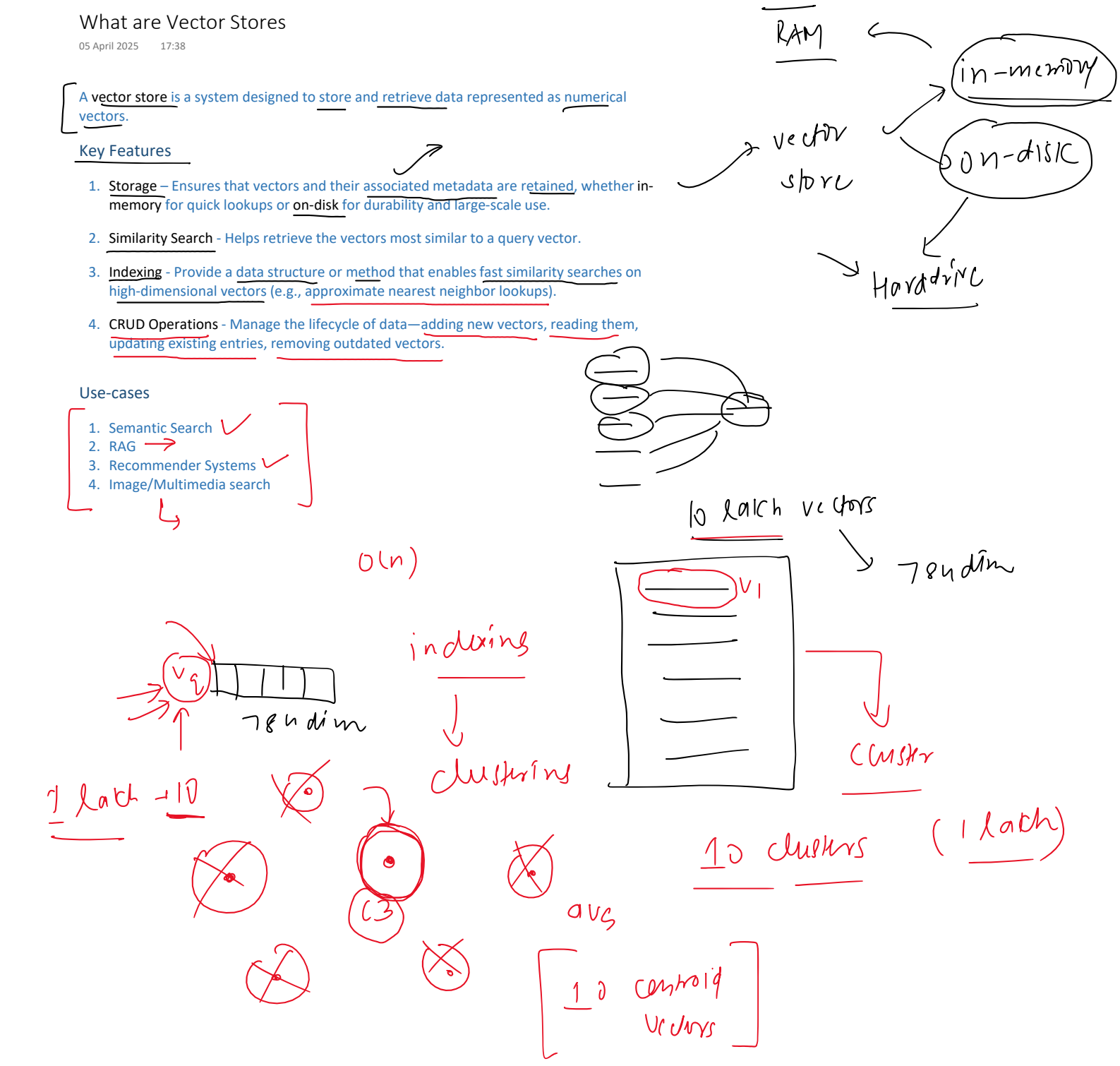
A vector store is a system designed to store and retrieve data represented as numerical vectors.

Key Features

1. Storage – Ensures that vectors and their associated metadata are retained, whether in-memory for quick lookups or on-disk for durability and large-scale use.
2. Similarity Search - Helps retrieve the vectors most similar to a query vector.
3. Indexing - Provide a data structure or method that enables fast similarity searches on high-dimensional vectors (e.g., approximate nearest neighbor lookups).
4. CRUD Operations - Manage the lifecycle of data—adding new vectors, reading them, updating existing entries, removing outdated vectors.

Use-cases

1. Semantic Search
2. RAG
3. Recommender Systems
4. Image/Multimedia search

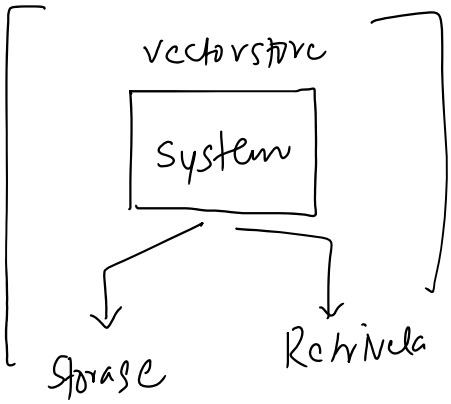


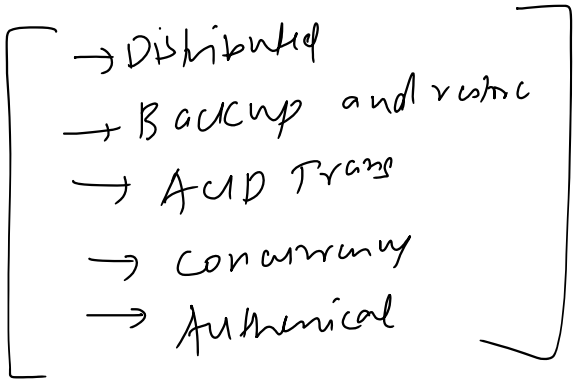
Vector Store Vs Vector Database

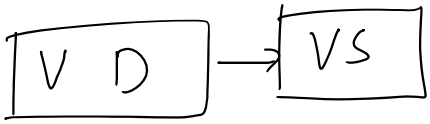
05 April 2025 17:40

- Vector Store
  - Typically refers to a lightweight library or service that focuses on storing vectors (embeddings) and performing similarity search.
  - May not include many traditional database features like transactions, rich query languages, or role-based access control.
  - Ideal for prototyping, smaller-scale applications
  - Examples: FAISS (where you store vectors and can query them by similarity, but you handle persistence and scaling separately).
- Vector Database
  - A full-fledged database system designed to store and query vectors.
  - Offers additional "database-like" features:
    - Distributed architecture for horizontal scaling
    - Durability and persistence (replication, backup/restore)
    - Metadata handling (schemas, filters)
    - Potential for ACID or near-ACID guarantees
    - Authentication/authorization and more advanced security
  - Geared for production environments with significant scaling, large datasets
  - Examples: Milvus, Qdrant, Weaviate. Pinecone

A vector database is effectively a vector store with extra database features (e.g., clustering, scaling, security, metadata filtering, and durability)



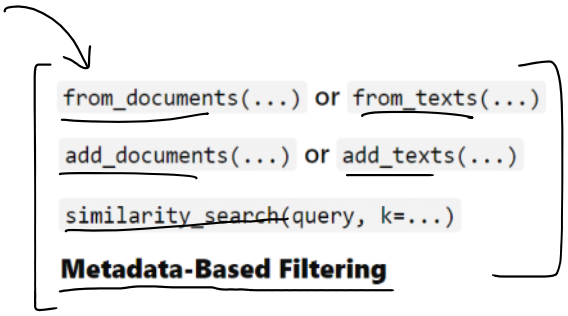


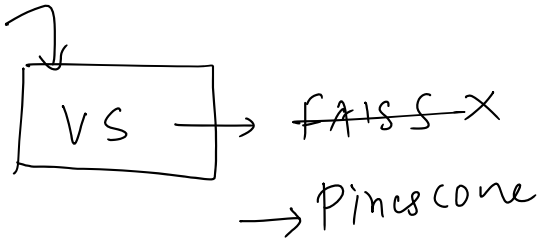


Vector Stores in LangChain

05 April 2025 17:41

- Supported Stores: LangChain integrates with multiple vector stores (FAISS, Pinecone, Chroma, Qdrant, Weaviate, etc.), giving you flexibility in scale, features, and deployment.
- Common Interface: A uniform Vector Store API lets you swap out one backend (e.g., FAISS) for another (e.g., Pinecone) with minimal code changes.
- Metadata Handling: Most vector stores in LangChain allow you to attach metadata (e.g., timestamps, authors) to each document, enabling filter-based retrieval.

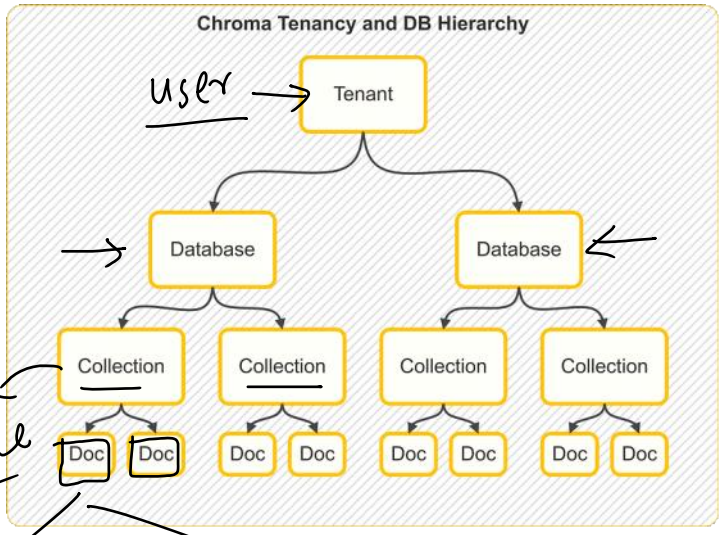




Chroma Vector Store

05 April 2025 17:41

Chroma is a lightweight, open-source vector database that is especially friendly for local development and small- to medium-scale production needs.

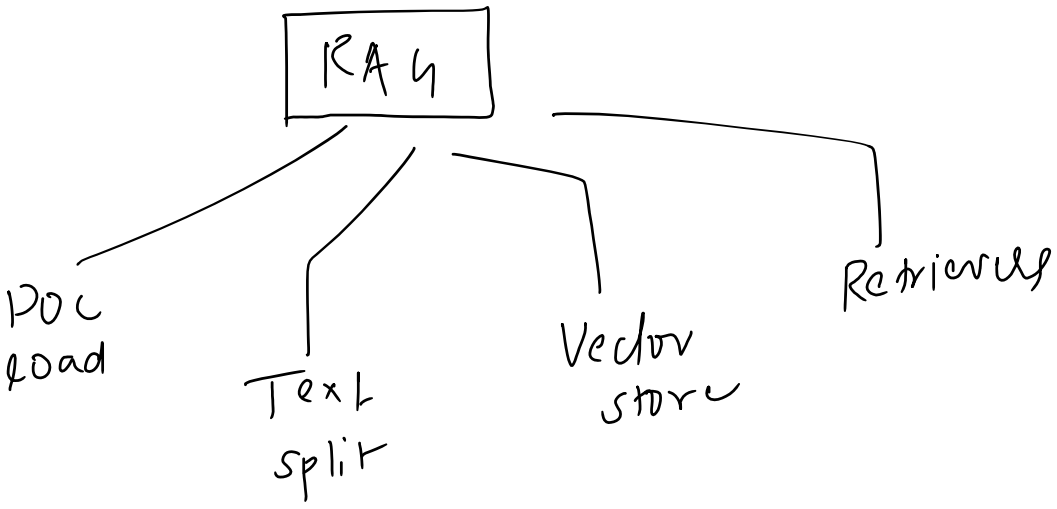


Code

05 April 2025 17:41

Recap

12 April 2025 13:51



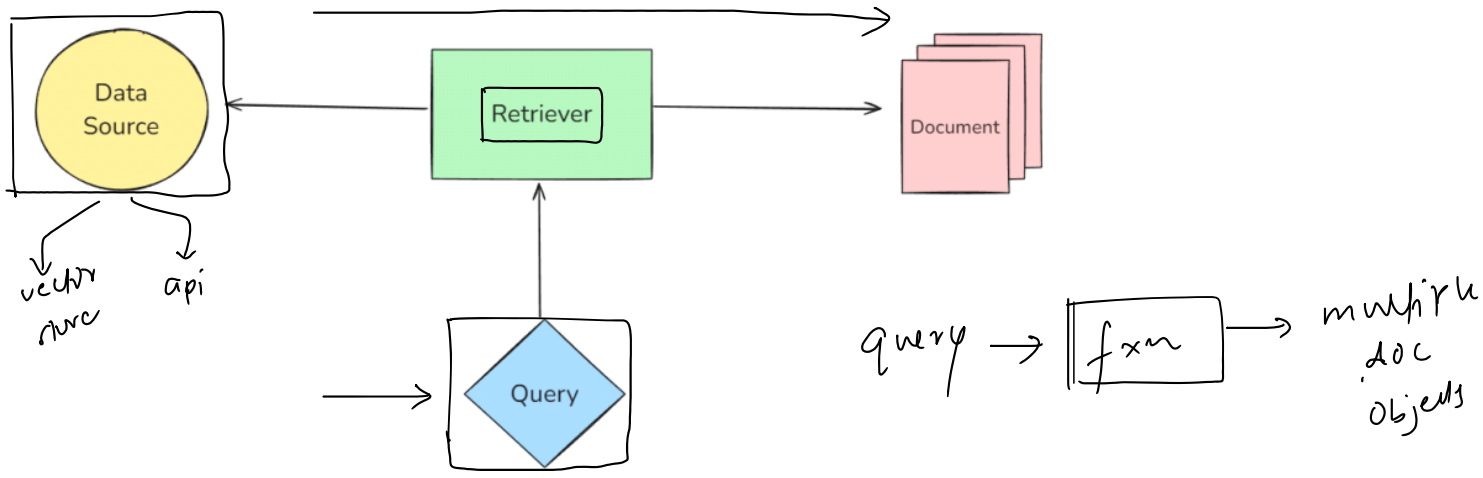
What are Retrievers

10 April 2025 07:56

A retriever is a component in LangChain that fetches relevant documents from a data source in response to a user's query.

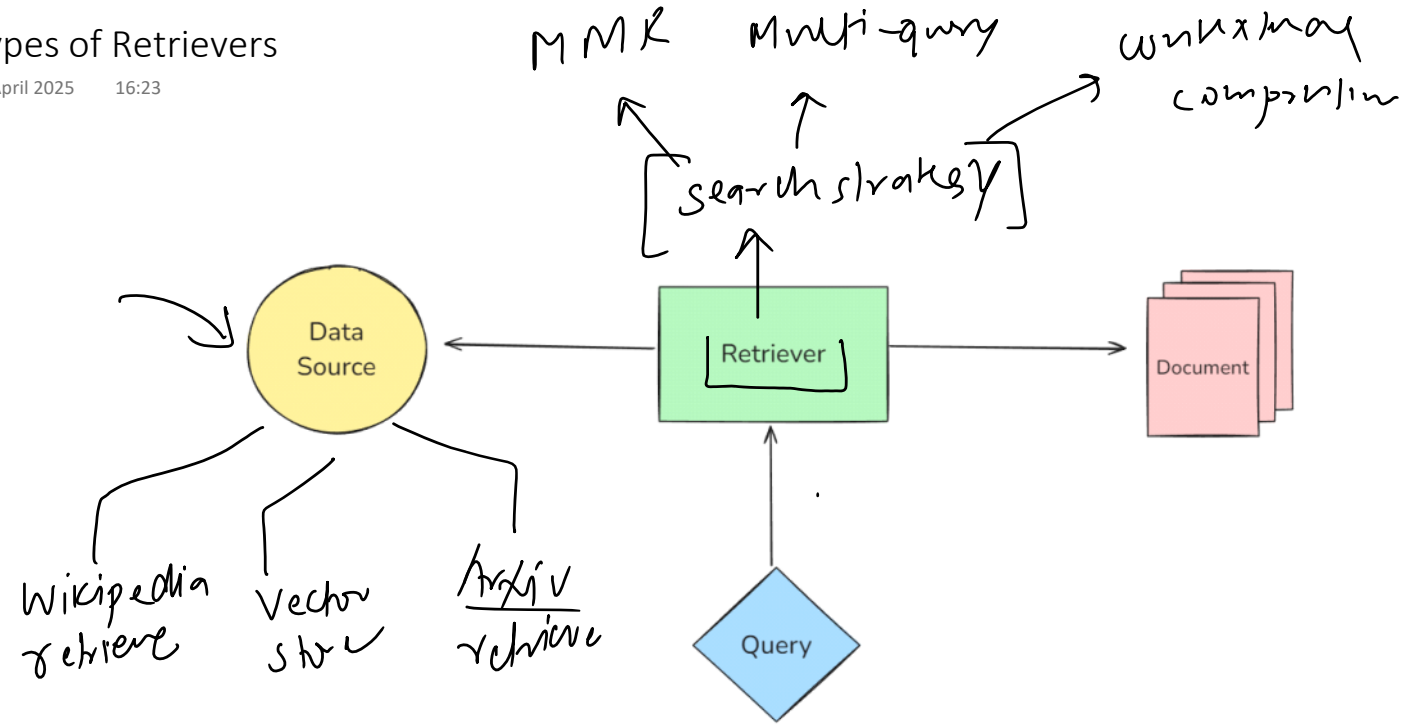
→ There are multiple types of retrievers

→ All retrievers in LangChain are runnables



Types of Retrievers

10 April 2025 16:23



Wikipedia Retriever → doc loader

10 April 2025 16:23

A Wikipedia Retriever is a retriever that queries the Wikipedia API to fetch relevant content for a given query.

How It Works

1. You give it a query (e.g., "Albert Einstein")
2. It sends the query to Wikipedia's API
3. It retrieves the **most relevant articles**
4. It returns them as LangChain `Document` objects

Vector Store Retriever

10 April 2025 16:24

A Vector Store Retriever in LangChain is the most common type of retriever that lets you search and fetch documents from a vector store based on semantic similarity using vector embeddings.

⚙️ How It Works

1. You store your documents in a **vector store** (like FAISS, Chroma, Weaviate)
2. Each document is converted into a **dense vector** using an **embedding model**
3. When the user enters a query:
    - It's also turned into a vector
    - The retriever compares the query vector with the stored vectors
    - It retrieves the top-k most similar ones

Maximal Marginal Relevance (MMR)

10 April 2025 16:24

> "How can we pick results that are not only relevant to the query but also different from each other?"

MMR is an information retrieval algorithm designed to reduce redundancy in the retrieved results while maintaining high relevance to the query.

🤔 Why MMR Retriever?

In regular similarity search, you may get documents that are:

- All very similar to each other
- Repeating the same info
- Lacking diverse perspectives

MMR Retriever avoids that by:

- Picking the **most relevant document** first
- Then picking the next most relevant **and least similar** to already selected docs
- And so on...

This helps especially in RAG pipelines where:

- You want your context window to contain **diverse but still relevant information**
- Especially useful when documents are semantically overlapping

Doc ID	Content
D1	"Climate change is causing glaciers to melt rapidly in the Arctic region."
D2	"Glaciers in the Arctic are melting at an alarming rate due to rising temperatures."
D3	"Deforestation in the Amazon is accelerating global climate change."
D4	"Climate change is increasing the frequency of wildfires in California."
D5	"Rising sea levels due to climate change threaten coastal cities like Mumbai and New York."

```
Top 3 results:
1. D1: Arctic glaciers melting
2. D2: Arctic glaciers melting
3. D3: Deforestation in Amazon
```

-diverse perspective

```
Top 3 results:
1. D1: Arctic glaciers melting
2. D4: Wildfires in California
3. D5: Rising sea levels in coastal cities
```

Multi-Query Retriever

10 April 2025 16:26

> **Sometimes a single query might not capture all the ways information is phrased in your documents.**

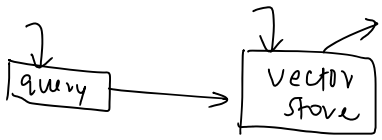
For example:

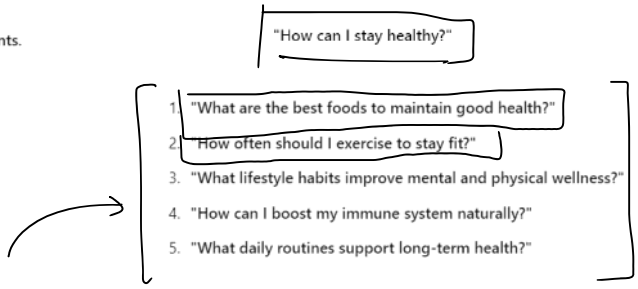
Query:

"How can I stay healthy?"

Could mean:

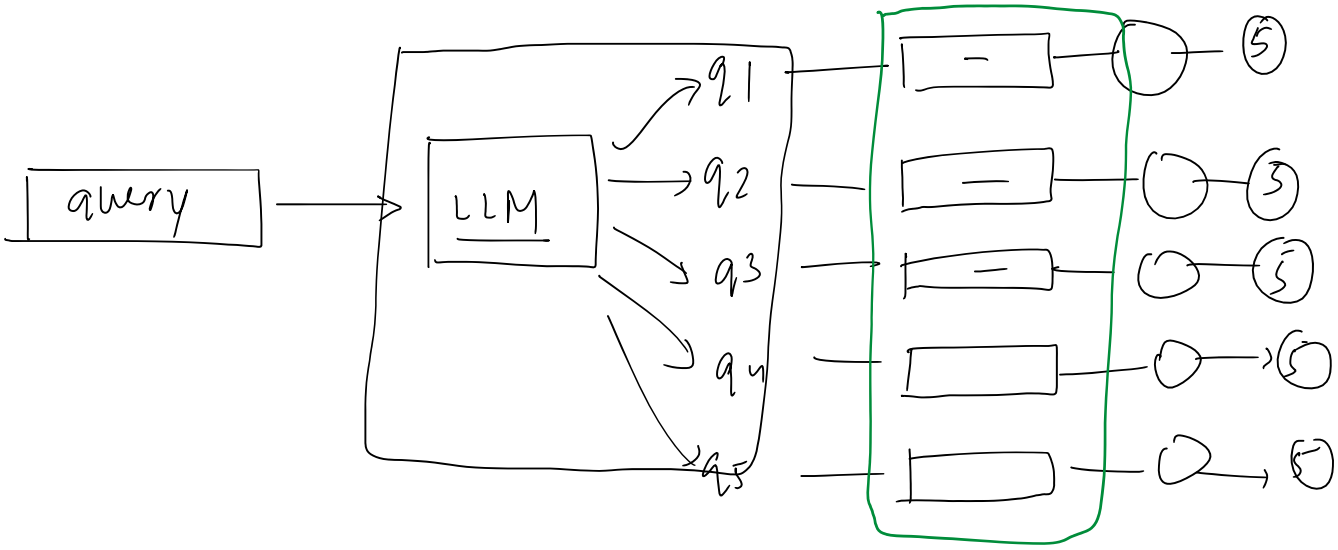
- What should I eat?
- How often should I exercise?
- How can I manage stress?





A simple similarity search might **miss documents** that talk about those things but don't use the word "healthy."

1. **Takes your original query**
2. **Uses an LLM (e.g., GPT-3.5)** to generate multiple semantically different versions of that query
3. **Performs retrieval for each sub-query**
4. **Combines and deduplicates the results**



Contextual Compression Retriever

10 April 2025 16:29

The Contextual Compression Retriever in LangChain is an advanced retriever that improves retrieval quality by compressing documents after retrieval — keeping only the relevant content based on the user's query.

❓ **Query:**

"What is photosynthesis?"

Retrieved Document (by a traditional retriever):

"The Grand Canyon is a famous natural site.
Photosynthesis is how plants convert light into energy.
Many tourists visit every year."

❌ **Problem:**

- The retriever returns the **entire paragraph**
- Only **one sentence** is actually relevant to the query
- The rest is **irrelevant noise** that wastes context window and may confuse the LLM

✅ **What Contextual Compression Retriever does:**

Returns only the relevant part, e.g.

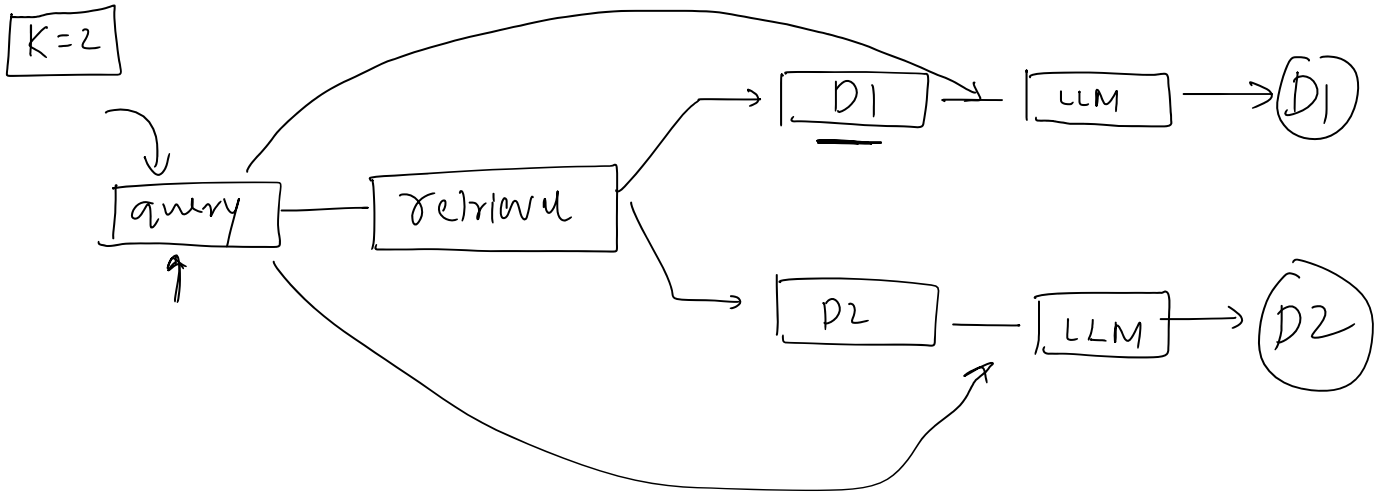
```
"Photosynthesis is how plants convert light into energy."
```

⚙️ **How It Works**

1. **Base Retriever** (e.g., FAISS, Chroma) retrieves N documents.
2. A **compressor** (usually an LLM) is applied to each document.
3. The compressor keeps **only the parts relevant to the query**.
4. Irrelevant content is discarded.

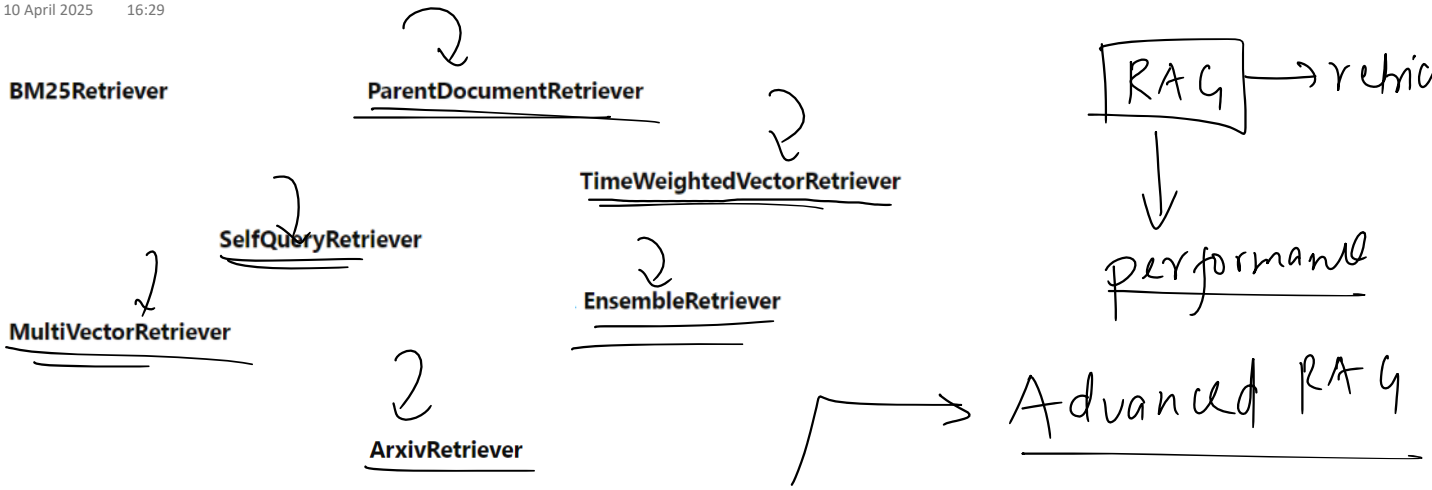
✅ **When to Use**

- Your documents are **long and contain mixed information**
- You want to **reduce context length** for LLMs
- You need to **improve answer accuracy** in RAG pipelines



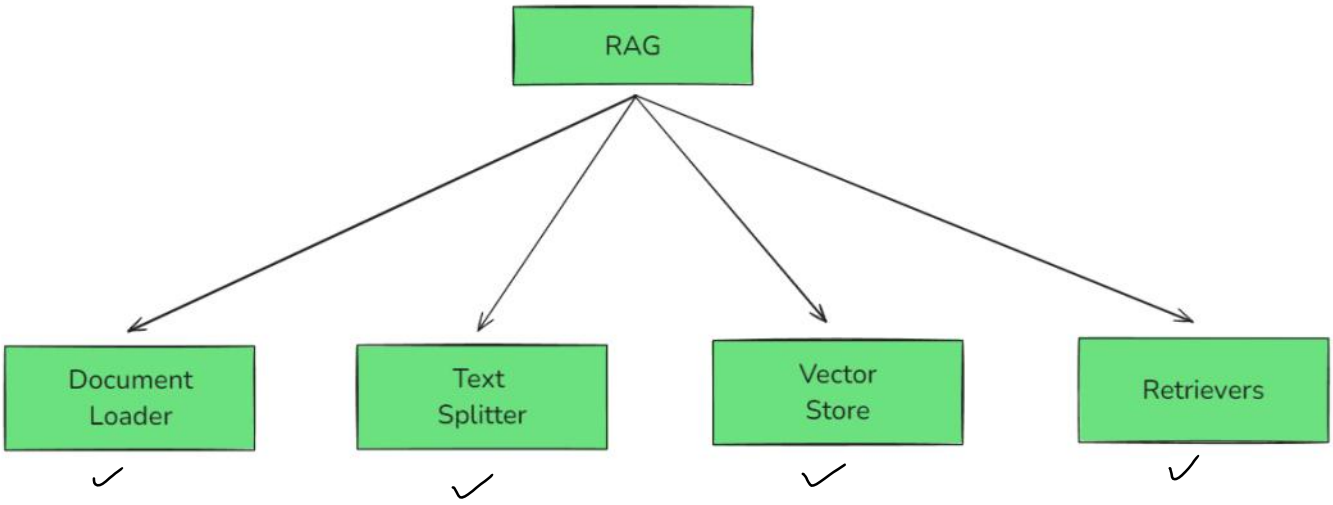
More Retrievers

10 April 2025 16:29



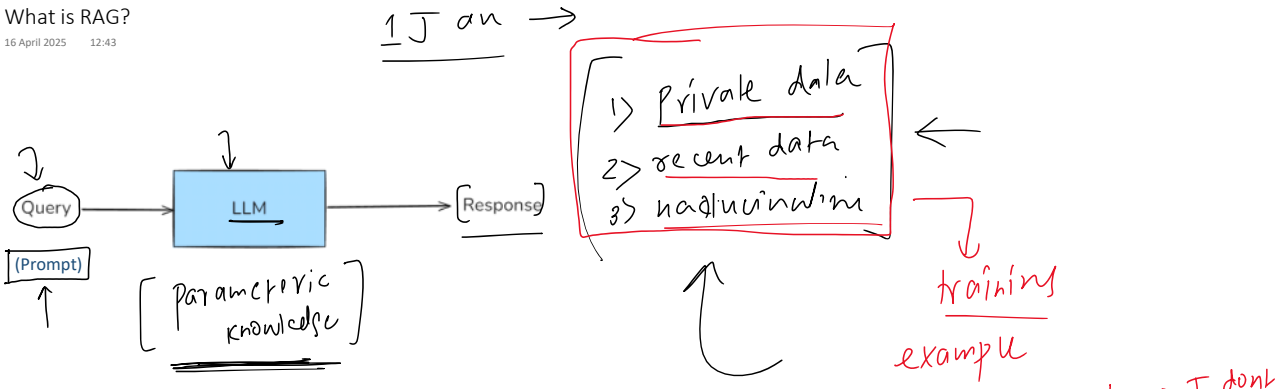
Recap

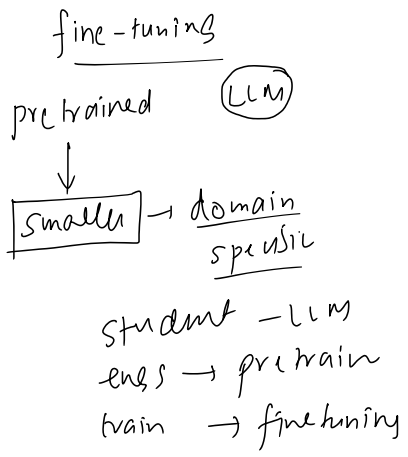
16 April 2025 12:45



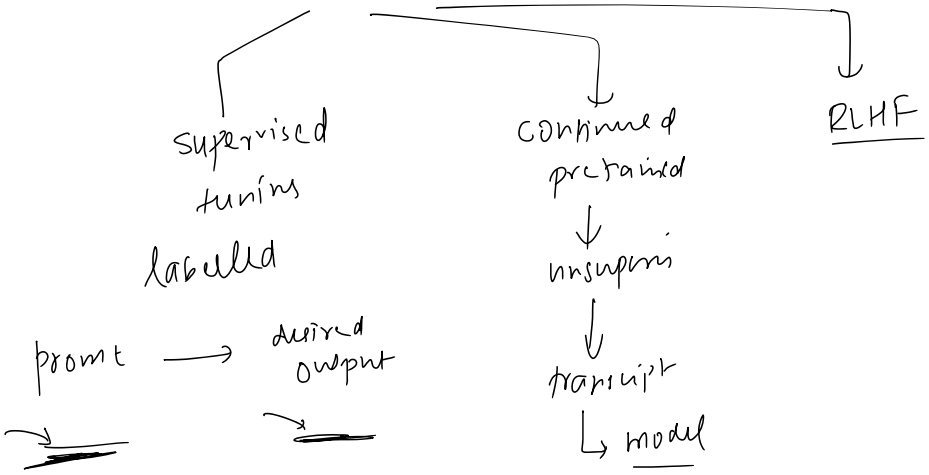
What is RAG?

16 April 2025 12:43





1. Collect data	A few hundred – few hundred-thousand carefully curated examples (prompts → desired outputs).
2. Choose a method	Full-parameter FT, LoRA/QLoRA, or parameter-efficient adapters.
3. Train for a few epochs	You keep the base weights frozen or partially frozen and update only a small subset (LoRA) **or** all weights (full FT).
4. Evaluate & safety-test	Measure exact-match, factuality, and hallucination rate against held-out data; red-team for safety.



1) LLM → train
2) technical expertise
3)

In context learning

In-Context Learning is a core capability of Large Language Models (LLMs) like GPT-3/4, Claude, and Llama, where the model learns to solve a task purely by seeing examples in the prompt—without updating its weights.

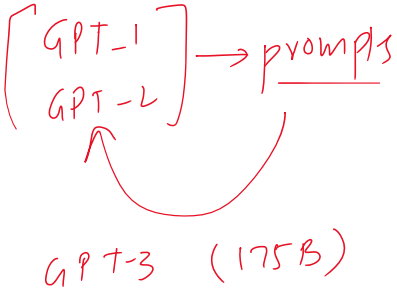
LLM → emergent property

```
Below are examples of texts labeled with their sentiment.
Use these examples to determine the sentiment of the final text.

Text: I love this phone. It's so smooth. → Positive
Text: This app crashes a lot. → Negative
Text: The camera is amazing! → Positive
```

LLM → emergent property

```
Use these examples to determine the sentiment of the final text.

Text: I love this phone. It's so smooth. → Positive
Text: This app crashes a lot. → Negative
Text: The camera is amazing! → Positive

Text: I hate the battery life. →
```

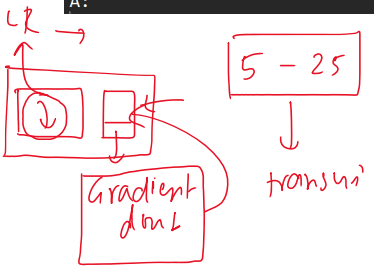
An **emergent property** is a behaviour or ability that suddenly appears in a system when it reaches a certain scale or complexity—even though it was not explicitly programmed or expected from the individual components.




```
Label the named entities in the sentences (Person, Location, Organization):

Sentence: Elon Musk founded SpaceX.
Entities: [Elon Musk → Person], [SpaceX → Organization]

Sentence: The Eiffel Tower is in Paris.
Entities: [Eiffel Tower → Location], [Paris → Location]

Sentence: Sundar Pichai leads Google.
Entities:
```

```
Solve the math problems step by step:

Q: John has 3 apples. He buys 4 more. How many apples does he have now?
A: John had 3 apples. He bought 4 more. 3 + 4 = 7. → 7

Q: Sarah has 10 pens. She gives away 3. How many does she have left?
A: Sarah had 10 pens. She gave away 3. 10 - 3 = 7. → 7

Q: A pizza is cut into 8 slices. Alex eats 3 slices. How many are left?
A:
```

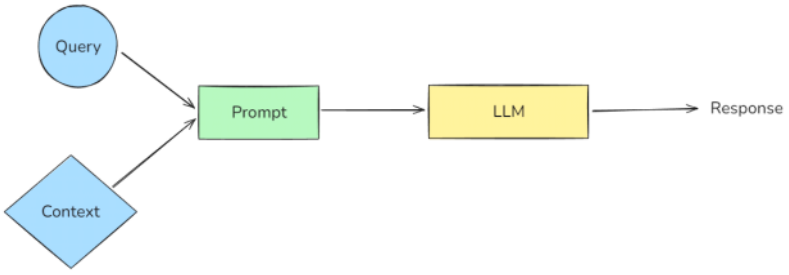
context



- Instead of just *example tasks*, retrieve **background information**, **facts**, **documents**, **product manuals**, etc.
- Inject that into the prompt to **augment the model's knowledge**

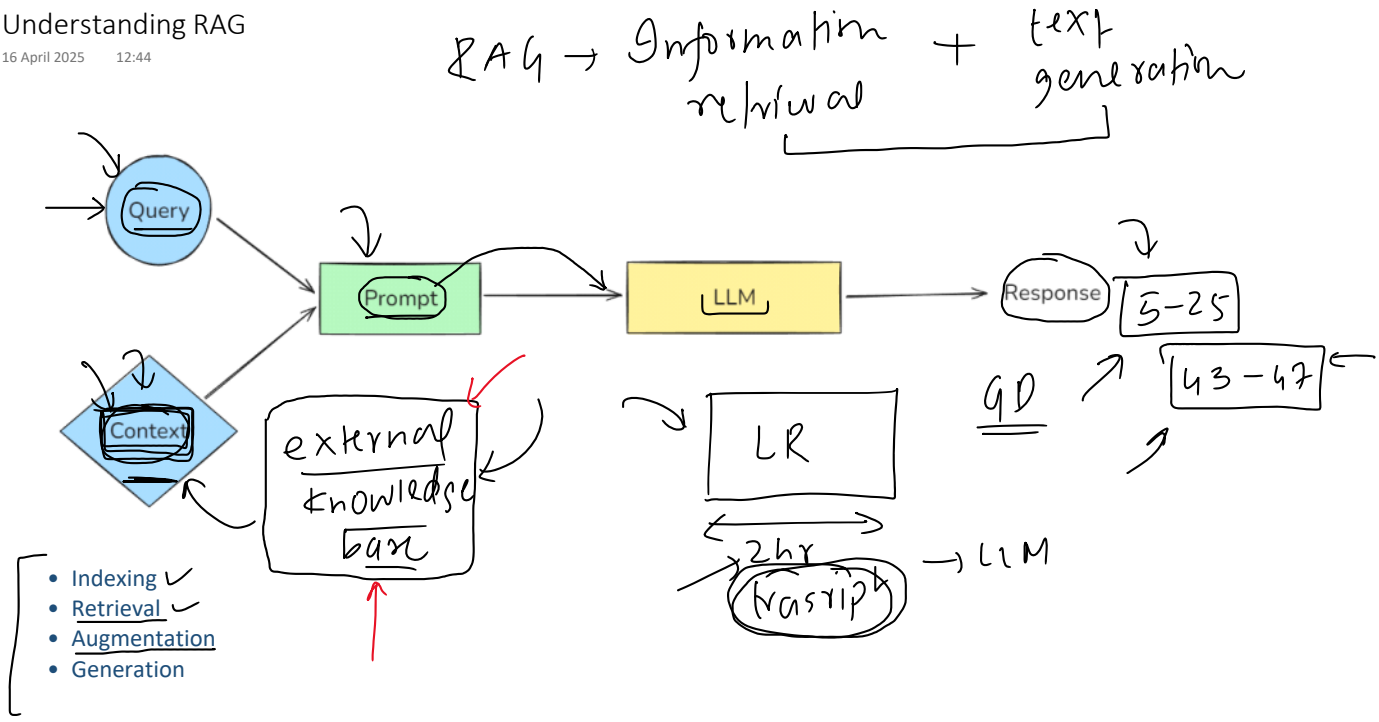
```
"""You are a helpful assistant.
Answer the question ONLY from the provided context.
If the context is insufficient, just say you don't know.

{context}
Question: {question}"""
```

RAG is a way to make a language model (like ChatGPT) smarter by giving it extra information at the time you ask your question.



Understanding RAG

16 April 2025 12:44



Indexing - **Indexing** is the process of **preparing your knowledge base so that it can be efficiently searched** at query time. This steps consists of 4 sub-steps.

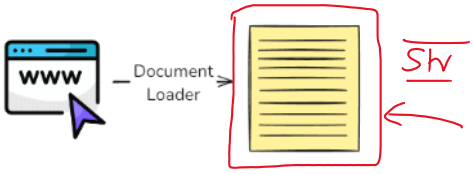
1. Document Ingestion - You load your source knowledge into memory.

   Examples:

   - PDF reports, Word documents
   - YouTube transcripts, blog pages
   - GitHub repos, internal wikis
   - SQL records, scraped webpages

   Tools:

   - LangChain loaders (`PyPDFLoader`, `YoutubeLoader`, `WebBaseLoader`, `GitLoader`, etc.)



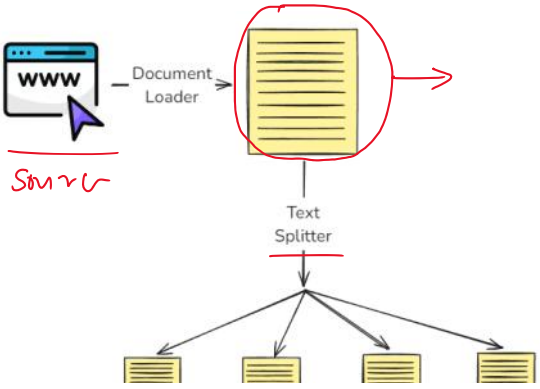
2. Text Chunking - Break large documents into small, semantically meaningful **chunks**

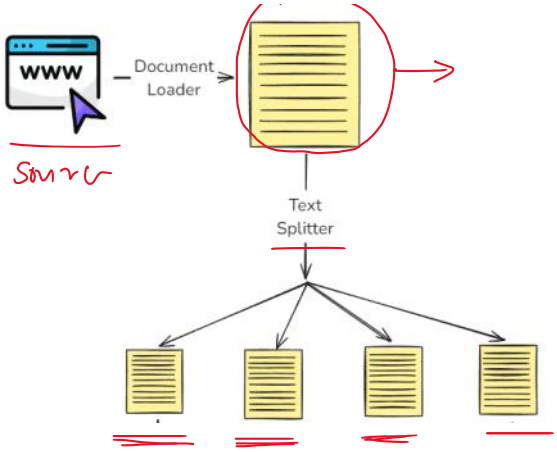
   Why chunk?

   - LLMs have context limits (e.g., 4K–32K tokens)
   - Smaller chunks are more focused → better semantic search

   Tools:

   - `RecursiveCharacterTextSplitter`, `MarkdownHeaderTextSplitter`, `SemanticChunker`





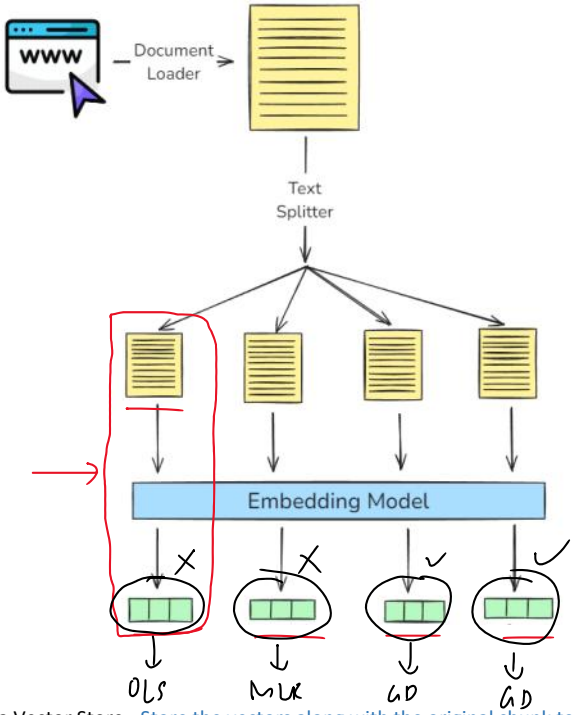
3. Embedding Generation - Convert each chunk into a **dense vector** (embedding) that captures its meaning.

   🔍 Why embeddings?

   - Similar ideas land close together in vector space
   - Allows fast, fuzzy semantic search

   🔧 Tools:

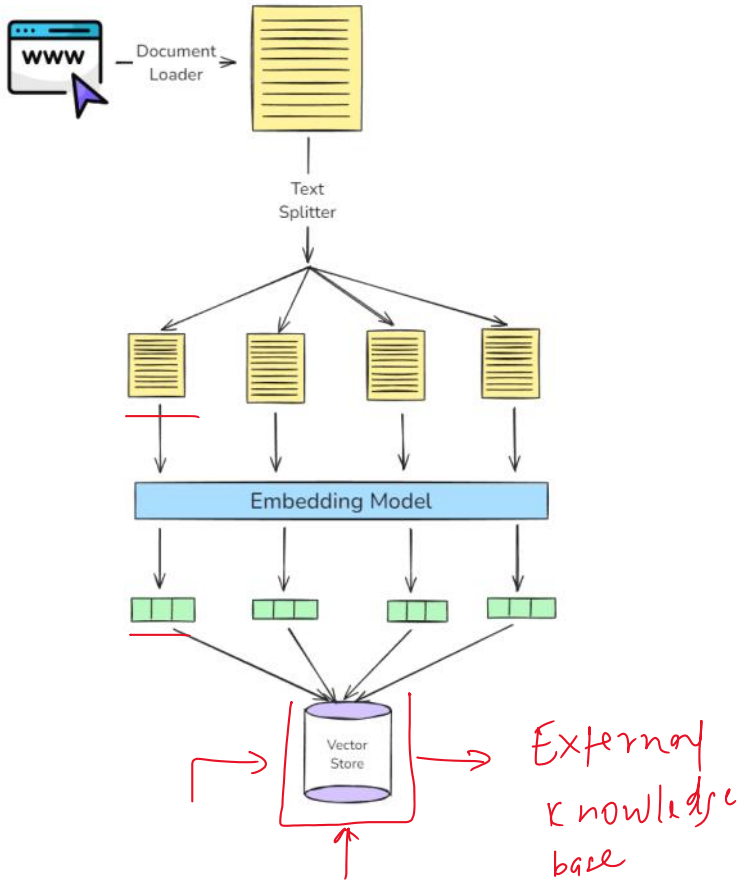
   - `OpenAIEmbeddings`, `SentenceTransformerEmbeddings`, `InstructorEmbeddings`, etc.



4. Storage in a Vector Store - Store the vectors along with the original chunk text + metadata in a **vector database**.

   🔧 Vector DB options:

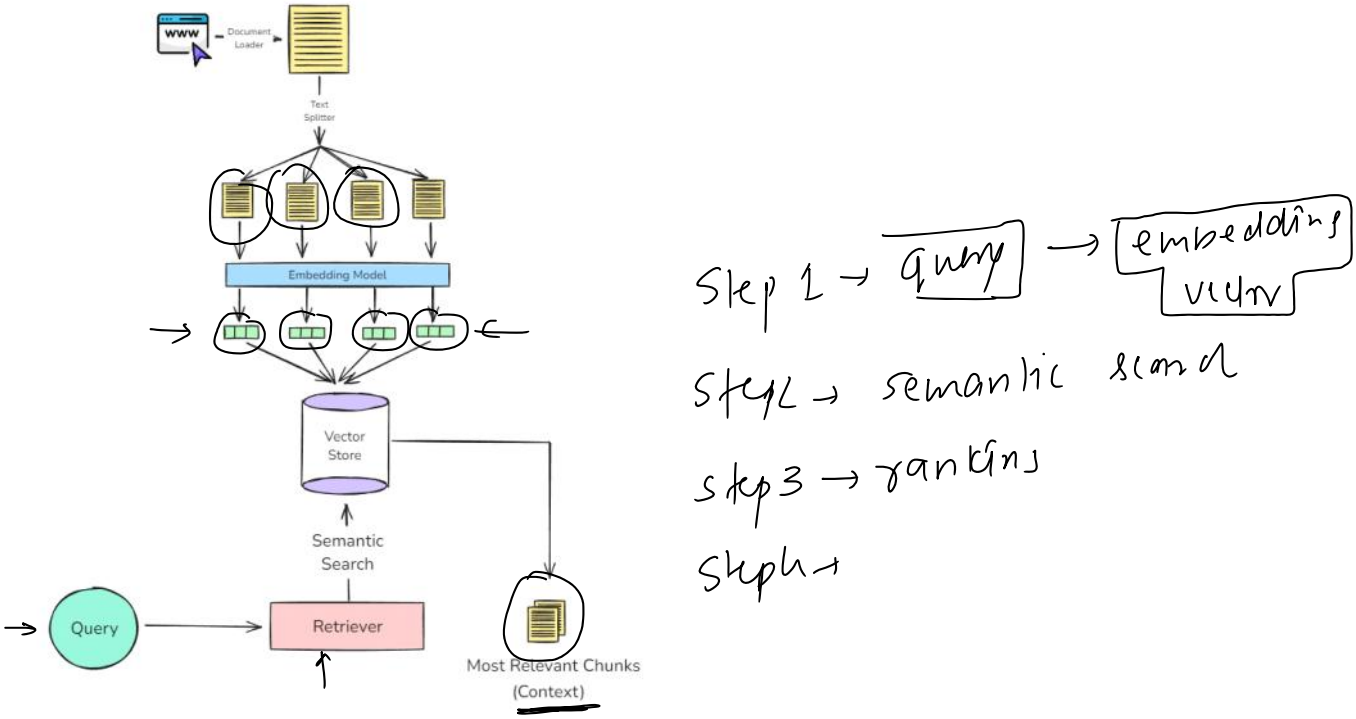
   - Local: `FAISS`, `Chroma`
   - Cloud: `Pinecone`, `Weaviate`, `Milvus`, `Qdrant`



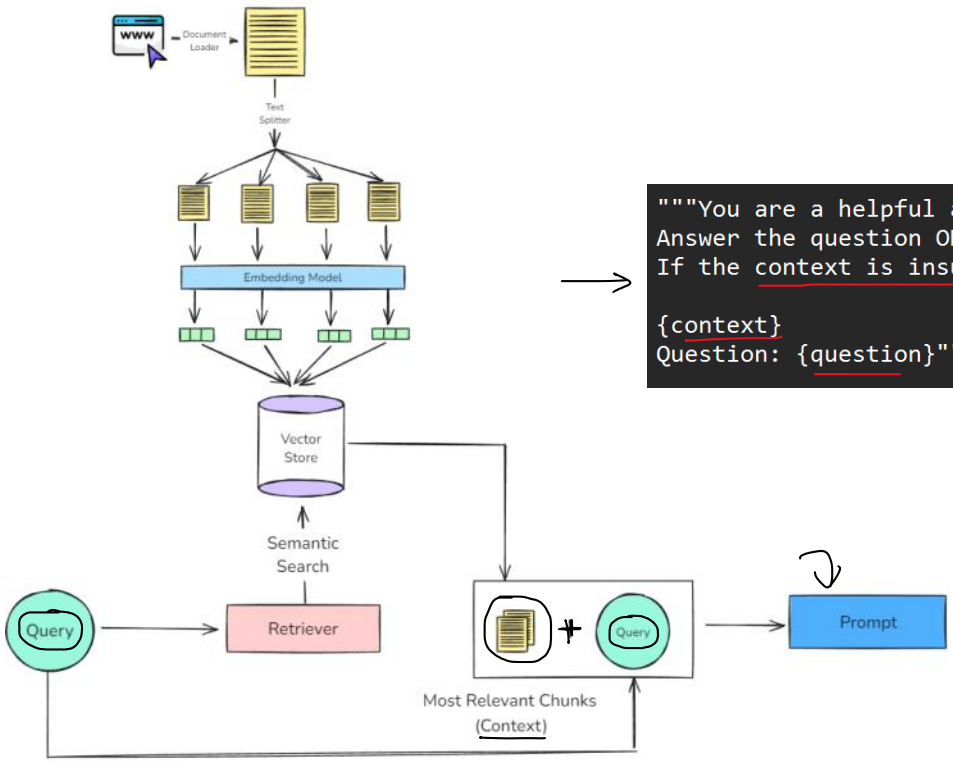
Retrieval - **Retrieval** is the *real-time* process of **finding the most relevant pieces of information** from a pre-built index (created during indexing) based on the user's question.

It's like asking:

> "From all the knowledge I have, which 3–5 chunks are most helpful to answer this query?"

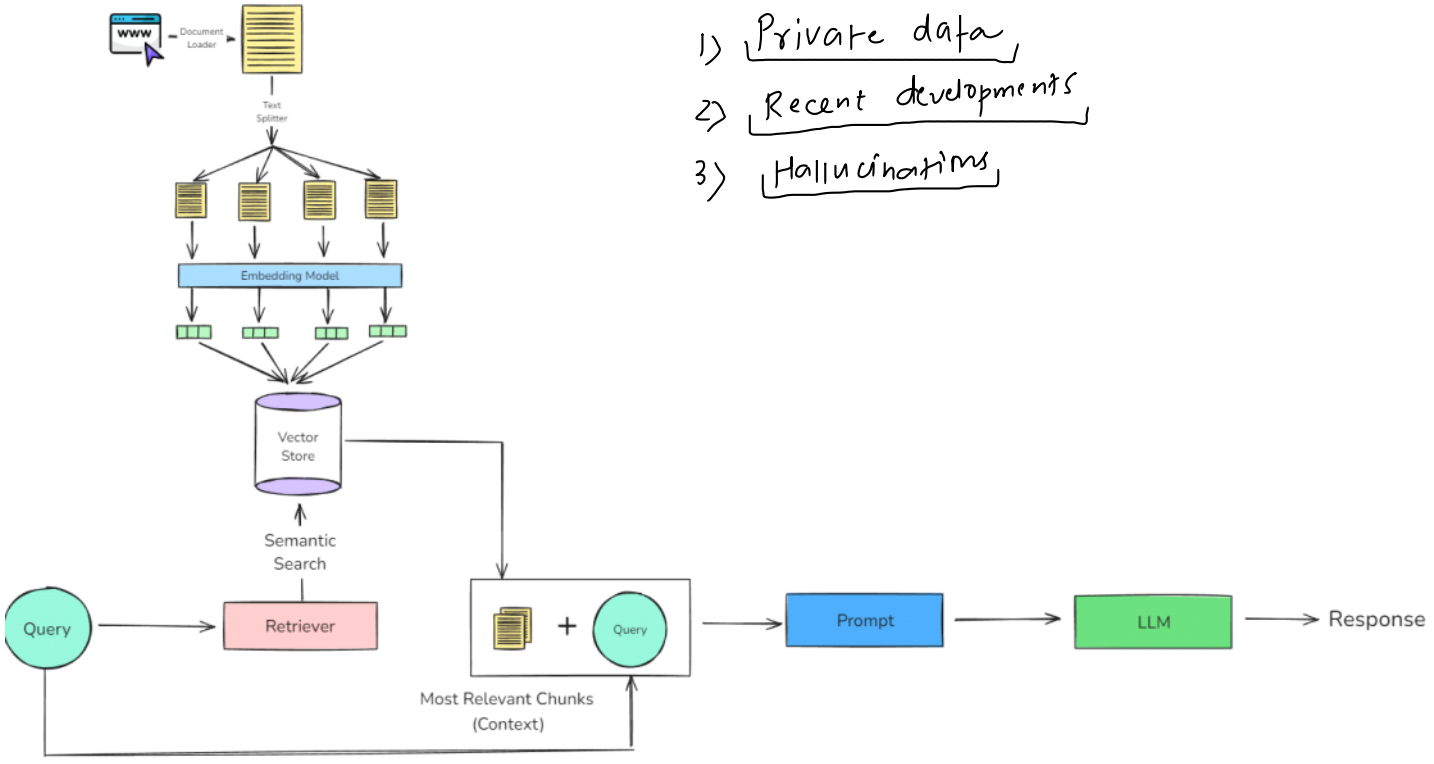


Augmentation - **Augmentation** refers to the step where the **retrieved documents** (chunks of relevant context) are **combined with the user's query** to form a new, enriched prompt for the LLM.




```
"""You are a helpful assistant.
Answer the question ONLY from the provided context.
If the context is insufficient, just say you don't know.

{context}
Question: {question}"""
```

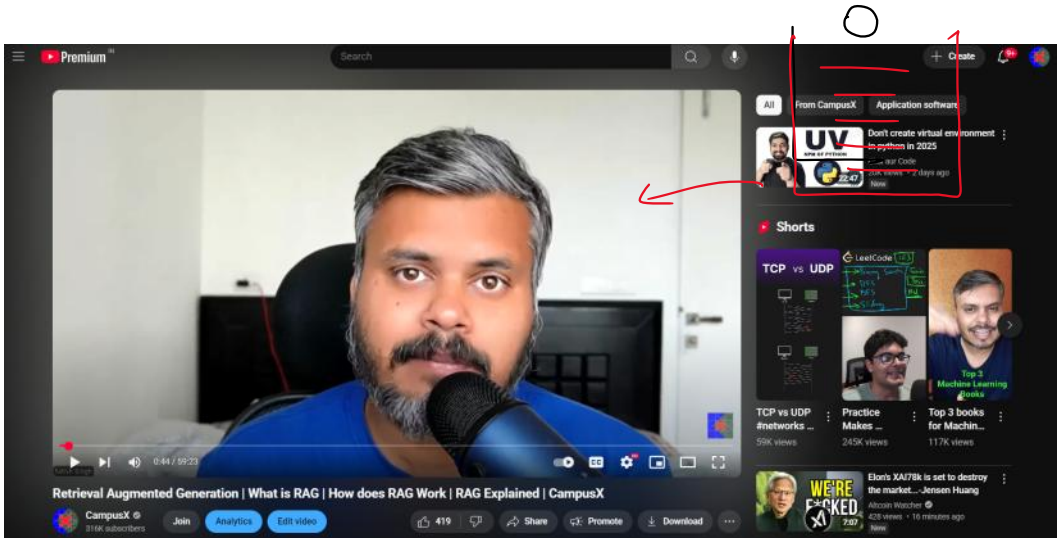
Generation - **Generation** is the final step where a **Large Language Model (LLM)** uses the **user's query** and the **retrieved & augmented context** to generate a response.



1) Private data
2) Recent developments
3) Hallucinations

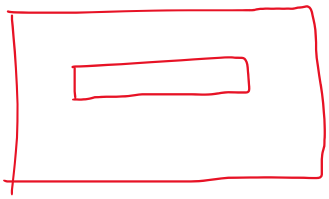
Problem Statement

21 April 2025 08:55



→ Chrome plugin

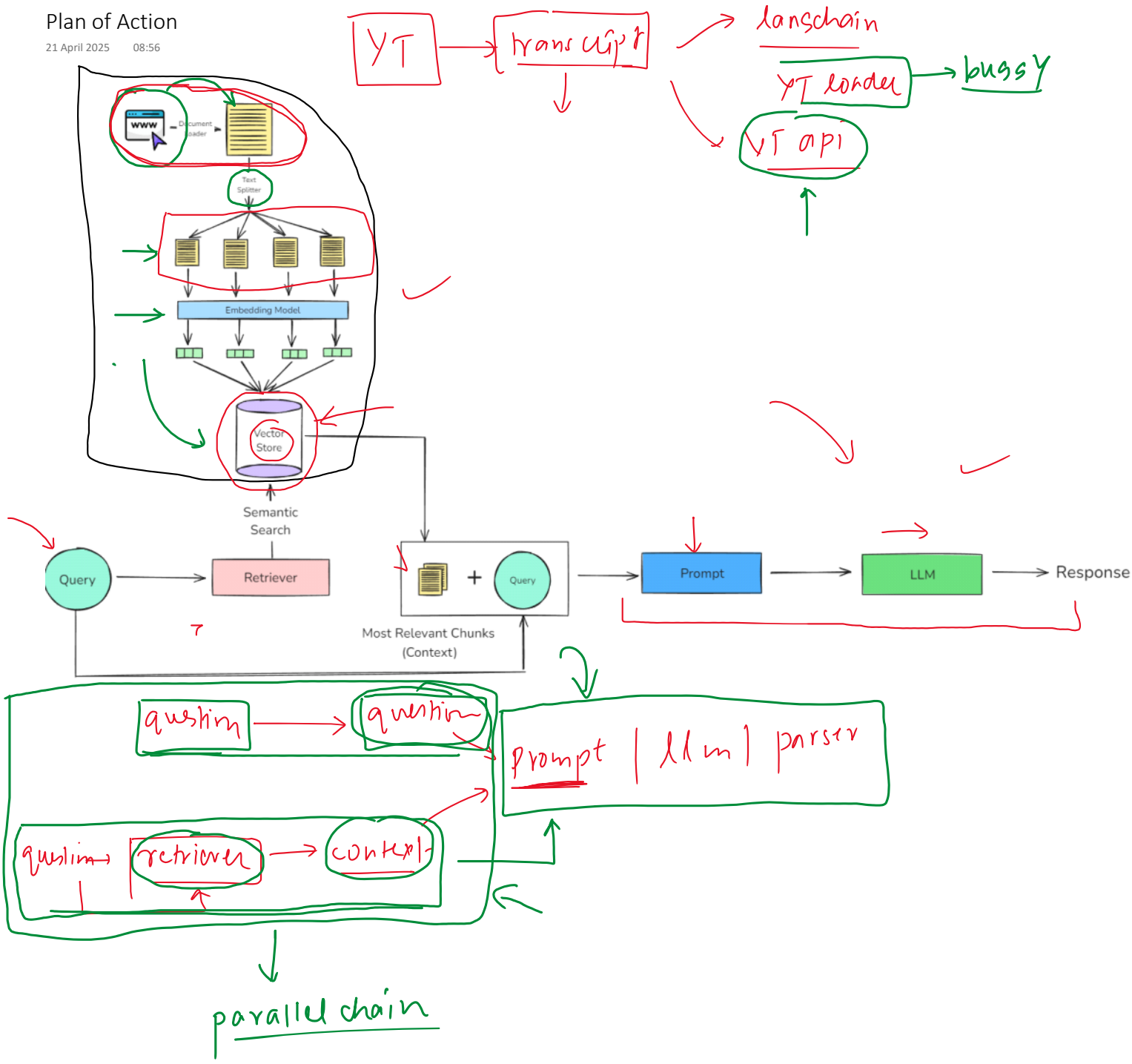
html/css/js



Streamlit →

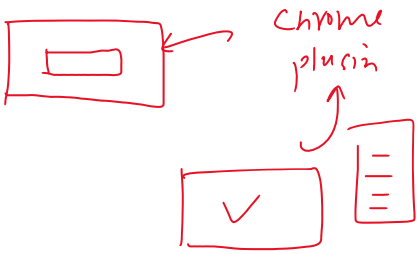
Plan of Action

21 April 2025 08:56



Improvements

22 April 2025 08:50

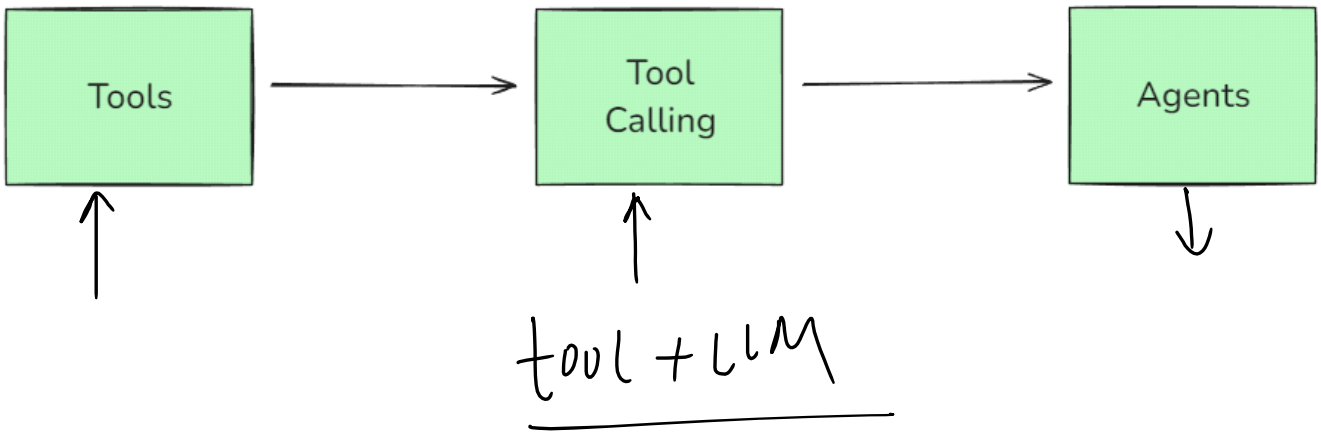


1. UI based enhancements
2. Evaluation
   a. Ragas
   b. LangSmith
3. Indexing
   a. Document Ingestion
   b. Text Splitting
   c. Vector Store — Pine Cone
4. Retrieval
   a. Pre-Retrieval
      i. Query rewriting using LLM
      ii. Multi-query generation
      iii. Domain aware routing
   b. During Retrieval
      i. MMR
      ii. Hybrid Retrieval
      iii. Reranking
   c. Post-Retrieval
      i. Contextual Compression
5. Augmentation
   a. Prompt Templating
   b. Answer grounding
   c. Context window optimization
6. Generation
   a. Answer with Citation
   b. Guard railing
7. System Design
   a. Multimodal
   b. Agentic
   c. Memory based

Metric	What It Measures
faithfulness	Is the answer grounded in the retrieved context?
answer_relevancy	Is the answer relevant to the user's question?
context_precision	How much of the retrieved context is actually useful?
context_recall	Did we retrieve all necessary information?

Overview

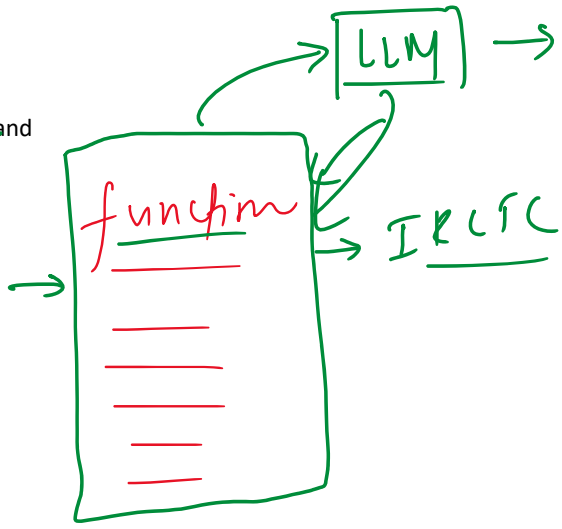
23 April 2025 15:51



What is a Tool?

23 April 2025 15:45

A tool is just a Python function (or API) that is packaged in a way the LLM can understand and call when needed.

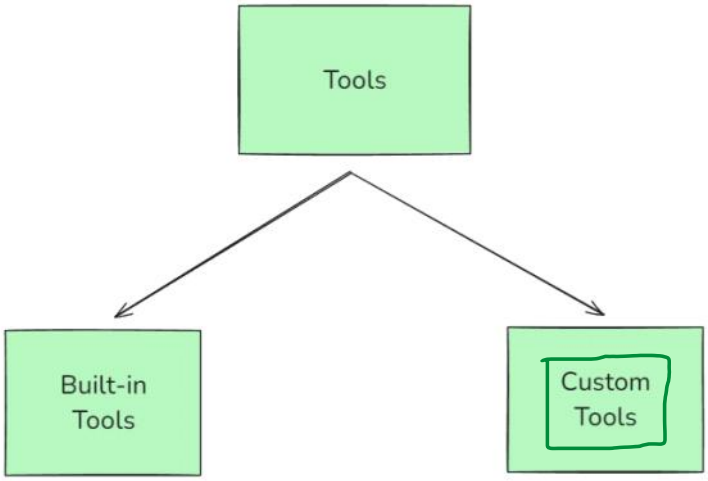


LLMs (like GPT) are great at:

- Reasoning (think)
- Language generation (speak)

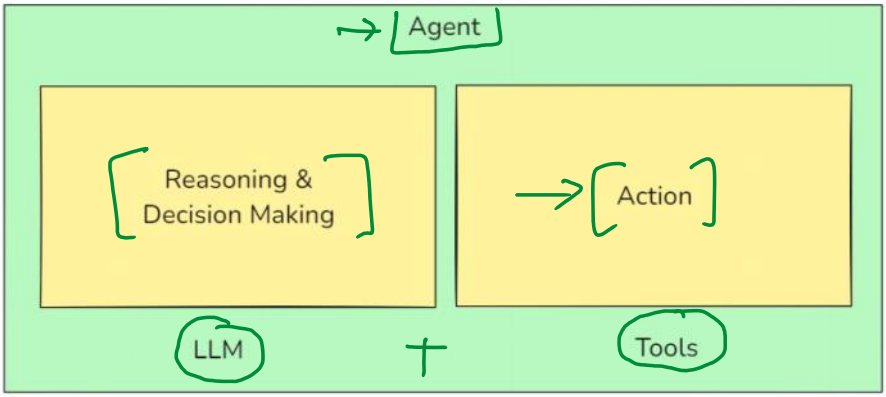
But they **can't do things like:**

- Access live data (weather, news)
- Do reliable math
- Call APIs
- Run code
- Interact with a database



How Tools fits into the Agent ecosystem

An AI agent is an LLM-powered system that can autonomously think, decide, and take actions using external tools or APIs to achieve a goal.



Built-in Tools

23 April 2025 15:46

A built-in tool is a tool that LangChain already provides for you —it's pre-built, production-ready, and requires minimal or no setup.

You don't have to write the function logic yourself — you just import and use it.

`DuckDuckGoSearchRun`	Web search via DuckDuckGo
`WikipediaQueryRun`	Wikipedia summary
`PythonREPLTool`	Run raw Python code
`ShellTool`	Run shell commands
`RequestsGetTool`	Make HTTP GET requests
`GmailSendMessageTool`	Send emails via Gmail
`SlackSendMessageTool`	Post message to Slack
`SQLDatabaseQueryTool`	Run SQL queries

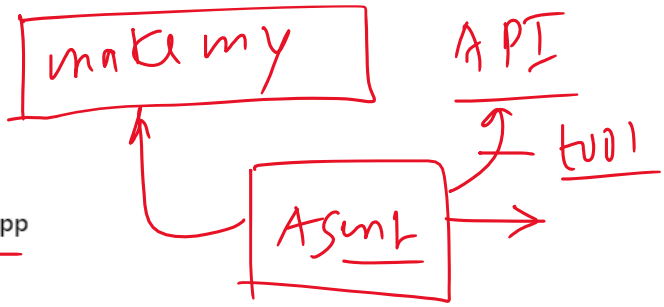
Custom Tools

23 April 2025 15:46

A custom tool is a tool that you define yourself.

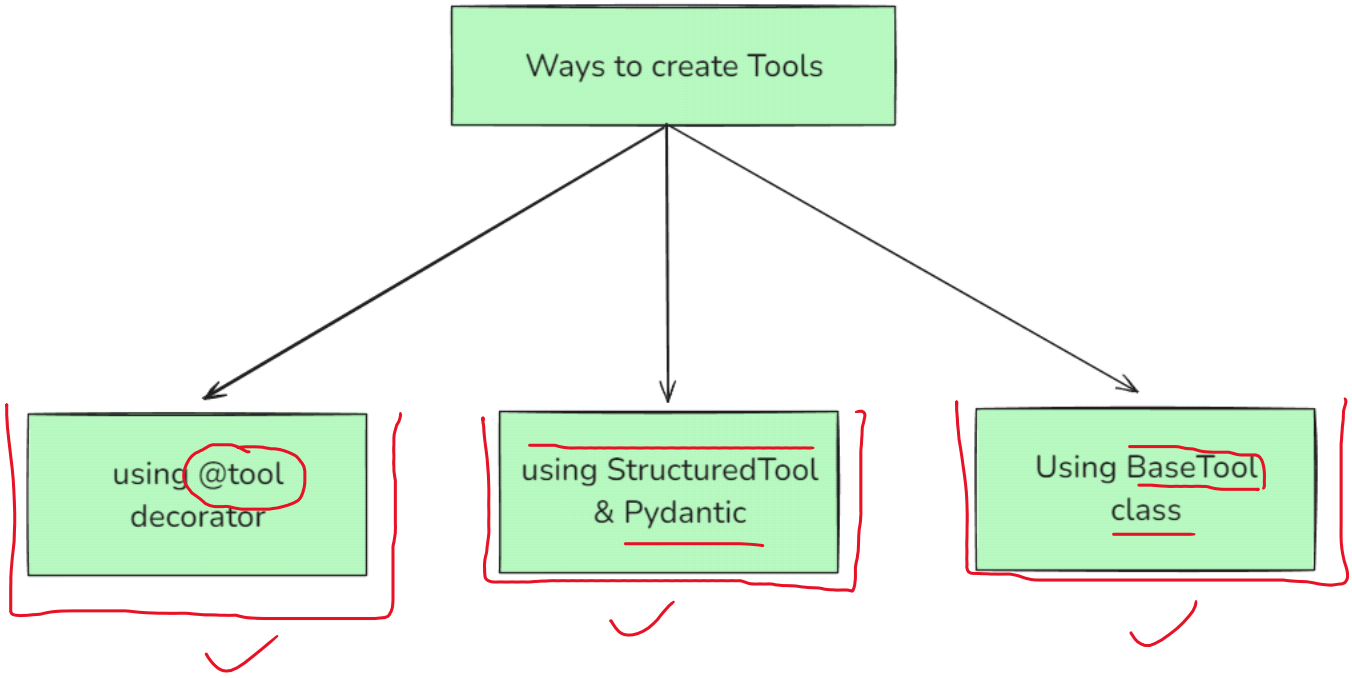
Use them when:

- You want to call **your own APIs**
- You want to **encapsulate business logic**
- You want the LLM to interact with **your database**, **product**, **or app**



Ways to create Custom Tools

23 April 2025 15:46



A **Structured Tool** in LangChain is a special type of tool where the input to the tool follows a structured schema, typically defined using a Pydantic model.

BaseTool is the abstract base class for all tools in LangChain.
It defines the core structure and interface that any tool must follow, whether it's a simple one-liner or a fully customized function.

All other tool types like **@tool, StructuredTool** are built on top of BaseTool

Toolkits

23 April 2025 15:47

A toolkit is just a collection (bundle) of related tools that serve a common purpose — packaged together for convenience and **reusability**.

In LangChain:

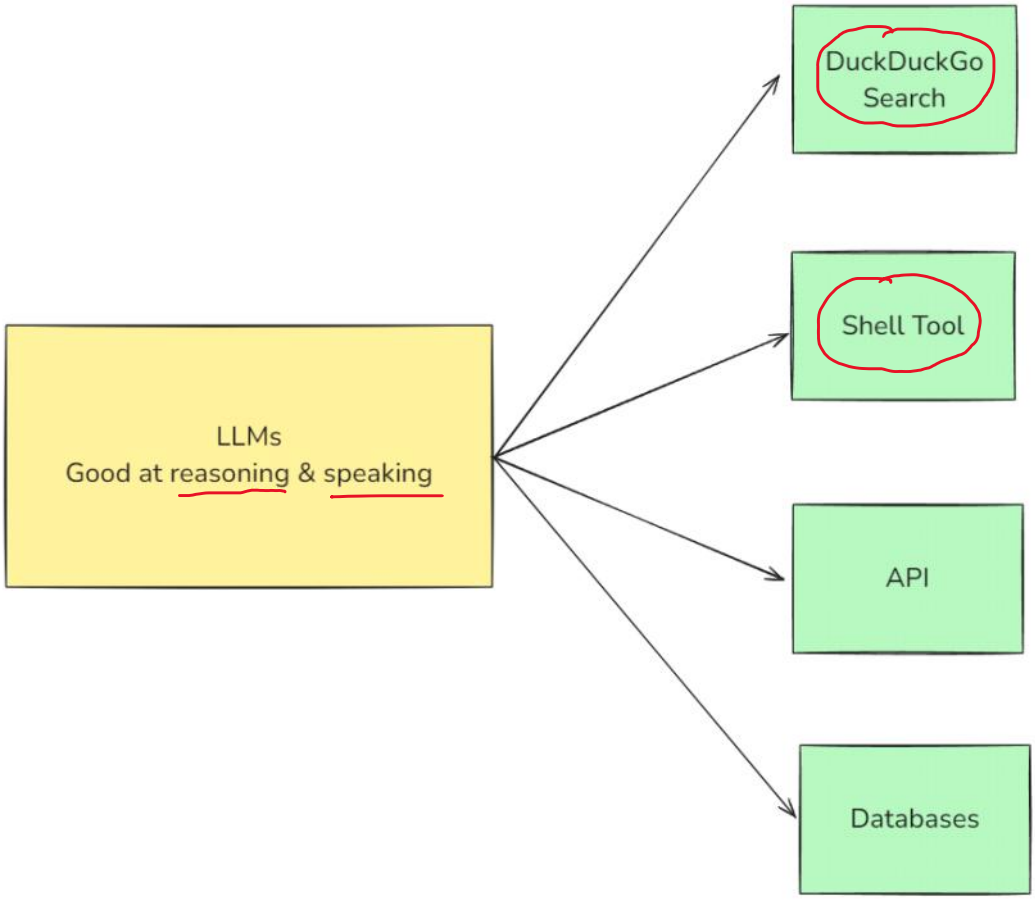
- A toolkit might be: GoogleDriveToolKit
- And it can contain the following tools

- `GoogleDriveCreateFileTool` : Upload a file
- `GoogleDriveSearchTool` : Search for a file by name/content
- `GoogleDriveReadFileTool` : Read contents of a file

Quick Revision

25 April 2025 16:18

TOOLS

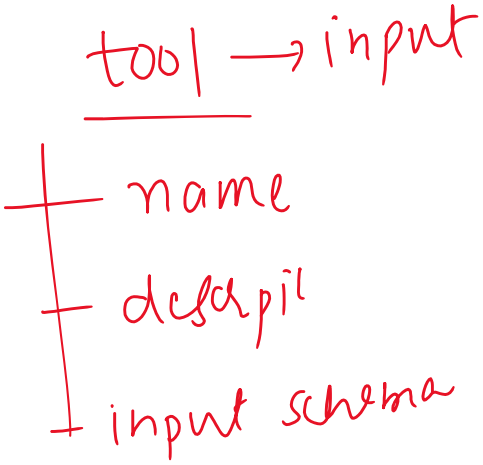


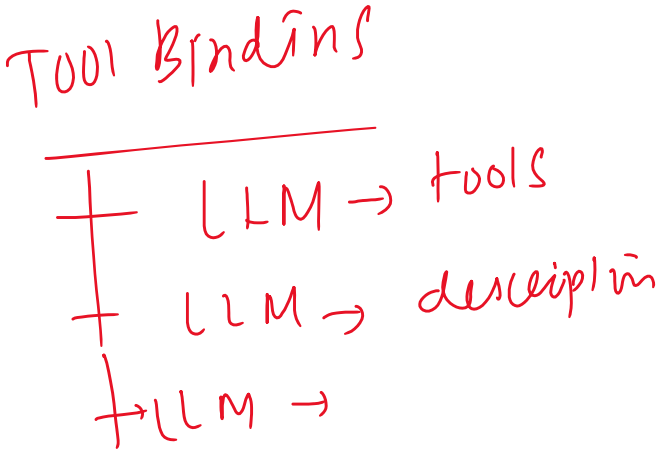
Tool Binding

25 April 2025 16:45

Tool Binding is the step where you **register tools with a Language Model (LLM)** so that:

1. The LLM knows **what tools are available**
2. It knows **what each tool does** (via description)
3. It knows **what input format to use** (via schema)





Tool Calling

25 April 2025 16:54

Tool Calling is the process where the **LLM (language model)** decides, during a conversation or task, that it needs to **use a specific tool (function)** — and generates a structured output with:

- the **name of the tool**
- and the **arguments** to call it with

🚨 The LLM **does not actually run the tool** — it just **suggests** the tool and the input arguments. The **actual execution is handled by LangChain or you**.

> "What's 8 multiplied by 7?"

The LLM responds with a **tool call**:

```json
{
  "tool": "multiply",
  "args": { "a": 8, "b": 7 }
}
```

Tool creation
Tool bindings

name : multiply
schema

Tool Execution

25 April 2025 17:12

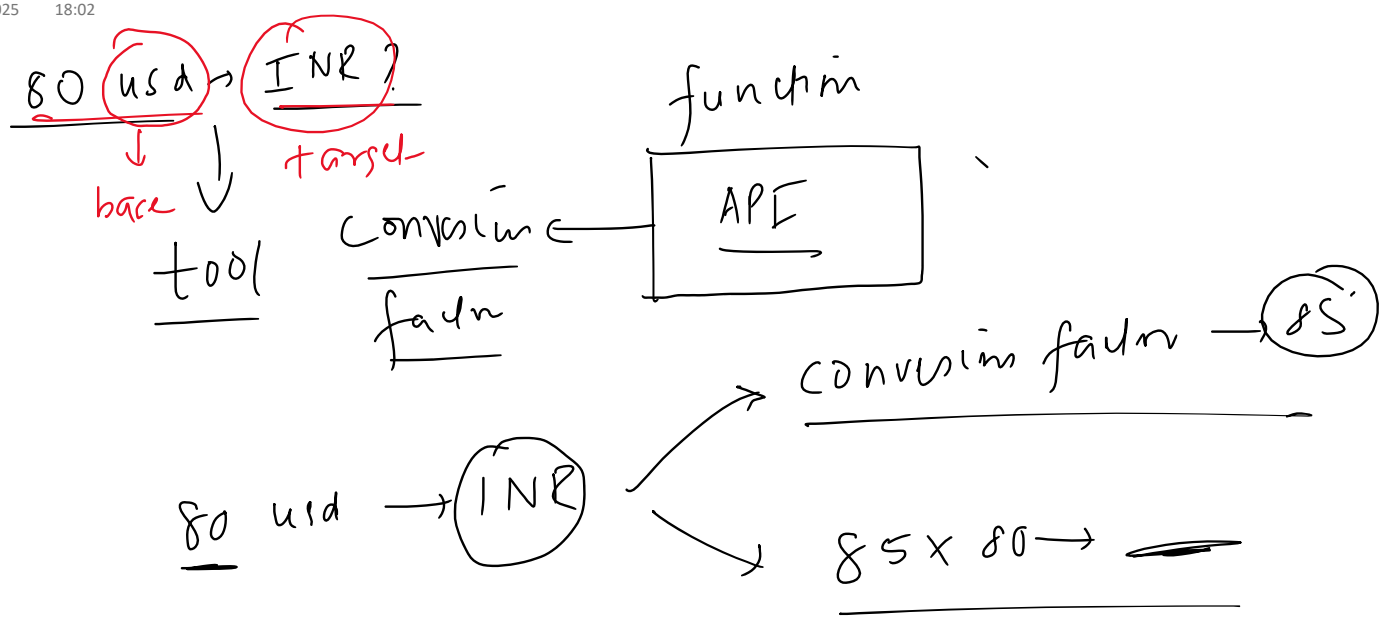
Tool Execution is the step where the **actual Python function (tool)** is run using the input arguments that the **LLM suggested during tool calling**.

In simpler words:

> 🧠 The LLM says:
> *"Hey, call the `multiply` tool with a=8 and b=7."*
>
> ⚙️ **Tool Execution** is when *you or LangChain actually run:*
> `multiply(a=8, b=7)`
> → and get the result: `56`

Currency Conversion Tool

25 April 2025 18:02



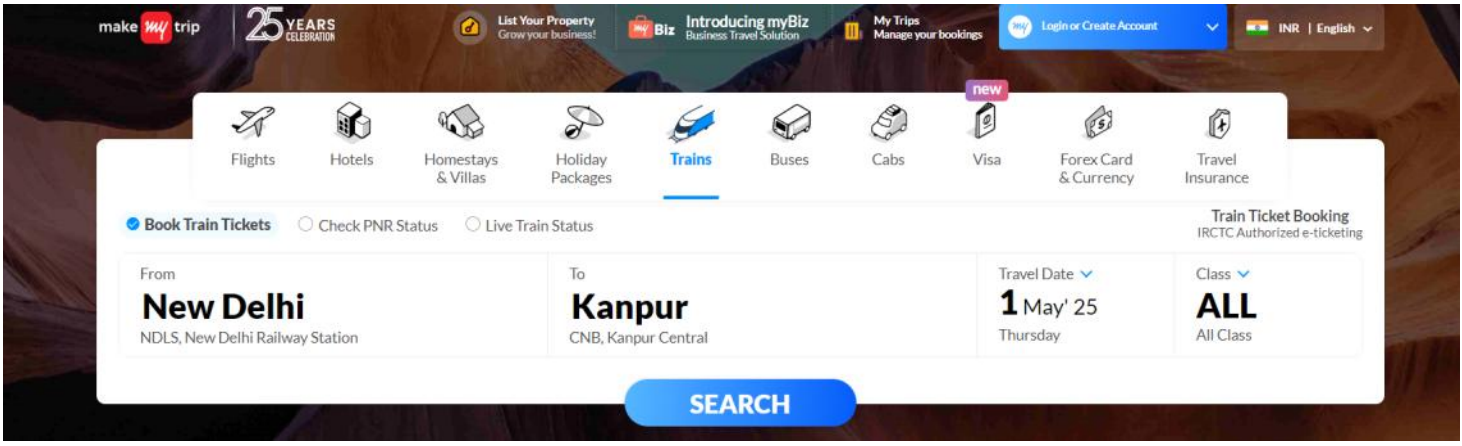
> ❌ "LLM, do **not** try to fill this argument." ✅ "I (the developer/runtime) will inject this value after running earlier tools."

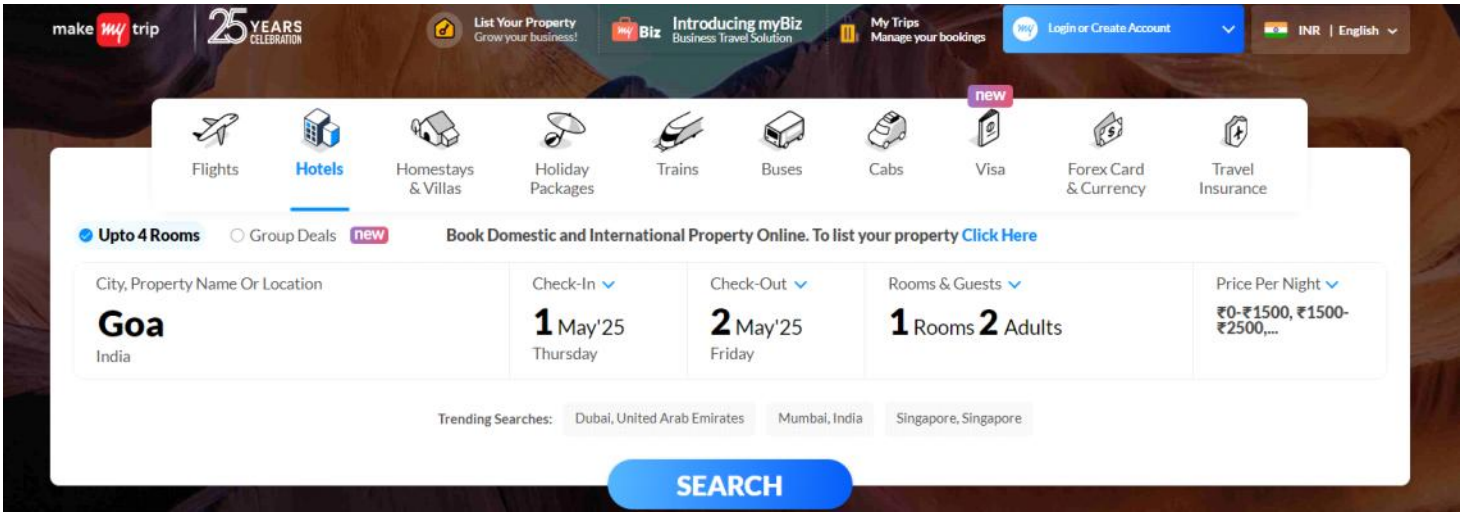
AI Agent → tools / tool calling (X)

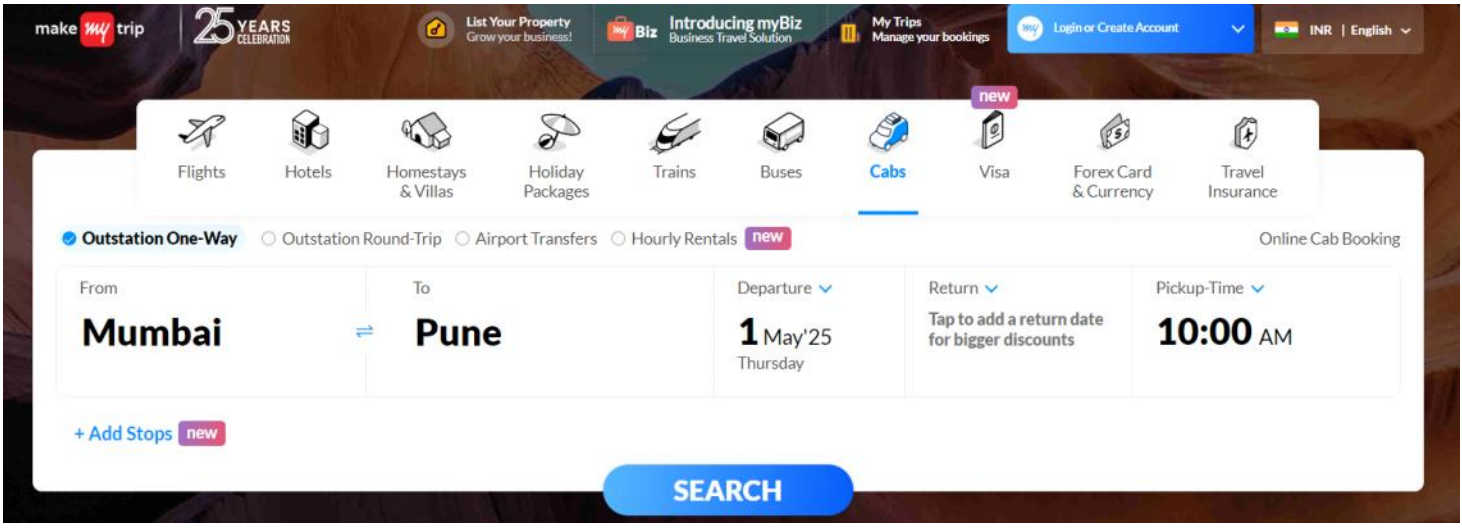
1. **User says**: "Convert 10 USD to INR."
2. **LLM thinks**: "I don't know the rate. First, let me call `get_conversion_factor`."
3. **Tool result comes**: 85.3415
4. **LLM looks at result, THINKS again**: "Now I know the rate, next I should call `convert` with 10 and 85.3415."
5. **Tool result comes**: 853.415 INR
6. **LLM summarizes**: "10 USD is 853.415 INR at current rate."

What are AI Agents

29 April 2025 22:53







✅ **User enters query:**

> "Can you create a budget travel itinerary from Delhi to Goa from 1st to 7th May?"

🧠 1. Understanding your intent

The agent internally interprets:

- ✅ Origin: Delhi
- ✅ Destination: Goa
- ✅ Dates: May 1–7
- ✅ Preference: Budget travel
- ✅ Goal: Plan complete itinerary + optimize cost

Agent thinks: *"I need to plan affordable travel, stay, local movement, and activities over 7 days."*

🔍 2. Research transport options

Agent (internally):

- Calls **IRCTC/train API** and **Flight APIs**
- Compares price, duration, and availability

Agent (to user):

> "The cheapest and most available option is the Goa Express from Delhi on April 30th night (sleeper class ₹800 or 3AC ₹1500).
> It arrives in Goa on May 1st evening. Would you like to go ahead?"

✅ User selects "3AC".

🏨 3. Search for affordable stay

Agent (internally):

- Uses hotel API or scraping tools
- Filters by:
  - Price < ₹700/night
  - Close to Baga or Anjuna beach
  - User reviews

Agent (to user):

> "I found a dorm room at The Hosteller, Baga for ₹650/night with good ratings. Shall I book this for 6 nights (May 1–7)?"

✅ User agrees.

🛵 4. **Plan local travel**

Agent (to user):

> "Scooter rentals are the most cost-effective in Goa.
> Should I pre-book a scooty for ₹300/day? (~₹1800 for 6 days)"

✅ User says yes.

🎯 5. **Generate activity plan**

Agent (internally):

- Calls sightseeing API / uses static local dataset
- Maps popular attractions across North & South Goa
- Builds a balanced plan:
  - Culture, relaxation, nightlife, shopping

Agent (to user):

> "Here's a draft plan:

- **May 2**: Baga, Calangute, Aguada Fort
- **May 3**: Old Goa churches, Fontainhas, local lunch
- **May 4**: South Goa day trip: Palolem, Cabo de Rama
- **May 5**: Anjuna flea market + sunset at Chapora
- **May 6**: Free day + beach cafe dinner"

✅ User loves it.

🛬 6. **Plan return**

Agent (to user):

> "Return options on May 7:

1. Train (₹800–₹1500, overnight)
2. Early morning flight (₹2800, 6:00 AM via SpiceJet)

Choose your preference."

✅ User picks train.

🧾 7. **Budget summary**

Agent (to user):

Item	Cost
Train (round trip)	₹3,000
Stay (6 nights)	₹3,900
Scooter + fuel	₹2,200
Food (local + budget)	₹2,500
Sightseeing & entry	₹1,000
Shopping/misc	₹1,500
Total	**~₹14,100**

> "Total trip cost: ₹14,100. Would you like to finalize the bookings?"

✅ User confirms.

📩 8. **Final Output**

Agent (to user):

> ✅ Bookings confirmed.
> 🧾 Receipt sent to your email.
> 🗓️ Calendar invite added for train departures.
> 📱 You'll get daily reminders and local recommendations during your stay.
> 💬 Need help anytime? Just ask!

> **An AI agent is an intelligent system that receives a high-level goal from a user, and autonomously plans, decides, and executes a sequence of actions by using external tools, APIs, or knowledge sources — all while maintaining context, reasoning over multiple steps, adapting to new information, and optimizing for the intended outcome.**

Goal-driven	You tell the agent *what you want*, not *how to do it*
Autonomous planning	Agent breaks down the problem and sequences tasks on its own
Tool-using	Agent calls APIs, calculators, search tools, etc.
Context-aware	Maintains memory across steps to inform future actions
Reasoning-capable	Makes decisions dynamically (e.g., "what to do next")
Adaptive	Rethinks plan when things change (e.g., API fails, no data)

Building an agent in LangChain

29 April 2025 22:53

Explanation

29 April 2025 22:53

1. ReAct

01 May 2025 14:17

ReAct is a design pattern used in AI agents that stands for Reasoning + Acting. It allows a language model (LLM) to interleave internal reasoning (Thought) with external actions (like tool use) in a structured, multi-step process.

Instead of generating an answer in one go, the model thinks step by step, deciding what it needs to do next and optionally calling tools (APIs, calculators, web search, etc.) to help it.

```
Thought: I need to find the capital of France.
Action: search_tool
Action Input: "capital of France"
Observation: Paris
Thought: Now I need the population of Paris.
Action: search_tool
Action Input: "population of Paris"
Observation: 2.1 million
Thought: I now know the final answer.
Final Answer: Paris is the capital of France and has a population of ~2.1 million.
```

ReAct is useful for:

- Multi-step problems
- Tool-augmented tasks (web search, database lookup, etc.)
- Making the agent's reasoning **transparent and auditable**

It was first introduced in the paper:

"*ReAct: Synergizing Reasoning and Acting in Language Models*" (Yao et al., 2022)

REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

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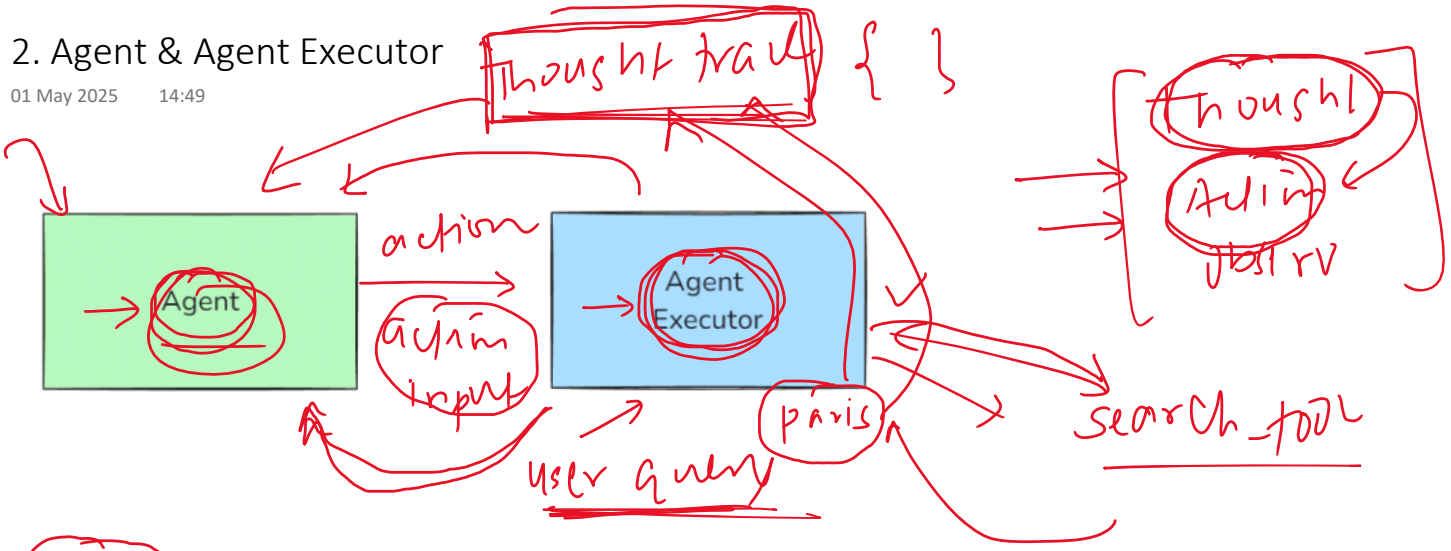
ABSTRACT

While large language models (LLMs) have demonstrated impressive performance across tasks in language understanding and interactive decision making, their abilities for reasoning (e.g. chain-of-thought prompting) and acting (e.g. action plan generation) have primarily been studied as separate topics. In this paper, we explore the use of LLMs to generate both reasoning traces and task-specific actions in an interleaved manner, allowing for greater synergy between the two: reasoning traces help the model induce, track, and update action plans as well as handle exceptions, while actions allow it to interface with and gather additional information from external sources such as knowledge bases or environments. We apply our approach, named ReAct, to a diverse set of language and decision making tasks and demonstrate its effectiveness over state-of-the-art baselines in addition to improved human interpretability and trustworthiness. Concretely, on question answering (HotpotQA) and fact verification (Fever), ReAct overcomes prevalent issues of hallucination and error propagation in chain-of-thought reasoning by interacting with a simple Wikipedia API, and generating human-like task-solving trajectories that are more interpretable than baselines without reasoning traces. Furthermore, on two interactive decision making benchmarks (ALFWorld and WebShop), ReAct outperforms imitation and reinforcement learning methods by an absolute success rate of 34% and 10% respectively, while being prompted with only one or two in-context examples.

2. Agent & Agent Executor

01 May 2025 14:49



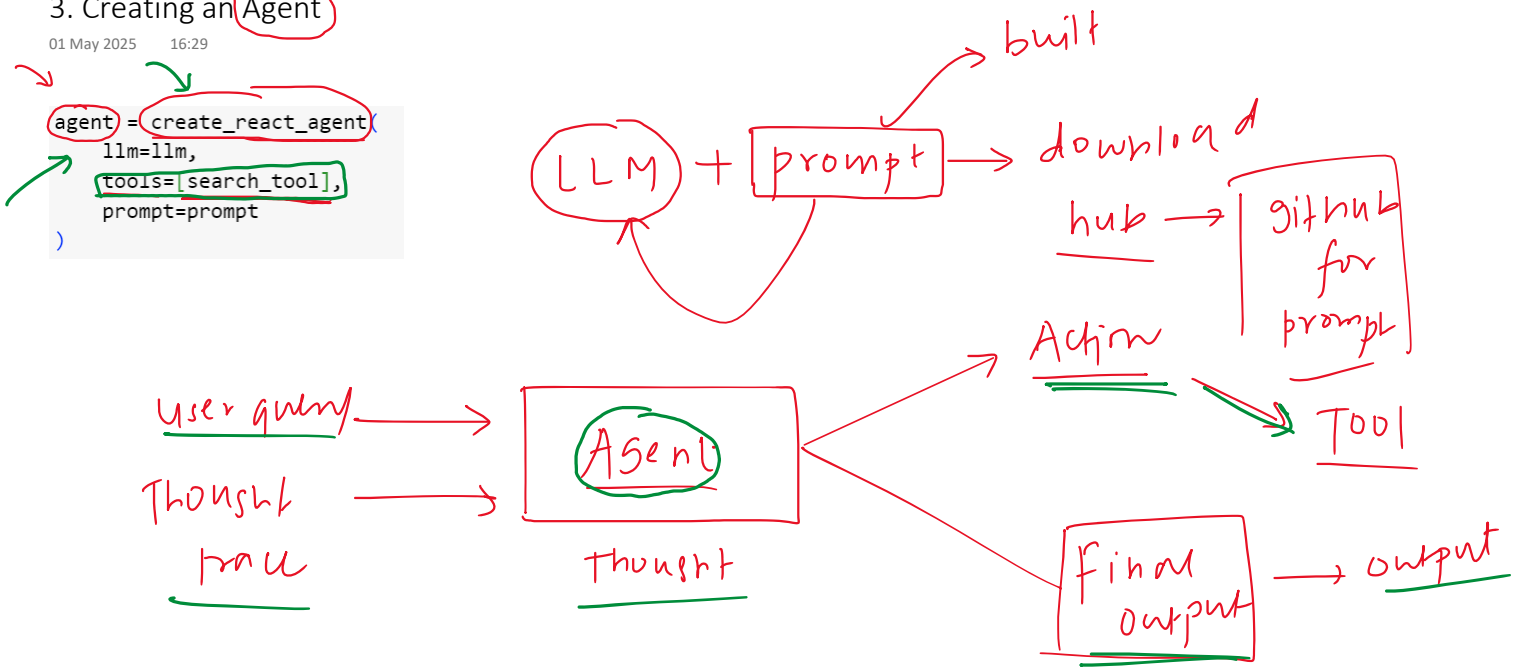
`AgentExecutor` orchestrates the **entire loop**:

1. Sends inputs and previous messages to the agent
2. Gets the next `action` from agent
3. Executes that tool with provided input
4. Adds the tool's `observation` back into the history
5. Loops again with updated history until the agent says `Final Answer`.

3. Creating an Agent

01 May 2025 16:29

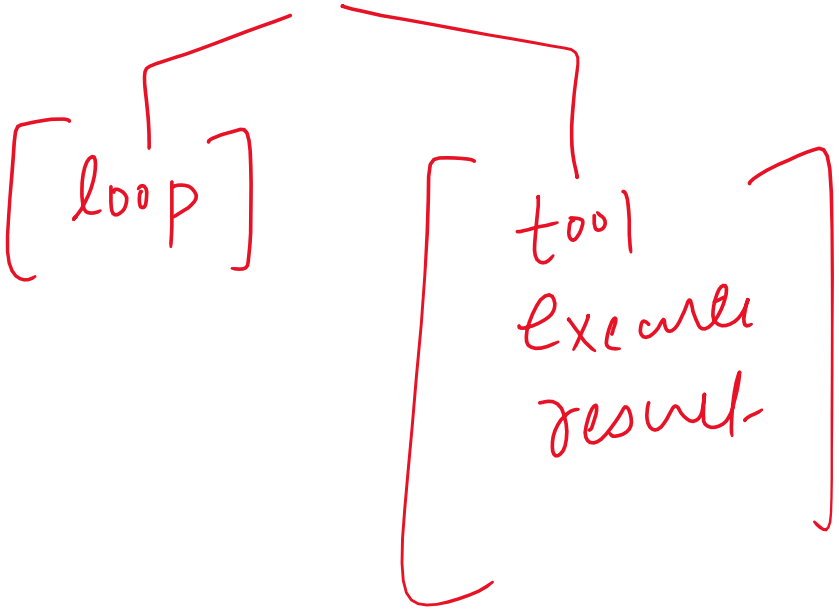
```
agent = create_react_agent(
    llm=llm,
    tools=[search_tool],
    prompt=prompt
)
```



4. Creating an Agent Executor

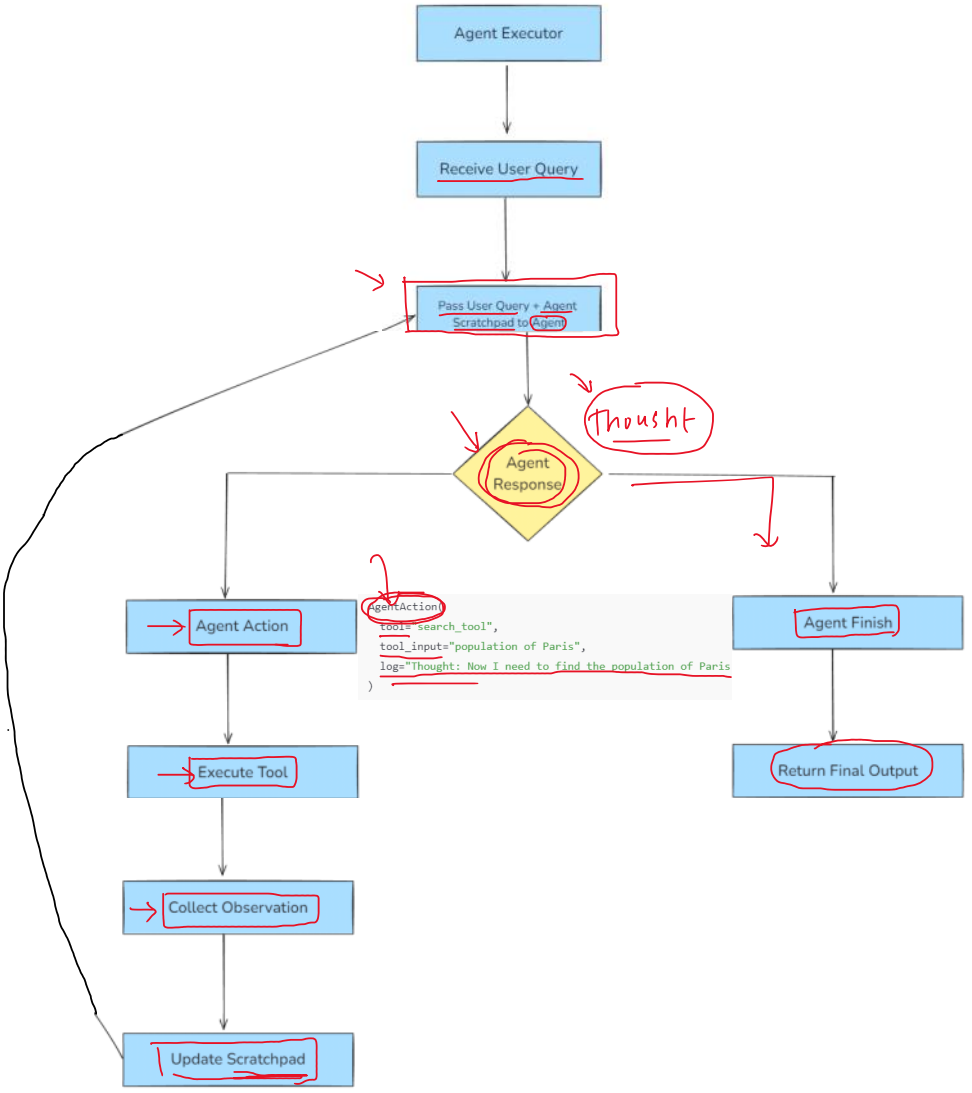
01 May 2025 16:30

```
agent_executor = AgentExecutor(
    agent=agent,
    tools=[search_tool],
    verbose=True
)
```



5. Flow Chart

01 May 2025 16:30



6. Example

01 May 2025 17:23

🧠 Input Query:

> "What is the capital of France and what is its population?"

```
Answer the following questions as best you can. You have access to the following tools:

search_tool: Useful for answering general knowledge questions by querying a search API.

Use the following format:

Question: the input question you must answer
Thought: you should always think about what to do
Action: the action to take, should be one of [search_tool]
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation can repeat N times)
Thought: I now know the final answer
Final Answer: the final answer to the original input question

Begin!

Question: What is the capital of France and what is its population?
Thought:
```

```
Thought: I need to find the capital of France first.
Action: search_tool
Action Input: "capital of France"
```

```
AgentAction(
  tool="search_tool",
  tool_input="capital of France",
  log="Thought: I need to find the capital of France first.
)
```

```
observation = search_tool("capital of France")
```

→ `"Paris is the capital of France."`

```
Thought: I need to find the capital of France first.
Action: search_tool
Action Input: "capital of France"
Observation: Paris is the capital of France.
```

📌 Current state so far:

- **User Input**:

`What is the capital of France and what is its population?`

- **Agent Scratchpad after Step 2**:

text

```
Thought: I need to find the capital of France first.
Action: search_tool
Action Input: "capital of France"
Observation: Paris is the capital of France.
```

```
Answer the following questions as best you can. You have access to the following tools

search_tool: Useful for answering general knowledge questions by querying a search API.

Use the following format:

Question: the input question you must answer
Thought: you should always think about what to do
Action: the action to take, should be one of [search_tool]
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation can repeat N times)
Thought: I now know the final answer
Final Answer: the final answer to the original input question

Begin!

Question: What is the capital of France and what is its population?
Thought: I need to find the capital of France first.
Action: search_tool
Action Input: "capital of France"
Observation: Paris is the capital of France.
Thought:
```

```
Thought: Now I need to find the population of Paris.
Action: search_tool
Action Input: "population of Paris"
```

```
AgentAction(
  tool="search_tool",
  tool_input="population of Paris",
```

```
AgentAction(
  tool="search_tool",
  tool_input="population of Paris",
  log="Thought: Now I need to find the population of Paris.
)
```

```
observation = search_tool("population of Paris")
```

```
"Paris has a population of approximately 2.1 million."
```

```
Thought: I need to find the capital of France first.
Action: search_tool
Action Input: "capital of France"
Observation: Paris is the capital of France.
Thought: Now I need to find the population of Paris.
Action: search_tool
Action Input: "population of Paris"
Observation: Paris has a population of approximately 2.1 million.
```

```
Answer the following questions as best you can. You have access to the following tools:

search_tool: Useful for answering general knowledge questions by querying a search API.

Use the following format:

Question: the input question you must answer
Thought: you should always think about what to do
Action: the action to take, should be one of [search_tool]
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation can repeat N times)
Thought: I now know the final answer
Final Answer: the final answer to the original input question

Begin!

Question: What is the capital of France and what is its population?
Thought: I need to find the capital of France first.
Action: search_tool
Action Input: "capital of France"
Observation: Paris is the capital of France.
Thought: Now I need to find the population of Paris.
Action: search_tool
Action Input: "population of Paris"
Observation: Paris has a population of approximately 2.1 million.
Thought:
```

```
Thought: I now know the final answer.
Final Answer: Paris is the capital of France and has a population of approximately 2.1 million.
```

```
Thought: I now know the final answer.
Final Answer: Paris is the capital of France and has a population of approximately 2.1 million.
```

```
AgentFinish(
  return_values={"output": "The capital of France is Paris..."},
  log="Thought: I now know the final answer.\nFinal Answer: ..."
)
```

```
{
  "output": "Paris is the capital of France and has a population of approximately 2.1 million."
}
```